# Recalibrating Reality: Sensory Reweighting and Cybersickness Susceptibility in Virtual Reality

by

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# Author's Declaration

I hereby declare that I am the sole author of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

# Abstract

This dissertation investigates how the brain dynamically reweights sensory cues to resolve multisensory conflicts in virtual reality (VR), with a specific focus on the etiology of cybersickness. The research characterises how visual, vestibular, and proprioceptive cues are integrated and recalibrated following VR gameplay while accounting for individual differences in these processes which dictate susceptibility to sickness.

Across three experiments, this work challenges the assumption that VR exposure triggers a uniform sensory shift. Experiment 1 utilised the Subjective Visual Vertical (SVV) task, finding that high-intensity exposure shifted participants' perceived upright toward gravitational vertical and away from the body axis. Crucially, individuals who demonstrated greater reweighting reported lower cybersickness, suggesting that rapid sensory adaptation acts as a protective mechanism. Experiment 2 employed the Oriented CHAracter Recognition Test (OCHART) to further tease apart the relative weighting of visual and gravitational cues. While VR exposure did not produce a uniform group-level shift in the Perceptual Upright (PU), exploratory analyses revealed that increased visual weighting post-exposure was associated with higher cybersickness, indicating that a failure to down-weight unreliable visual inputs may be maladaptive. Experiment 3 attempted to manipulate these weights by degrading vestibular reliability through stochastic electrical vestibular stimulation (EVS). Contrary to the hypothesis that reducing vestibular certainty would force a beneficial reweighting, EVS exacerbated cybersickness and increased attrition, while cue-weighting models remained unstable.

Altogether, this work demonstrates that cybersickness is not merely a product of sensory conflict, but a failure of the central nervous system to successfully reweight unreliable cues. The take-home message is twofold: first, that sensory plasticity is a predictor of tolerance, where the ability to rapidly de-emphasise body-referenced or visual cues in favour of stable gravitational inputs reduces distress. Second, the results suggest that external disruption of vestibular signals (via EVS) may hinder rather than help adaptation. These findings provide a framework for developing personalised VR protocols and inform the design of habituation-based counter-measures to enhance user experience in immersive virtual environments.

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This dissertation would not exist without Dr. Michael Barnett-Cowan. You took a chance on me and gave me the space to develop as a researcher while knowing exactly when to push me forward. Your patience through the long stretches, your willingness to sit with messy data and help me find the story in it, and your genuine investment in my growth as a scientist shaped this work at every stage. I am a better thinker because of you, and I hope this dissertation reflects that.

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To my parents, everything I have accomplished traces back to a decision you made before I could understand what it meant. You left behind the life you knew, your home, your family, your careers, and your language and moved to a new country because you believed your children deserved a better future. That is the most selfless thing I can imagine anyone doing. Every word of this dissertation exists because of that sacrifice. I will never be able to repay it, but I hope this makes you proud.

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# Dedication

For my sister, Sara. My fiercest supporter, for as long as I can remember.

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# Chapter 1
# Introduction

## 1.1 Overview

What we call "reality" is not a direct read-out of the external world but a brain-derived model that fuses multisensory data with prior expectations to generate a coherent, actionable scene (Fetsch et al., 2013; Seth & Friston, 2016). Contemporary work in perceptual neuroscience describes this process as a form of controlled hallucination, whereby the brain's predictive signals are continuously compared with incoming sensory evidence and updated to minimise error (Clark, 2013; Manning et al., 2024; Sprevak & Smith, 2023). Multisensory integration research shows that the weighting of visual, vestibular, and proprioceptive cues changes dynamically with their moment-to-moment reliability, allowing a percept that is stable and plastic (Ernst & Bülthoff, 2004; Fetsch et al., 2013). As Troje (2019) notes, virtual reality (VR) technologies exploit this constructive nature of perception by replacing reliable sensory regularities with artificial ones, thereby offering a unique laboratory for observing how the brain recalibrates its model of reality when confronted with persistent cue conflicts (Troje, 2019).

VR makes this problem experimentally tractable because it can impose sustained mismatches between normally correlated cues (e.g., visual optic flow signalling self-motion while vestibular and body cues signal relative stasis), often producing cybersickness. A major unresolved issue is why susceptibility varies so widely across individuals and, mechanistically, whether that variability reflects differences in the capacity to reweight multisensory cues under conflict. Although sensory conflict accounts describe why VR can provoke sickness (Reason & Brand, 1975), we still lack a clear account of how VR exposure alters multisensory weighting in spatial-orientation judgments and whether those changes systematically relate to symptom severity (Saredakis et al., 2020; Weech et al., 2020; Kourtesis et al., 2024). This dissertation targets that gap by quantifying cue weighting and its individual variability before and after sickening VR exposure.

The central nervous system (CNS) is challenged by having to make decisions and initiate actions based on constantly changing information from different sensory systems, each detecting different types of energies in the environment. This is known as the "binding problem" (Treisman, 1996). It is thought that multisensory integration is one step that the CNS takes to resolve this binding problem. Multisensory integration is a critical CNS strategy to address this issue, enhancing decision quality through the

synthesis of sensory data, thus leveraging redundancies and reliability across modalities (Nashner & Berthoz, 1978). A key CNS capability is sensory reweighting, the adjustment of sensory input significance according to task demands and information reliability, facilitated by short-term neural plasticity (Maurer et al., 2006; Peterka, 2002). This flexibility is crucial for tasks like balance maintenance under perturbations (Assländer & Peterka, 2014; Oie et al., 2002; Peterka, 2002).

Sensory reweighting has also been identified as a process that may protect an organism from harm in situations where sensory information may disagree (Peterka, 2002). For example, during self-motion, there is usually correspondence between incoming visual information and our movement. In human-made environments, such as in a vehicle, the driver of a calibrated vehicle has good correspondence between their movement of the controls and the consequent movement of the vehicle and self. However, passengers receive incongruent information about self-motion. Here, passengers do not exert movements that correspond with vehicle and self-motion. Worse, passengers who receive little information about self-motion from one sensory modality and more information from another, such as at the back of the vehicle with little visual input corresponding with motion but constant stimulation from the vestibular system, are in a state of sensory conflict.

Critically, this reweighting process is not uniform across individuals. People differ substantially in how quickly and effectively they adjust sensory weights in response to conflict, and these individual differences have meaningful consequences for susceptibility to motion sickness and cybersickness (Weech et al., 2020; Kourtesis et al., 2024). Factors such as sex, age, prior motion sickness history, gaming experience, and inherent sensory dominance profiles all contribute to variability in how the CNS resolves conflicting inputs (Saredakis et al., 2020; Stanney, Fidopiastis, et al., 2020). Some individuals appear to be highly visually dependent, relying on visual cues even when they conflict with vestibular and proprioceptive signals, while others are more vestibular- or body-dominant in their weighting strategy (Witkin et al., 1954; Josupeit, 2024). Populations with specialized sensory training, such as dancers and athletes, may exhibit distinct cue weighting profiles owing to years of refined proprioceptive and vestibular calibration (Barnett-Cowan et al., 2010; Beck et al., 2020). This variability is not merely noise in the data; it is central to understanding why some individuals tolerate VR with minimal discomfort while others experience debilitating sickness from the same exposure. A core aim of this dissertation is to characterize this variability and test whether individual differences in sensory reweighting capacity can explain differential susceptibility to cybersickness.

This dissertation investigates CNS strategies for resolving sensory conflicts, through experiments that examine sensory weighting changes following VR-induced cybersickness, aiming to illuminate

countermeasures for such sensory conflicts in artificial environments. It presents three unique experiments that each measure the relative weighting of visual, vestibular, and bodily cues before and after exposure to VR games that are known to make people sick from sensory conflict (cybersickness). The first two experiments assess the effects of VR exposure on two separate measures of multisensory integration and sensory reweighting, while the third experiment uses electrical vestibular stimulation (EVS) to provoke sensory reweighting directly. Across all three studies, a unifying focus is on individual variability in cue weighting as a predictor of cybersickness outcomes, rather than relying solely on group-level averages. By identifying the underlying processes by which the CNS may reweight sensory information, we may be able to develop better tools or countermeasures to prevent cybersickness and sensory conflict in general within human-made environments.

This introduction outlines the sensory systems involved in the proposed studies, delves into motion and cybersickness theories, and discusses multisensory integration and sensory reweighting processes. The introduction will conclude with a summary of each experimental chapter and its specific hypotheses.

## 1.2 Visual System

### 1.2.1 Overview

The visual system is the part of the CNS which gives organisms the ability to process visual detail by detecting and interpreting information from visible light to build a representation of the surrounding environment. Every object in the environment is made of material substances, and it is because of the way the light is reflected from different surfaces that visual perception is possible. The light reflected from the surfaces in the environment forms a densely structured optic array at a point of observation (Gibson, 1955, 1957; Niehorster, 2021) and becomes focused before reaching the photoreceptors in the retina as light passes through the cornea and lens of the eye (Kandel et al., 2000). The neural signals are initially processed by the retina. The conversion of light into electrical signals, or phototransduction, occurs when light alters photopigments in the rods and cones of the retina, triggering a chemical change that modifies the membrane potential. These electrical signals are refined by horizontal, bipolar, and amacrine cells and then conveyed via ganglion cells whose axons form the optic nerve. At the optic chiasm, fibres from the nasal halves of each retina cross over, ensuring information from each visual field is processed by the opposite brain hemisphere. The signals reach the lateral geniculate nucleus (LGN) in the thalamus, where they are further processed and sent through the optic radiations to the primary visual cortex (V1) in the occipital lobe. V1 processes these signals, recognizing simple visual elements and creating a retinotopic map of the visual field, marking the beginning of complex visual scene perception (Kandel et al., 2000).

This intricate pathway, from light perception to cortical processing, underscores the complexity and efficiency of the visual system in the brain.

V1 is the earliest cortical visual area, which is located in and around the calcarine fissure in the occipital lobes of mammals. It is highly specialized for processing information about static and moving objects and pattern recognition. The two streams of visual processing, known as the dorsal and ventral streams, represent a fundamental organizational principle in the visual system, beginning in V1 and projecting to different areas of the brain to support distinct aspects of visual perception and action. The ventral stream, often referred to as the "what" pathway, extends from V1 into the temporal lobe, specifically targeting the inferotemporal cortex. This pathway is crucial for object recognition, facial recognition, and the processing of colour and fine detail, enabling the identification of objects within the visual field. In contrast, the dorsal stream, or the "where/how" pathway, projects from V1 to the parietal lobe, particularly focusing on the posterior parietal cortex. It is involved in processing the spatial location of objects, the movement of objects and the observer, and coordinating visual information with motor actions, facilitating spatial awareness, and guiding interactions with objects. This dual-stream model, initially proposed by Ungerleider and Mishkin in the 1980s (Mishkin et al., 1982; Ungerleider et al., 1983, 1984), highlights the division of labour in the visual system, where the ventral stream deciphers the content of what we see, while the dorsal stream interprets the spatial context and supports visually guided actions, underlying our ability to understand and navigate our environment effectively.

Although this dual-stream framework is a useful organizing principle, the dorsal and ventral streams are not independent or mutually exclusive, and their integration is important for binding object identity with spatial context and action-relevant information. Three broad interaction architectures have been proposed, including reintegration within shared target regions, recurrent feedback loops between pathways, and direct transfer of information between the streams at multiple stages (Cloutman, 2013). More recent perspectives similarly emphasize that dorsal and ventral processing should be treated as highly interconnected networks that cooperate to support perception and visually guided behaviour, rather than as strictly segregated streams (Goodale & Milner, 2018; Milner, 2017; Rolls, 2024). The dorsal stream can carry identity-relevant object information and can predict subsequent ventral responses, consistent with dorsal-to-ventral information transfer during object recognition (Ayzenberg et al., 2023; Ayzenberg & Behrmann, 2022). Anatomically, these interactions are supported by vertical occipital pathways that bridge dorsal and ventral visual cortex, including the vertical occipital fasciculus identified with combined functional mapping and tractography, corroborated by dissection-based structural evidence, and elaborated by recent work on developmental and comparative primate anatomy (Jitsuishi et al., 2020; Takemura et al., 2016, 2024; Vinci-Booher et al., 2022).

### 1.2.2 Optic Flow

Optic flow can be defined as the motion of all the surface elements in the visual world. Humans are constantly surrounded by moving people and objects, and as individuals move through the world, the objects, and surfaces within the visual environment flow around them (Lee & Kalmus, 1980). A visualization of optic flow can be seen in Figure *1.1*. When an individual moves through their environment, the optic flow generates a complex field of visual motion cues that the brain interprets to determine direction, speed of movement, and to navigate effectively. The human visual system can determine the current direction of travel from the focus of expansion of visual input. As objects move past the visual field in the opposite direction to the individual's movement, the brain can deduce that the body must be moving in the opposite direction to the movement of the visual field (Zhai et al., 2021). For example, as an individual turns their head to the right, their visual field moves to the left relative to their head. The brain will use this information from optic flow to deduce the true direction of the head. However, discrepancies in optic flow, especially when visual information conflicts with vestibular and proprioceptive inputs, can lead to motion sickness (Adhanom et al., 2022). This phenomenon, often triggered in environments where movement is perceived visually without corresponding physical motion (as in virtual reality settings or while reading in a moving vehicle), arises from the brain's difficulty in reconciling mismatched sensory information. Optic flow's crucial role in motion perception thus links to the experience of motion sickness through the mechanism of sensory conflict, where the brain's expectations based on visual input clash with the lack of congruent physical motion cues, leading to symptoms such as dizziness, nausea, and disorientation. This relationship underscores the importance of coherent sensory integration for comfortable and accurate perception of motion and spatial orientation (Adhanom et al., 2022).

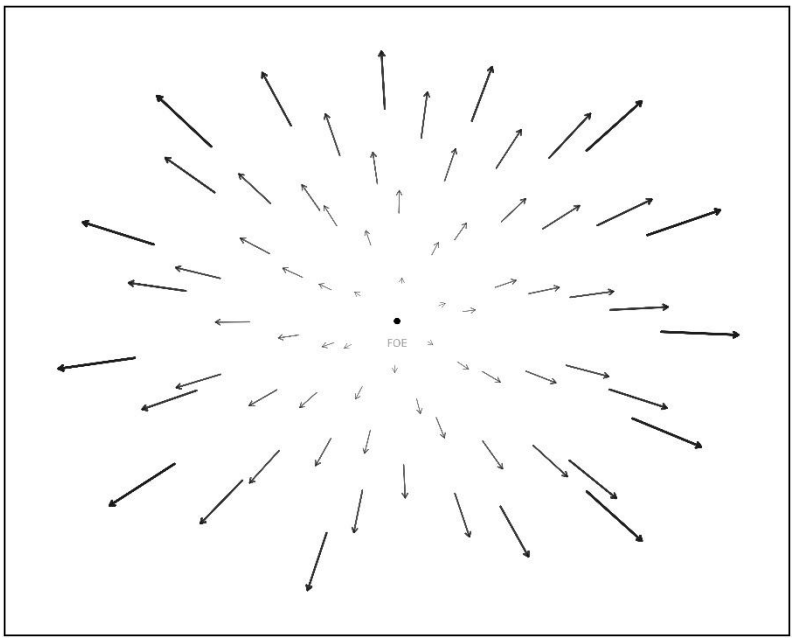


Figure 1.1: A visualization of the objects and surfaces within the visual environment that flow around a point.

In a virtual environment, where the acceleration of the optic flow can be adjusted, a higher acceleration of visual cues may cause increased cue conflict between the visual and vestibular systems. This is due to incongruent sensory information presented to the CNS from the two modalities; the visual system is presented with cues that portray movement, while the vestibular system does not perceive movement to be occurring.

## 1.3 Vestibular System

### 1.3.1 Overview

The vestibular system is a key component of the inner ear and CNS. The vestibular system plays a crucial role in maintaining balance, posture, and the body's orientation in space. Comprising the semicircular canals, which detect rotational movements, and the otolith organs (the utricle and saccule) that sense linear accelerations and gravity, this system provides continuous sensory feedback about motion and equilibrium (Kandel et al., 2000). When the head moves, fluid within the semicircular canals shifts, stimulating hair cells that convert mechanical movements into electrical signals sent to the brain. Similarly, the otolith organs respond to changes in head angle and linear movement, allowing for the detection of orientation relative to gravity and acceleration. These signals are integrated with information from the visual and proprioceptive systems in the brain, enabling the vestibular system to contribute to

reflexes that maintain gaze stabilization through the vestibulo-ocular reflex (VOR) and to adjust posture to preserve balance. Dysfunction in the vestibular system can lead to symptoms such as dizziness, vertigo, balance issues, and visual disturbances, highlighting its essential function in daily activities and overall well-being (Kandel et al., 2000).

### 1.3.2 Vestibular Anatomy

The vestibular system is a structure of organs that are located within the inner ear, known as the vestibular labyrinth. It is comprised of three tubes, the semicircular canals and two sacs, the utricle and saccule. The semicircular canals are primarily responsible for detecting rotational acceleration, with two oriented vertically and one horizontally. All three are orthogonal to each other, allowing detection across all axes of rotation. The utricle and saccule detect linear acceleration in the horizontal and vertical planes, respectively. The vestibular labyrinth functions based on a mechanoelectrical framework: inertia from acceleration of the head and gravity causes the fluid found within the labyrinth to flow, which displaces the hair bundles (kinocilium and stereocilia) attached to the hair cells; this triggers the activation and firing of those cells, generating a signal that is transduced through the nervous system (Goldberg et al., 2012).

### 1.3.3 Vestibular Nerve

Axons originating from the superior and inferior divisions of the vestibular ganglion join to form the vestibular nerve, which subsequently combines with the cochlear nerve to create the vestibulocochlear nerve. The nerve fibres make their way into the brainstem at the pontomedullary junction. At this juncture, the vestibular nerve diverges from the cochlear nerve. Most of the afferent vestibular fibres project to the ipsilateral vestibular nuclear complex located in the pons. Additionally, some nerve fibres extend their projections to the flocculonodular lobe of the cerebellum and the adjacent vermis (Hain & Helminski, 2014).

### 1.3.4 Vestibular Nuclei

The vestibular nuclei complex is the primary processor of vestibular input and consists of four major nuclei: medial, superior, lateral, and inferior. They are located under the floor of the fourth ventricle and extend from the rostral medulla to the caudal pons in two major columns (Tascioglu, 2005).

The superior vestibular nucleus receives strong canal-related input, particularly from the crista ampullaris of the superior and posterior semicircular ducts, and plays a key role in the vestibulo-ocular reflex (VOR).

Its ascending efferent fibres travel in the medial longitudinal fasciculus to the extraocular motor nuclei, supporting stable vision during head motion (Hain & Helminski, 2014; Lee, 2016). The inferior vestibular nucleus receives prominent otolith input from the maculae of the utricle and saccule and contributes to integrating head position signals with downstream projections to the cerebellum and other vestibular nuclei, supporting balance-related reflexes (Lee, 2016). The lateral vestibular nucleus (the largest of the complex) integrates inputs from canal and otolith organs as well as the vestibulocerebellum, and gives rise to the lateral vestibulospinal tract in the ipsilateral spinal cord. Through this pathway it contributes to the vestibulospinal reflex by modulating tone in trunk muscles and proximal limb extensors to maintain posture and balance (Lee, 2016). The medial vestibular nucleus comprises a substantial portion of the vestibular nuclear complex and receives prominent input from the crista ampullaris of the lateral semicircular duct. It supports the VOR via ascending projections through the medial longitudinal fasciculus to extraocular motor nuclei, and also contributes to postural control through bilateral descending projections in the medial vestibulospinal tract that terminate in the cervical spinal cord, supporting head and neck stabilization (Hain & Helminski, 2014; Khan & Chang, 2013; Lee, 2016).

### 1.3.5 Cortical Regions

The processing of vestibular signals in the cerebral cortex is unique from other senses in that there is no specific region dedicated to vestibular information. After the signal from the vestibular afferents is integrated in the vestibular nuclei within the brain stem, the information is broadly distributed to various regions of the cortex (Cheng & Gu, 2018).

Specifics regarding the complex cortical vestibular connections are not yet fully understood, and there is still no agreement on the precise location of a dedicated vestibular cortex. In primate studies, the primary cortical regions responsive to vestibular stimuli include the parietoinsular vestibular cortex (PIVC), area 2v in the intraparietal sulcus, and area 3av in the central sulcus. Among these, the PIVC seems to play a vital role, serving as the primary vestibular cortical region where information from other vestibular cortical areas is integrated (Cullen & Sadeghi, 2008).

Human studies propose that the primary cortical processing region for vestibular information is located in (or near) the parietal or insular cortex. Through electrically induced vestibular symptoms, researchers have identified a lateral cortical temporoparietal area, suggesting its correspondence to the PIVC in primates (Lopez et al., 2012). In response to various vestibular stimuli, Lopez and colleagues (2012) found convergence of cortical activity in the retroinsular cortex, alongside the parietal operculum and the posterior insula. Another study by zu Eulenburg (2012) employed Positron Emission Tomography (PET)

and Functional Magnetic Resonance Imaging (fMRI) activation and demonstrated that the right hemispheric parietal opercular area consistently showed activation in response to vestibular stimuli (zu Eulenburg et al., 2012). Studies involving extracellular recordings in primates have revealed several cortical regions that respond to both visual and vestibular signals during heading tasks, including the medial superior temporal sulcus (MSTd) (Fetsch et al., 2012; Gu, 2018), ventral parietal area (VIP) (Bremmer et al., 2002), frontal eye fields (FEF) (Gu et al., 2016) and visual posterior sylvian area (VPS) (Chen et al., 2011).

It is believed that vestibular connections extend to the thalamus and hippocampus, as animal studies reveal the presence of vestibular neurons in various thalamic regions. In humans, it is suggested that some ascending vestibular fibres make connections in the ventral posterior nucleus of the thalamus before reaching the cortex (Lopez et al., 2012). The hippocampus is thought to be crucial for processing spatial orientation and memory, and vestibular input related to head and body movement is considered crucial for this role (Cullen & Sadeghi, 2008). Additional research is required to broaden our understanding of the cortical connections and functions at higher levels within the vestibular system.

### 1.3.6 Vestibulo-ocular Reflex

The visual and the vestibular systems are heavily integrated, allowing the CNS to receive additional sensory information to form a more meaningful representation of the environment. The VOR is responsible for coordinating eye movements to stabilize retinal images during head rotation. It plays a crucial role in stabilizing images on the retina, allowing individuals to see clearly despite changes in head position. VOR is particularly important for activities such as reading while walking or maintaining visual focus during head movements. The VOR operates through a three-neuron pathway, starting from the semicircular ducts, then passing through the vestibular nuclei, and finally reaching the extraocular muscles. This sequence results in coordinated eye movements that counteract head turning (Sadeghi et al., 2007, 2009).

For instance, when the head turns to the right, the flow of endolymph in the semicircular ducts deflects the cupula to the left (Figure *1.2*). This leads to depolarization of the hair cells on the right and hyperpolarization of the left hair cells, causing increased nerve signals in the right vestibular nerve. These impulses are then directed to the corresponding vestibular nuclei and the cerebellum. Excitatory impulses are then transmitted through the medial longitudinal fasciculus to the right oculomotor nuclei and the ascending tracts from the lateral vestibular nucleus (Deiters nucleus) to the left abducens nuclei.

Dysfunction of the VOR can lead to symptoms such as dizziness, blurred vision, and difficulty maintaining balance.

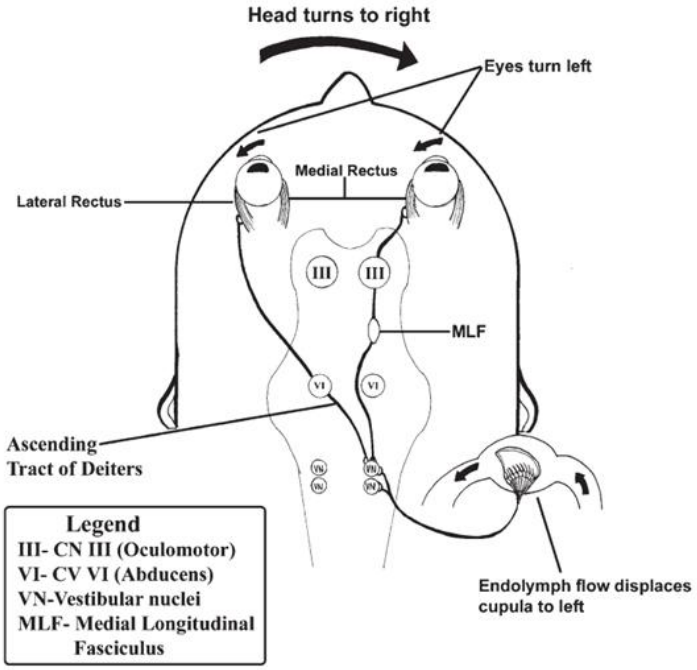


Figure 1.2: Vestibulo-ocular Reflex (VOR). Representation of VOR pathways during a right head turn (Khan & Chang, 2013).

### 1.3.7 Visual/Vestibular Integration Sites

Integration of visual and vestibular information in the central nervous system occurs across multiple areas to maintain balance and spatial orientation. Key integration sites include the vestibular nuclei in the brainstem, which receive direct inputs from both the vestibular organs and the visual system. The thalamus, particularly the ventral posterior nucleus, also plays a critical role in this sensory integration, acting as a relay centre that processes and forwards information to higher brain areas. The cerebellum is essential for coordinating movement and balance, integrating vestibular and visual inputs to fine-tune motor responses. In the cerebral cortex, multisensory integration involves areas like the PIVC, the temporo-parietal junction, and the posterior parietal cortex, which are crucial for spatial awareness and navigation. These cortical regions work in concert with the medial superior temporal area (MST) and the VIP, known for processing motion and spatial relationships. Collectively, these networks allow the CNS to construct a coherent perception of bodily orientation and movement, combining visual and vestibular cues to support activities such as gaze stabilization, postural control, and navigation in a three-dimensional environment (Cullen, 2012; Maciokas & Britten, 2010).

### 1.3.8 Electrical Vestibular Stimulation

EVS involves the application of electrical currents to the vestibular system to modulate its activity, primarily for research and therapeutic purposes. This technique can be administered non-invasively through methods like Galvanic Vestibular Stimulation (GVS), where surface electrodes deliver electrical currents transcutaneously, or invasively via vestibular implants that directly stimulate the vestibular nerve pathways (McLaren et al., 2022). This method has been widely used in both basic research and clinical settings to investigate vestibular function and its interactions with other sensory systems (Fitzpatrick & Day, 2004).

Research has demonstrated that EVS can selectively activate the vestibular afferents, generating perceptual effects such as illusory body tilt or rotation without actual head movement (Curthoys, 2012; Pastras et al., 2018; Wardman et al., 2003). These induced vestibular signals provide a unique opportunity to study how the brain integrates sensory information to maintain spatial awareness and posture. Furthermore, EVS has been instrumental in understanding the functional contributions of the otolith organs and semicircular canals to vestibular perception (Curthoys, 2012).

Transcutaneous EVS does not stimulate the vestibular nuclei directly. Instead, it modulates the activity of vestibular nerve afferents at the periphery, and those signals then propagate into the brainstem and are processed by the vestibular nuclei complex (Curthoys, 2012; Kim & Curthoys, 2004). Neurons within the vestibular nuclei integrate vestibular input with somatosensory, visual, and descending cortical influences, and then distribute this information through vestibulo-ocular and vestibulospinal pathways, as well as ascending projections through the thalamus to cortical multisensory networks (Fitzpatrick & Day, 2004; Hitier et al., 2014; Lopez & Blanke, 2011). This anatomy is central to the rationale for EVS in VR, because altering afferent firing at the vestibular nerve changes the reliability and content of the signal entering the vestibular nuclei, which is the same node that normally mediates reflexive stabilization and contributes to higher-order perceptual integration (Lajoie et al., 2021; Marchand et al., 2025).

Another compelling mechanism by which EVS may influence perception and motor control is through stochastic resonance (SR). SR is a counterintuitive phenomenon in which the addition of a moderate amount of noise to a nonlinear system can enhance the detection and processing of weak signals (Moss et al., 2004). In the context of vestibular function, noisy (i.e., stochastic) EVS introduces a zero-mean, band-limited electrical current that artificially increases variability in vestibular afferent firing, without generating a coherent illusory motion (Fitzpatrick & Day, 2004). In practical terms, stochastic EVS increases variability in vestibular afferent firing that arrives at the vestibular nuclei, which can alter

detection thresholds and reflex gains when delivered at subthreshold intensities (Moss et al., 2004; Mulavara et al., 2011; McLaren et al., 2023). This added noise can paradoxically improve balance and orientation by amplifying otherwise subthreshold signals, thereby enhancing perceptual sensitivity and postural responses (Mulavara et al., 2011).

However, the relationship between EVS amplitude and functional benefit is not monotonic. The stochastic resonance framework predicts an inverted-U function in which subthreshold noise enhances signal detection, but suprathreshold noise degrades it by reducing the effective signal-to-noise ratio (Moss et al., 2004). In the vestibular literature, reported optimal amplitudes for stochastic vestibular stimulation fall within the range of approximately 100 to 500 µA, commonly calibrated to 35 to 55% of each individual's perceptual threshold (Galvan-Garza et al., 2018; Goel et al., 2015). At higher amplitudes, effects can become inconsistent or detrimental. Assländer and colleagues (2021) found no reliable stochastic resonance effects on standing balance in young healthy adults across a range of noisy GVS amplitudes, with only 10 to 33% of trials showing the expected inverted-U curve. This parameter sensitivity is important for the present dissertation, because the effectiveness of any vestibular noise intervention depends critically on whether stimulation falls within or outside the facilitative window.

In VR research, GVS has been proposed as a tool to reduce cybersickness by promoting adaptive sensory reweighting. GVS delivers a bipolar current between electrodes placed over the mastoid processes, generating artificial vestibular signals that can interfere with natural head motion cues (Fitzpatrick & Day, 2004). By using noisy GVS, researchers can selectively degrade the reliability of vestibular input. According to principles of multisensory integration, the brain should respond by reducing reliance on the now-unreliable vestibular signal and upweighting other modalities, such as proprioception or vision (Alais & Burr, 2019). This reweighting may reduce the intersensory conflict that gives rise to cybersickness.

Weech and colleagues (2020) demonstrated that noisy GVS can attenuate cybersickness symptoms during immersive VR exposure. In their study, participants received continuous bilateral GVS (± 1750 µA) while navigating a VR environment. Compared to sham stimulation, participants receiving GVS reported significantly lower sickness levels as measured by the Fast Motion Sickness scale during and immediately after VR exposure. However, the effect was transient, with no significant differences observed on Simulator Sickness Questionnaire scores taken post-exposure. These findings suggest that GVS may provide short-term mitigation of cybersickness through mechanisms like sensory reweighting or stochastic resonance.

## 1.4 Cybersickness

### 1.4.1 Overview

Sensory conflict often emerges in human-made settings such as airplanes, trains, boats, and cars, and is particularly pronounced in virtual environments where there is a mismatch between visual and vestibular-proprioceptive inputs. The advent of VR technology, which creates immersive, computer-generated environments engaging multiple senses, has led to a unique form of motion sickness known as cybersickness. Despite VR's innovative advantages, it can cause discomfort, with symptoms varying widely among individuals. These symptoms, which intensify with longer VR exposure, range from mild (e.g., cold sweats, fatigue) to severe (e.g., nausea, vomiting), potentially affecting balance and mobility (Keshavarz & Hecht, 2011; McCauley & Sharkey, 1992; Miller et al., 1993; Oman et al., 1986). The concept of sensory reweighting, a mechanism proposed to account for individual differences in motion sickness susceptibility, involves the dynamic adjustment of sensory inputs to maintain posture and balance during environmental changes, relying on short-term neuroplastic changes in neural excitability (Horak, 2006; Horak & MacPherson, 1996; Laboissière et al., 2015; Stevens & Wang, 1995). The formal basis for sensory reweighting in postural control was established by Peterka (2002), who showed that the central nervous system dynamically adjusts the relative contributions of visual, vestibular, and proprioceptive inputs using a feedback control model in which cue weights shift with stimulus amplitude, such that increased reliance on one cue occurs at the expense of the others. Critically, vestibular-loss patients in that study showed linear, non-reweighting responses, establishing that intact vestibular function is necessary for adaptive cue redistribution. This process underscores the brain's ability to adapt sensory processing in response to changing conditions, highlighting the importance of sensory integration in maintaining stability and orientation.

Recent work has pinpointed visual-vestibular mismatch as a core trigger of cybersickness, showing that neural activity patterns can predict or correlate with cybersickness onset (Chang et al., 2023). Large-scale studies also reveal that individual susceptibility to motion sickness can be explained partly by genetic variants linked to vestibular function, providing further evidence that innate factors significantly influence cybersickness severity (Hromatka et al., 2015).

### 1.4.2 Postural Stability

Balance, or postural stability, is defined as the capacity to keep the body's centre of mass aligned with its support base (Pollock et al., 2000). In contrast, postural instability, a key risk factor for falls, refers to the

challenge in maintaining a stable stance, often exacerbated by immersive virtual environments. This instability is linked to motion sickness; the degree of postural instability directly influences motion sickness severity (Nishiike et al., 2013). Research by Stoffregen and Smart (1998) demonstrated that increased postural sway, or horizontal movement around the centre of gravity, often precedes the onset of motion sickness symptoms, supporting the postural instability theory. Measuring the Centre of Pressure (COP) path length, the total distance the COP moves horizontally while standing still, serves as a reliable indicator of postural stability (Faude et al., 2012).

Effective balance control, essential for daily activities, depends on the CNS's ability to integrate sensory information from visual, proprioceptive, and vestibular inputs, adapting posture to maintain stability under varying conditions. This integration supports both postural orientation, which involves aligning the body with respect to gravity and the environment, and postural equilibrium, which involves maintaining the centre of mass during stability disturbances. The balance system's role, although often overlooked, is crucial for providing a clear vision during movement and recognizing the direction and speed of motion (Horak, 2006; Manchester et al., 1989). Beyond physical inputs, psychological factors also significantly affect balance, underscoring the complex interplay of mental and sensory inputs in maintaining and adjusting posture.

### 1.4.3 Measures of Sickness

#### 1.4.3.1 Motion Sickness Susceptibility Questionnaire

To assess discomfort in virtual environments, commonly used subjective methods include the Motion Sickness Susceptibility Questionnaire (MSSQ), Simulator Sickness Questionnaire (SSQ), and Fast Motion Sickness Scale (FMS). The MSSQ gauges an individual's proneness to motion sickness across various transport modes, emphasizing the sensory conflict encountered during movement. It features both long and short versions to detail the frequency of motion sickness symptoms experienced in different vehicles throughout one's life (Golding, 1998).

#### 1.4.3.2 Simulator Sickness Questionnaire

The SSQ, a reliable tool for measuring motion simulator sickness, has been widely adopted to evaluate cybersickness across diverse virtual settings. It categorizes 16 symptoms into three groups: oculomotor disturbances, disorientation, and nausea, allowing participants to rate their severity on a Likert scale. However, its limitation lies in its static pre- or post-exposure application, missing dynamic changes in symptoms (Kennedy et al., 1993).

#### 1.4.3.3 Fast Motion Sickness Scale

Meanwhile, the FMS offers a rapid assessment of motion sickness severity, especially in environments featuring swift or sudden movements. Participants quantify their discomfort using a 20-point scale, where higher scores indicate greater sickness. This scale enables immediate evaluation of responses to motion stimuli, providing a direct measure of symptom intensity (Keshavarz & Hecht, 2011).

Despite longstanding reliance on questionnaires (e.g., SSQ), recent research has begun to pair continuous physiological monitoring (electroencephalogram, heart rate, galvanic skin response) with these subjective measures, aiming to predict cybersickness in real time (Chang et al., 2023). These predictive models could enable VR systems to adapt on the fly, potentially adjusting display parameters to reduce conflict and mitigate symptom onset.

### 1.4.4 Sensory Conflict Theory and Multisensory Integration

The leading theory behind cybersickness is the sensory-conflict theory (Reason & Brand, 1975). Under everyday conditions, the brain expects visual, vestibular, and proprioceptive inputs to agree about self-motion. In VR, this congruence is broken where optic-flow signals imply movement while the vestibular organs signal stasis. This mismatch is interpreted as an error and triggers autonomic responses such as nausea, dizziness, and disorientation leading to cybersickness (Bles et al., 1998). Yet conflict magnitude alone does not explain why susceptibility varies widely across individuals. Factors such as prior VR exposure, gaming habits, sex, age, and genetic vestibular variants modulate vulnerability, suggesting personal sensory-integration profiles play a key role (Hromatka et al., 2015).

Within the framework of Bayesian integration, the central nervous system combines sensory inputs by weighting each cue in proportion to its reliability (Ernst & Banks, 2002; Körding & Wolpert, 2004). This process allows for optimal perceptual estimates under uncertainty and helps resolve discrepancies between competing sensory information. For example, in an upright orientation task, visual and body cues may be down weighted when conflicting or unreliable, and more consistent gravity-derived vestibular cues may dominate.

This reweighting is not fixed but can adapt over time and in response to new environments or sensory conflicts. In VR, where artificial visual motion can conflict with inertial vestibular cues, the brain may resolve these inconsistencies by shifting reliance away from the vestibular system and toward visual or proprioceptive inputs (Zaidel et al., 2011). This adaptability plays a critical role in determining susceptibility to cybersickness (Reason & Brand, 1975). Recent work by Chung and Barnett-Cowan

(2023) supports this view: individuals who demonstrated greater shifts in sensory weighting (as measured by subjective visual vertical) following intense VR exposure reported lower levels of sickness, suggesting that efficient reweighting may be a protective mechanism. Similarly, studies using tasks like the Oriented Character Recognition Test have shown that reweighting can be quantified through changes in the influence of visual, gravity, and body cues on perceived upright (Dyde et al., 2006).

However, not all individuals reweigh cues in the same way or to the same extent. Visual dependency, the tendency to over-rely on visual cues even when they conflict with other modalities, has been linked to increased susceptibility to VR sickness, though recent evidence has challenged its predictive power (Josupeit, 2024). In particular, static measures of visual dependence such as rod-and-frame performance, including virtual rod-and-frame tasks, show only limited associations with cybersickness and likely have little value as a stand-alone screening tool. By contrast, findings from tasks that capture how people adjust sensory weights suggest that greater down weighting of visual input, for example in visual–postural coupling paradigms, is associated with lower cybersickness. Taken together, this points to adaptive reweighting capacity, rather than a fixed level of visual dependence, as the more plausible marker of resilience in VR.

A critical but often overlooked property of sensory reweighting is that it is temporally asymmetric. Assländer and Peterka (2014) showed that transitions from high-amplitude to low-amplitude sensory perturbations produce slower reweighting dynamics than the reverse direction. This asymmetry has implications for understanding cybersickness: entering VR represents a rapid increase in visual-vestibular conflict (prompting fast reweighting), whereas exiting VR requires the slower reverse transition (Assländer & Peterka, 2014). The observation that cybersickness symptoms often persist after headset removal is consistent with this asymmetric time course, because the nervous system may require substantially more time to restore pre-exposure cue weights than it took to initially shift them.

Accordingly, interventions designed to encourage sensory reweighting, including exposure training, vestibular noise stimulation, and cognitive strategies, may hold promise for mitigating cybersickness. This dissertation focuses on that adaptive capacity across multiple perceptual tasks and neural contexts, asking whether perceptual realignment (e.g., shifts in Perceptual Upright or Subjective Visual Vertical) reflects successful reweighting and whether those changes are associated with reduced discomfort in VR.

## 1.5 Perceptual Tests

### 1.5.1 Subjective Visual Vertical

The subjective visual vertical test is an established method for evaluating an individual's perception of verticality, where participants adjust a line to align with their perceived gravitational direction. Importantly, the Subjective Visual Vertical (SVV) is not a purely visual judgement, but a multisensory estimate of earth-vertical that emerges from the integration of vestibular graviceptive signals, visual contextual information, and body-referenced somatosensory cues (Dieterich & Brandt, 2019; Howard, 1982, 1997; Schoenmaekers et al., 2025). Typically, in a dark environment with the participant upright, the adjusted line closely aligns with true gravity, usually within a two-degree margin of error (Lacquaniti et al., 2017; Mann et al., 1949; Witkin & Asch, 1948). Factors influencing SVV accuracy include the physical tilt of the participant (Asch & Witkin, 1948; Aubert H, 1861; Beck et al., 2020; Müller, 1915), alterations in the visual environment (Asch & Witkin, 1948; Beck et al., 2020; Friederici & Levelt, 1990; Mittelstaedt, 1986; Mittelstaedt & Fricke, 1988; Witkin & Asch, 1948; Zoccolotti et al., 1992), and concurrent adjustments of both the observer and the visual field relative to gravity (Dyde et al., 2006; Mittelstaedt, 1986; Poquin et al., 1995; Witkin & Asch, 1948; Yardley, 1990). Classic work by Howard and colleagues showed that tilted frames, visual polarity cues, and large-field visual motion can systematically bias judgements of vertical and, under some conditions, even alter the observer’s perceived self-orientation, highlighting the strong interaction between visual and vestibular signals in upright perception (Howard, 1997; Howard & Childerson, 1994; P. Howard & Templeton, 1966).

Individual and group variations in SVV responses, particularly to visual environment tilts, have been noted, with those highly influenced by visual cues termed 'visually dependent' (Witkin et al., 1954). Notably, women (Barnett-Cowan et al., 2010; Groberg et al., 1969; Gross, 1959; Hyde et al., 1975; Linn & Petersen, 1985) and people with Parkinson’s disease (Barnett-Cowan et al., 2010) show a greater tendency towards visual dependence.

In the context of VR, SVV has been employed to investigate the impact of head tilts within virtual settings (Cheng et al., 2022; Klėgėris et al., 2025; Wang et al., 2021; Yagi et al., 2021). This literature is especially relevant to immersive VR, where wide-field optic flow, tilted scenes, and visually polarised environments can create compelling illusions of self-tilt or reorientation even when the body remains stationary, thereby challenging the normal weighting of vestibular and visual cues (Allison et al., 1999; Howard & Hu, 2001). Further, recent research has applied the SVV to ascertain perceptions of verticality before and after VR exposure, offering insights into how virtual environments may influence spatial

orientation. For instance, recent findings indicate that participants who show greater shifts in their SVV estimates after VR exposure often report fewer cybersickness symptoms, suggesting an underlying sensory reweighting mechanism that helps resolve conflicts (Chung & Barnett-Cowan, 2023).

### 1.5.2 The Oriented CHAracter Recognition Test (OCHART)

The Oriented CHAracter Recognition Test (OCHART) serves as an indirect assessment tool for the Perceptual Upright (PU), the observer's perceived most upright orientation of objects, crucial for effective recognition of objects, faces, and characters (Dyde et al., 2006). Conceptually, the OCHART complements the SVV by measuring the orientation at which an object is most readily recognised rather than asking participants to make an explicit judgement about gravity, thereby providing an indirect measure of the perceptual upright (Dyde et al., 2006). This orientation is vital for interpreting objects correctly in space (Jolicoeur, 1985; Rock & Heimer, 1957).

OCHART tests perception by requiring participants to differentiate between the letters 'p' and 'd'. By calculating the average transition points from 'p' to 'd' and vice versa, PU is determined based on the 180-degree rotational relationship of these letters (Dyde et al., 2006). Studies have shown that PU typically aligns with the gravitational direction in an upright stance but shifts closer to the body's orientation than to gravity when the body tilts 90 degrees from gravity (Dyde et al., 2006).

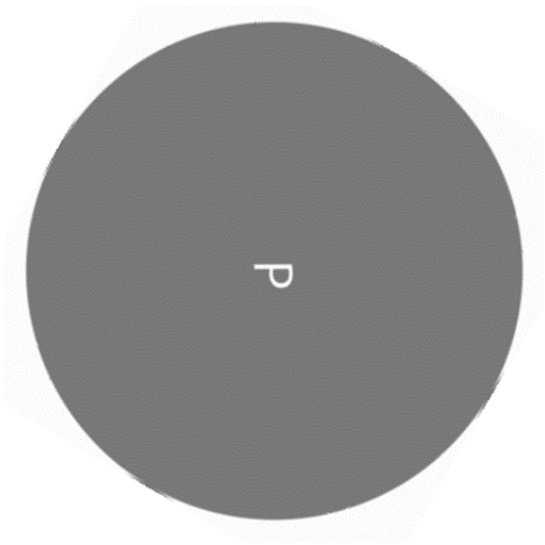


Figure 1.3: The OCHART probe: The letter 'p' is easily recognizable when upright, but often unclear when tilted 90° relative to the observer. Similarly, the letter 'd' is most readily identified when the 'p' is upside down relative to the observer.

The perceptual upright is commonly modelled as the vectorial sum of three orientation signals corresponding to vision, gravity, and the body axis. In this framework, PU reflects the resultant of these competing reference frames, and the magnitude of each vector reflects the relative contribution of that cue to the final estimate of upright. In the original OCHART study, Dyde et al. (2006) showed that PU was influenced by all three cues, with body orientation exerting a stronger influence on PU than on the SVV, and that the relative cue contributions could be captured by a weighted vector sum model. Subsequent work extended this framework by showing that the relative influence of visual cues on PU changes across altered gravity states (Dyde et al., 2009), that dynamic scenes can strengthen visual contributions to PU (Jenkin et al., 2011), that the contribution of gravity itself varies with gravitational magnitude (Harris et al., 2014), and that body-related contributions to PU can show systematic asymmetries (Barnett-Cowan et al., 2013).

More recently, OCHART has been used to examine perceptual upright under strong visual-vestibular conflict and in visual reorientation illusions. In this context, OCHART-derived cue weights appear to capture more than simple visual dependence alone. For example, McManus and Harris (2023) found that individuals vulnerable to visual reorientation illusions showed enhanced visually induced self-motion, yet weighted visual cues less and gravity cues more than resistant individuals, a pattern interpreted within a robust integration framework in which highly discrepant cues may be selectively discounted. Because the PU is more strongly weighted toward the body axis than the SVV, which in turn is dominated by gravitational cues (Dyde et al., 2006), comparing the two measures before and after VR exposure offers a way to dissociate changes in gravitational weighting from changes in body-referenced orientation, which may help clarify the sensory reweighting processes underlying individual differences in cybersickness.

### 1.5.3 A-effect and E-effect

The A-effect and E-effect are well-documented biases in perceived verticality that are most commonly described in SVV and related spatial orientation tasks performed during head or body roll tilt. These effects illustrate that estimates of upright are not determined by gravity alone, but instead reflect the combined influence of vestibular, body-based, and prior orientation signals. At larger roll tilts, observers often show an A-effect, in which perceived vertical is biased toward the direction of body tilt, consistent with an underestimation of tilt. At smaller roll tilts, observers may instead show an E-effect, in which perceived vertical is biased away from the direction of tilt, consistent with overestimation of tilt (Hoppenbrouwers et al., 2004; Tarnutzer et al., 2009; Kim et al., 2018). These tilt-dependent biases have often been interpreted using models that include an idiotropic or upright prior, whereby the nervous

system favours the most common body orientation relative to gravity (Mittelstaedt, 1986; Cuturi & Gori, 2019).

Although the A-effect and E-effect are classically associated with verticality judgements such as the SVV rather than with OCHART directly, they are relevant here because they demonstrate a broader principle that also underlies perceptual upright: when visual, vestibular, and body-based cues disagree, the brain does not recover upright veridically, but instead produces a weighted estimate that depends on cue reliability and task demands. This is particularly important in virtual reality, where immersive visual cues can perturb the normal balance between these sensory references.

## 1.6 Weighted Vector Sum Model of Sensory Integration

The relative importance the brain assigns to sensory information in spatial perception tasks can be well explained by treating gravity's orientation, visual cues, and the internal representation of the body as vectors. This concept was first proposed by Mittelstaedt (1983) in a model explaining the body's position relative to gravity. At that time, the non-linear interactions of these cues in the perception of the SVV were poorly defined. Mittelstaedt believed this resulted from the brain's difficulty in updating visual information from vestibular and proprioceptive inputs when the body is tilted and observing a vertical line (Mittelstaedt, 1983).

Mittelstaedt suggested that the SVV tends to shift toward the body's longitudinal axis, attributed to an imbalance caused by the unequal number of hair cells in the otoliths. He termed the body's internal representation as the 'idiotropic vector' and confirmed its role in predicting the SVV through experiments that isolated the unique contributions of different senses (Mittelstaedt, 1983, 1986, 1992, 1996, 1996; Mittelstaedt & Fricke, 1988). According to his model, the SVV is determined by the vector sum of the body's orientation relative to gravity (Mittelstaedt, 1983).

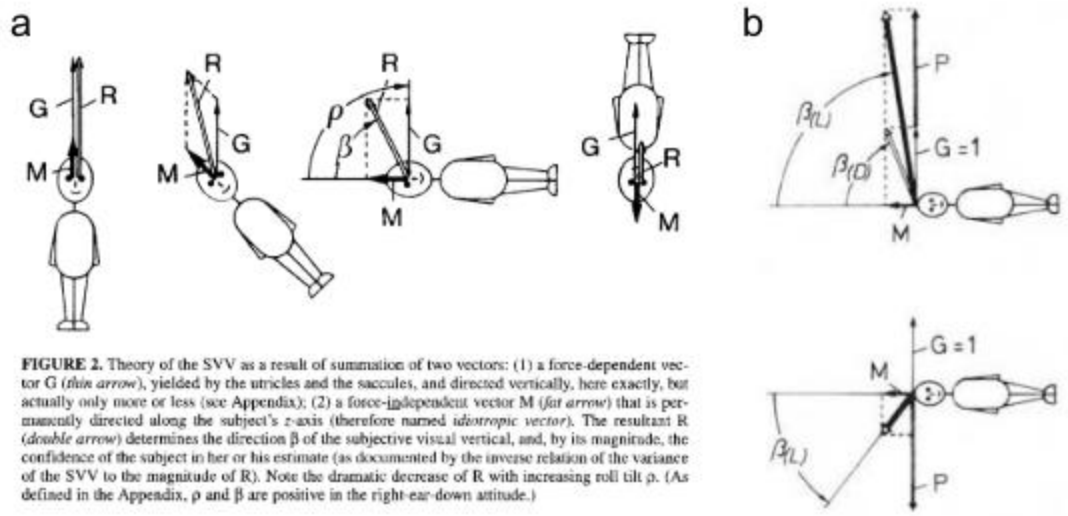


**FIGURE 2.** Theory of the SVV as a result of summation of two vectors: (1) a force-dependent vector G (*thin arrow*), yielded by the utricles and the saccules, and directed vertically, here exactly, but actually only more or less (see Appendix); (2) a force-independent vector M (*fat arrow*) that is permanently directed along the subject's *z*-axis (therefore named *idiotropic vector*). The resultant R (*double arrow*) determines the direction β of the subjective visual vertical, and, by its magnitude, the confidence of the subject in her or his estimate (as documented by the inverse relation of the variance of the SVV to the magnitude of R). Note the dramatic decrease of R with increasing roll tilt ρ. (As defined in the Appendix, ρ and β are positive in the right-ear-down attitude.)

Figure 1.4: Mittelstaedt's simple vector models of subjective visual vertical (SVV). a) Illustration of a two-vector model in which perceived vertical is determined by the combination of a gravity-based vector (G) and an idiotropic, body-based vector (M) under different body orientations. The resultant vector (R) predicts the subjective visual vertical. b) Predicted SVV under a simple two-vector model (gravity and body cues) and a three-vector model that additionally incorporates a visual cue (P) (from Mittelstaedt, 1986).

Dyde et al. (2006) later used this technique to predict the SVV and the perceived upright (PU). They found that the relative contributions of the internal body representation and visual surround to gravity differ between these tasks. The SVV was more influenced by gravity (weight of 1.0), while the internal body representation had a relative length of 0.2 and vision of 0.1. For the PU, the contributions were 1.0:2.6:1.2 (gravity, body, vision), indicating a greater influence of the body and vision compared to the SVV. This weighting framework is grounded in the Bayesian model of multisensory integration (Alais & Burr, 2019). Although individual weighting ratios vary, the findings indicate that, on average, bodily cues contribute more to perceptual upright judgments than visual or gravitational cues in the OCHART task (Dyde et al., 2006; Fraser et al., 2018).

PU and sensory cue weighting vary with factors such as sensory input type, context, and clinical status. For example, dynamic visual backgrounds increase visual influence on PU compared to static scenes (Jenkin et al., 2011). Astronauts returning from a 168-day spaceflight showed reduced reliance on gravitational cues for PU judgments, due to prolonged exposure to absent or unreliable gravity in microgravity environments (Dyde et al., 2010). Recent VR research confirms that a vector-sum approach effectively captures how the CNS combines body, vision, and gravity signals (Chung & Barnett-Cowan, 2023). In cases of extreme conflict, the model suggests that users who can alter these vector weights more quickly experience less cybersickness.

It is important to note that the weighted vector sum model assumes obligatory integration of all available cues with fixed weights within a condition. While this framework has proven useful for describing group-level patterns in orientation perception (Dyde et al., 2006; Harris et al., 2022), it may not fully capture behaviour under the large sensory conflicts imposed by VR, where the nervous system may switch between integration and segregation depending on inferred causal structure (de Winkel et al., 2018; Körding et al., 2007). This limitation is acknowledged throughout the experimental chapters and addressed in the General Discussion (Section 5.4).

## 1.7 Experimental Chapter Summaries and Hypotheses

The first experimental chapter of this dissertation examines individual differences in cybersickness development and the role of sensory reweighting in virtual reality. This first experiment builds off of a previous study from our lab that showed a significant change of the SVV (i.e., sensory reweighting) following high-intensity VR gameplay, whereby participants overall showed a shift of the SVV towards gravity and away from the body, and that those who changed their SVV more reported less sickness and participants who changed their SVV less reported more sickness (Chung & Barnett-Cowan, 2023). While promising, this study only examined SVV using a gray background; thus, the interpretation of sensory reweighting was restricted to bodily and gravitational cues, without considering the role of vision (a dominant sensory cue in VR). In the first experiment of this dissertation (Chapter 2), I will be investigating the relationship between sensory reweighting and cybersickness susceptibility, focusing on the SVV with rotated visual backgrounds relative to gravity and the body. In this study, the participants' SVV is measured before (upright and lying right side down) and after (lying right side down) a sickening VR session. My hypotheses are that larger changes in SVV should be associated with lower sickness scores, and that the relative weighting of visual cues will be reduced following VR, while the weightings of body and gravity will be reduced and increased, respectively.

Chapter 3 of this dissertation will continue to examine individual differences in cybersickness development and the role of sensory reweighting in virtual reality. Chapter 3's study will be identical to that of Chapter 2, where, instead of the SVV task here I will assess sensory reweighting using the OCHART, which assesses the role of visual, bodily, and gravitational cues more evenly (Barnett-Cowan, 2013). My hypotheses are like those of Chapter 2, where larger changes in the PU should be associated with lower sickness scores, and that the relative weighting of visual cues will be reduced following VR, while the weightings of body and gravity will be reduced and increased, respectively.

Chapter 4 of this dissertation further examines the effects of sensory reweighting in VR. In this study, we will apply noisy EVS to perturb natural sensation of motion from the vestibular organs (i.e., make the vestibular signal less reliable). The OCHART (similar setup as in Chapter 3) will be administered before and after EVS/noEVS VR sessions to measure the relative weighting of visual, bodily, and gravitational cues to orientation perception. I hypothesize that stochastic EVS applied during VR will degrade the reliability of vestibular afferent signals, which are the primary carriers of gravitational information to the central nervous system. According to reliability-based models of multisensory integration, this degradation should lead to a down-weighting of gravitational contributions to orientation perception, with a compensatory upweighting of bodily and visual cues that remain unperturbed by the stimulation.

This approach aligns with broader findings indicating that individuals who more readily shift reliance from one sensory cue to another during and after VR exposure show increased tolerance and reduced cybersickness (Chung & Barnett-Cowan, 2023). We also draw on the notion that repeated or gradual exposure (often termed "training") can enhance adaptation, as prior work on incremental exposure protocols has demonstrated generalized reductions in sickness across different VR scenarios (Adhanom et al., 2022).

# Chapter 2
# Reweighting of Visual, Bodily, and Gravitational Cues to the Subjective Visual Vertical and Cybersickness Susceptibility

## 2.1 Abstract

Virtual reality (VR) often provokes sensory conflicts that lead to cybersickness, a major obstacle to widespread VR adoption. To maintain spatial orientation in virtual environments, the central nervous system (CNS) is thought to resolve intersensory discrepancies through multisensory integration, potentially by dynamically reweighting visual, vestibular, and somatosensory cues based on their relative reliability. At the neural level, adaptive reweighting is thought to be implemented through coordinated computations spanning vestibular brainstem nuclei and cerebellar circuits, together with cortical arbitration within vestibular, parietal, and insular multisensory networks. This study investigated how intense VR exposure modulates these weightings by assessing shifts in the subjective visual vertical (SVV), a perceptual task used to measure the brain's estimate of gravitational upright based on the integration of visual, vestibular, and bodily cues. Twenty participants completed the SVV task in upright and right side-lying postures both before and after a 20-minute session in a high-intensity VR environment. Results revealed a post-exposure shift in SVV estimates away from the body axis and toward the gravitational vertical, indicating a decreased reliance on body cues and an increased weighting of vestibular input. Crucially, the extent of this SVV shift was inversely related to self-reported cybersickness severity, where participants who exhibited greater reweighting reported fewer symptoms, while those who maintained their pre-VR cue balance experienced more pronounced sickness. These findings suggest that flexible sensory reweighting, particularly in response to visual-vestibular-bodily conflict, is an important adaptive mechanism protecting against cybersickness. By quantifying cue contributions through vector sum modelling, this work provides new insight into the neural strategies CNS employs to preserve orientation under conditions of sensory discordance, with implications for individual susceptibility and VR design.

## 2.2 Introduction

Virtual reality (VR) technology has undergone significant growth over the past decade and is now widely accessible for entertainment, research, and therapeutic applications (Bohil et al., 2011; Kouijzer et al., 2023; Li et al., 2023). Despite the advancements made in VR technology, displays, and systems, cybersickness remains a persistent concern and a significant obstacle to widespread VR adoption among users. Cybersickness consists of a range of unpleasant symptoms, including nausea, disorientation, eye strain, drowsiness, cold sweating, and fatigue, which can vary in intensity and manifestation from one individual to another (Ang & Quarles, 2023; Caserman et al., 2021; Oman, 1990). This variability complicates efforts to accurately measure and predict cybersickness in VR users. Meta-analytic data show that 60–95 % of users experience at least one cybersickness symptom during high-intensity VR, and that sensory mismatch and self-motion cues are the primary predictors of severity (Caserman et al., 2021). Various modelling approaches have been successful in predicting cybersickness incidences, ranging from 34% to 55%, through considering factors such as sickness ratings, physiological responses, postural stability, sensory perception, behaviour, demographic characteristics, and features of the hardware and software used in VR systems (Arcioni et al., 2019; Chang et al., 2021; Dennison et al., 2016; Jasper et al., 2023).

Although modern head-mounted displays offer higher refresh rates, improved tracking, and refined rendering pipelines, cybersickness remains a persistent barrier to widespread VR use and is highly variable across individuals (Ang & Quarles, 2023; Caserman et al., 2021; Tian et al., 2022). This variability motivates behavioural markers that can index underlying sensory integration processes, rather than relying solely on symptom scales. The Subjective Visual Vertical (SVV) provides a well-established perceptual assay of how visual, vestibular, and body-related cues are combined to estimate gravitational vertical, and recent work suggests that SVV-based indices can relate meaningfully to cybersickness susceptibility (Chung & Barnett-Cowan, 2023). In this chapter, I use SVV before and after VR exposure, together with model-based estimates of cue weighting, to test whether individual differences in verticality perception are linked to real-time sickness development.

The sensory mismatch theory explains how sickness arises when there is an incongruency in the feedback received from sensory inputs indicating self-motion (Oman, 1990; Reason & Brand, 1975; Reason, 1978). This conflict becomes apparent in VR environments, where there's abundant visual motion but minimal signals from the vestibular and proprioceptive systems (Cobb et al., 1999; Howarth & Costello, 1997; Keshavarz et al., 2015; Kim et al., 2020). Some researchers have argued against this theory due to difficulties in predicting sickness occurrence and defining sensory mismatch (Stoffregen & Riccio, 1991),

the core concept of cybersickness seems rooted in the integration of sensory cues related to self-motion. Taking a multisensory perspective might offer a better understanding of the neural mechanisms underlying sickness (Gallagher & Ferrè, 2018).

Multisensory integration is a strategy used by the Central Nervous System (CNS) to address the issue of processing and making decisions based on ever changing incoming information – "the binding problem" (Treisman, 1996). The CNS addresses this issue through multisensory integration, which can leverage redundancies to enhance decision quality (Nashner & Berthoz, 1978).

The CNS can adjust the sensory input significance according to task demands and information reliability, facilitated by short-term neural plasticity (Maurer et al., 2006; Peterka, 2002). Sensory reweighting has been identified as a protective mechanism that helps an organism navigate situations where sensory inputs conflict. Under natural self-motion, what you see typically matches how your body moves. Inside a car, for example, the driver feels this match directly, where turning the steering wheel leads to a corresponding change in vehicle motion that is immediately confirmed by vestibular and proprioceptive cues in the driver's own body. Conversely, passengers often encounter conflicting sensory information regarding self-motion.

The SVV is a perceptual measure of verticality, requiring individuals to align a visually presented line with their perceived direction of gravity or upright. This task reflects the integration of multisensory signals, including visual, vestibular, and body cues (Dyde et al., 2006). The vestibular system contributes to the perception of upright through the mechanical properties of hair cells in the otolith organs and gravitational inertia, while the body's orientation provides a reference known as the idiotropic vector (Mittelstaedt, 1983). When upright, the vestibular and body signals are aligned, resulting in accurate SVV estimates of true vertical (Dyde et al., 2006).

However, tilting the head and body disrupts the alignment of vestibular and body cues, causing the SVV estimate to shift towards the longitudinal body axis, known as the A-effect, indicating a bias in the perception of upright (De Vrijer et al., 2009; Dyde et al., 2006). This bias depends on the degree of head tilt: larger tilts beyond 60° exhibit the A-effect, while smaller tilts result in a shift in the opposite direction, known as the E-effect, due to ocular counter-torsion. These shifts in perceived verticality are indicative of changes in the relative contributions of vestibular and somatosensory cues and can be modelled using reweighting or Bayesian frameworks (Dyde et al., 2006).

Additionally, the SVV task is sensitive to individual differences, including sex, and specific populations such as dancers, astronauts with highly trained vestibular systems, and patients with vestibular deficits

(Barnett-Cowan et al., 2010; Beck et al., 2020). This sensitivity suggests that the SVV task is a reliable measure for observing the sensory reweighting mechanism between vestibular and somatosensory signals.

Chung and Barnett-Cowan (2023) investigated sensory reweighting and motion sickness susceptibility using the SVV in a comfortable VR environment and a sickening environment. Their results showed that the SVV was significantly shifted away from the body representation of upright and towards the vestibular signal after exposure to a high-intensity VR game (Figure *2.1*). Their results highlight the aftereffects of VR for sensory perception and suggest a potential relationship between susceptibility to cybersickness and sensory reweighting.

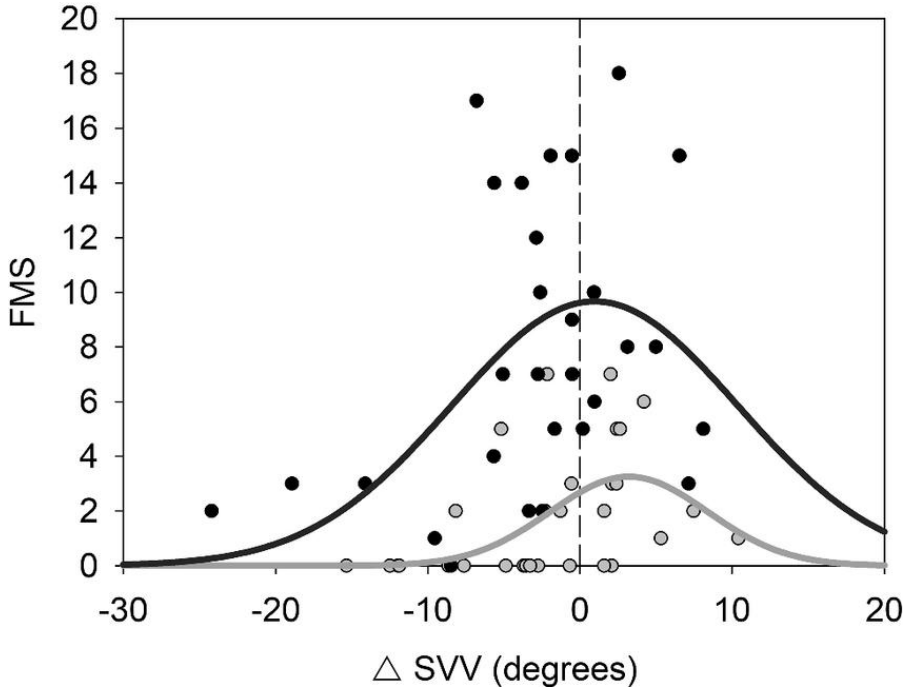


Figure 2.1: Gaussian model of FMS ratings relative to the change in SVV for both low and high intensity VR exposures from baseline. The grey-filled dots represent the data points from the low intensity VR exposure and the black-filled dots from the high intensity VR and the black line for the high intensity VR. The dashed line indicates the upper specification of the regression of no difference in the SVV from baseline (From Figure 5, Chung and Barnett-Cowan, 2023).

Focusing on the black dots (which are high-intensity VR) those who self-reported as feeling the sickest had the least change in their SVVs. While promising, Chung and Barnett-Cowan did not measure the influence of vision, and since VR is a vision-dominated medium, it is important to consider the effects of adding visual backgrounds to their SVV task.

The present experiment aims to investigate the aftereffects of VR on sensory perception and to determine whether the occurrence of cybersickness is associated with the sensory reweighting mechanism. Using the

SVV task, we intend to replicate previous findings by investigating if exposure to high-intensity virtual content alters the relative contributions of vestibular and body cues in the perception of upright. Based on the sensory conflict theory and the findings of Chung and Barnett-Cowan (2023), if visual and bodily signals are down weighted relative to vestibular signals to resolve conflicts induced by VR exposure, we expect SVV to shift away from the longitudinal body axis and toward gravitational vertical. Furthermore, if cybersickness depends on an individual's ability to dynamically reweight their sensory signals, we expect to observe changes in the SVV estimate that reflect alterations in the relative contributions of these sensory signals, correlating with the severity of symptoms experienced.

We hypothesize that an observer experiencing mild or no cybersickness symptoms will exhibit a larger shift in the SVV estimate compared to an observer with more severe symptoms, as observed in the study by Chung and Barnett-Cowan (2023). We predict that those with lower cybersickness scores will exhibit a down weighting of visual signals and body cues, and upweighting of vestibular.

This is because users navigating virtual environments do not receive physical body cues, and optic flow signals motion while vestibular and proprioceptive systems detect no actual acceleration (sensory conflict). To reduce this mismatch, the brain will adapt by down weighting misleading visual and bodily signals and reweighting toward the vestibular system, which continues to provide trustworthy gravitational information. This is supported by strong empirical evidence from Tian and colleagues (2022). They highlight that when visual and vestibular cues conflict, the brain adaptively decreases reliance on misleading visual and proprioceptive inputs while increasing the weighting of vestibular signals, resulting in more reliable integration (Tian et al., 2022).

At the systems level, sensory reweighting is typically framed as reliability-based updating of an internal estimate, supported by coupled brainstem, cerebellar, and cortical loops. Vestibular nuclei integrate canal and otolith afference with proprioceptive and efference-copy information, while vestibulo-cerebellar circuitry is well positioned to adaptively tune the gain assigned to discordant channels as prediction errors persist (Cullen, 2012). Cortically, vestibular signals converge in posterior insula and parietal operculum and are integrated with visual and somatosensory cues in posterior parietal regions that support state estimation for self-motion and verticality, with top-down modulation from fronto-cingulate control networks likely contributing to inter-individual differences in reweighting behaviour (Lopez et al., 2012; Shadmehr & Krakauer, 2008).

Operationally, we treat Point of Subjective Equality (PSE) (SVV bias) as an index of the relative weighting of body-referenced versus inertial cues in roll-tilt, and JND (SVV precision) as an index of

sensory uncertainty in the verticality estimate (Wichmann & Hill, 2001). For the tilted-room manipulation, we additionally quantify visual attraction as the periodic modulation of PSE as a function of background orientation, which reflects the influence of visual orientation priors on verticality judgments (Dyde et al., 2006; Haji-Khamneh & Harris, 2010). We predicted that (i) SVV bias and precision would change from pre to post VR in roll-tilt, consistent with altered cue weighting, and (ii) individual differences in cybersickness severity would relate to the magnitude of post VR change in SVV metrics, consistent with sensory reweighting as a tolerance mechanism (Chung & Barnett-Cowan, 2023; Weech et al., 2020).

## 2.3 Methods

### 2.3.1 Participants

Data were collected from twenty participants (11 males and 9 females; mean age =23.2, SD = ±4.9) recruited from the University of Waterloo. Twelve of the participants were volunteers, while the remaining participants were remunerated $10/h for their participation. All participants reported having no auditory or vestibular disorders and had normal or corrected-to-normal vision. This experiment was reviewed and approved by the University of Waterloo Research Ethics Committee, and all participants provided informed written consent in accordance with the committee's guidelines.

#### 2.3.1.1 A-priori Power Analysis

An a priori power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2007) to estimate the minimum sample size required to detect within-factor effects in a repeated-measures ANOVA. Input parameters included a medium effect size ($f = 0.25$), $\alpha = .05$, desired power of 0.95, a correlation among repeated measures of $r = 0.50$, and a nonsphericity correction $\varepsilon = 0.75$. The analysis returned a noncentrality parameter $\lambda = 15.19$, a critical $F(1.50, 78) = 3.44$, and a required total sample size of N=54, yielding an actual power of approximately 0.954. The required sample sizes changes to N=44 for a power of 0.90 and N=34 for a power of 0.80. Although the required sample size was larger, data collection was constrained by the unexpected discontinuation of the Echo VR game servers, necessitating early termination of recruitment at N=20. Consequently, the study remains powered to detect medium-to-large effects but may be underpowered for small-to-medium effect sizes.

#### 2.3.1.2 Post-hoc Power Analysis

A separate post-hoc power analysis was performed to assess the sensitivity of the study to detect the main effect of Condition (Pre-VR vs Post-VR). From the two-way repeated-measures ANOVA (Section 2.4.2) the relevant statistic was F(1, 19) = 1.44. The associated within-subjects effect size was derived as

(Eq 2.1) $$\eta_p^2 = \frac{Fdf_{effect}}{Fdf_{effect}+df_{error}} = \frac{1.44*1}{1.44*1+19} = 0.07,$$

which converts to Cohen's $f$ via

(Eq 2.2) $$f = \sqrt{\frac{\eta_p^2}{1-\eta_p^2}} = \sqrt{\frac{0.07}{1-0.07}} = 0.28.$$

G*Power input parameters were the calculated effect size (f = 0.28), α = .05, desired power of 0.80, and a nonsphericity correction ε =1. The analysis returned a noncentrality parameter λ =8.78. The analysis indicated that a minimum total sample of N=28.

### 2.3.2 Apparatus

The VR content for this experiment was delivered using the Oculus Quest system. The Oculus Quest has a dual OLED display with a resolution of 1600x1440 (per eye) at a 72Hz refresh rate. It is a completely wireless system featuring 6 degrees-of-freedom (6DOF) inside-out tracking, allowing the user to freely move around. Here, we allowed participants to move within a 2.4x2.4m designated play area.

One commercially available VR game at the time was Echo VR (Figure *2.2*). This game is no longer available as the servers were taken down by Meta. This game was chosen based on user reviews on the Oculus Quest store, as it was rated as "intense." Echo VR puts the user in a zero-gravity arena in which they must navigate using the provided controllers in combination with rotation of their physical body to catch and throw a disc into an opposing team's goal.

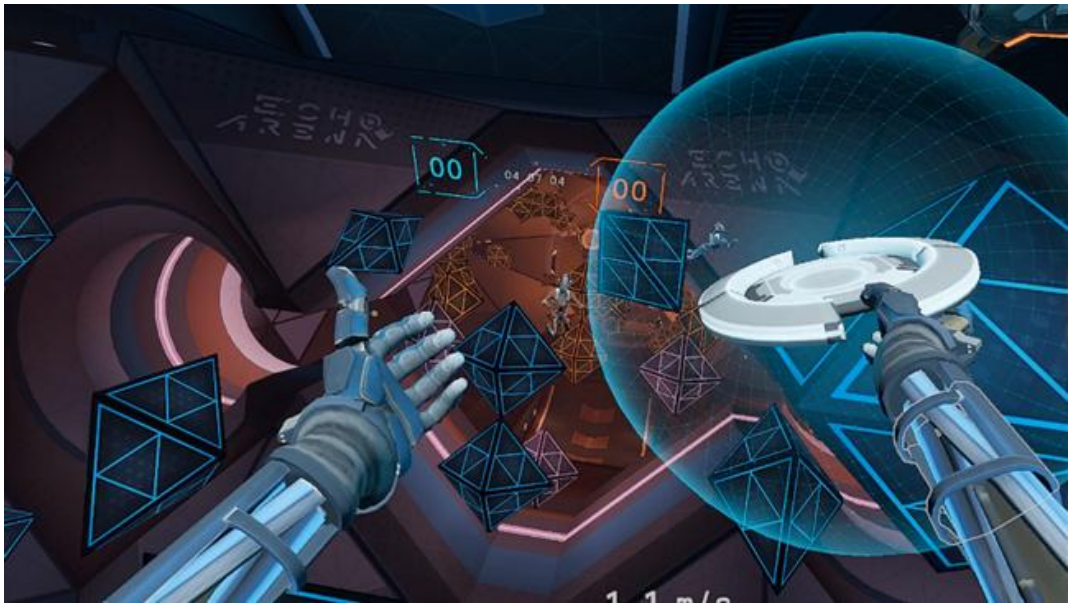


Figure 2.2: Screen capture of VR environment of the game used in the VR exposure, Echo VR.

### 2.3.3 Stimulus

To assess participants' reliance on vestibular cues, the "luminous line" technique (Dyde et al., 2006; Friedmann, 1970) was used within the SVV task and the method of constant stimuli. Specifically, a line measuring 3 x 0.5° was briefly displayed (500 ms) against a neutral grey background on a MacBook Pro laptop. To minimise peripheral distractions, a shroud was installed around the laptop screen, confining participants' view. Consequently, the screen appeared as a 35° diameter circle at a distance of 25 cm.

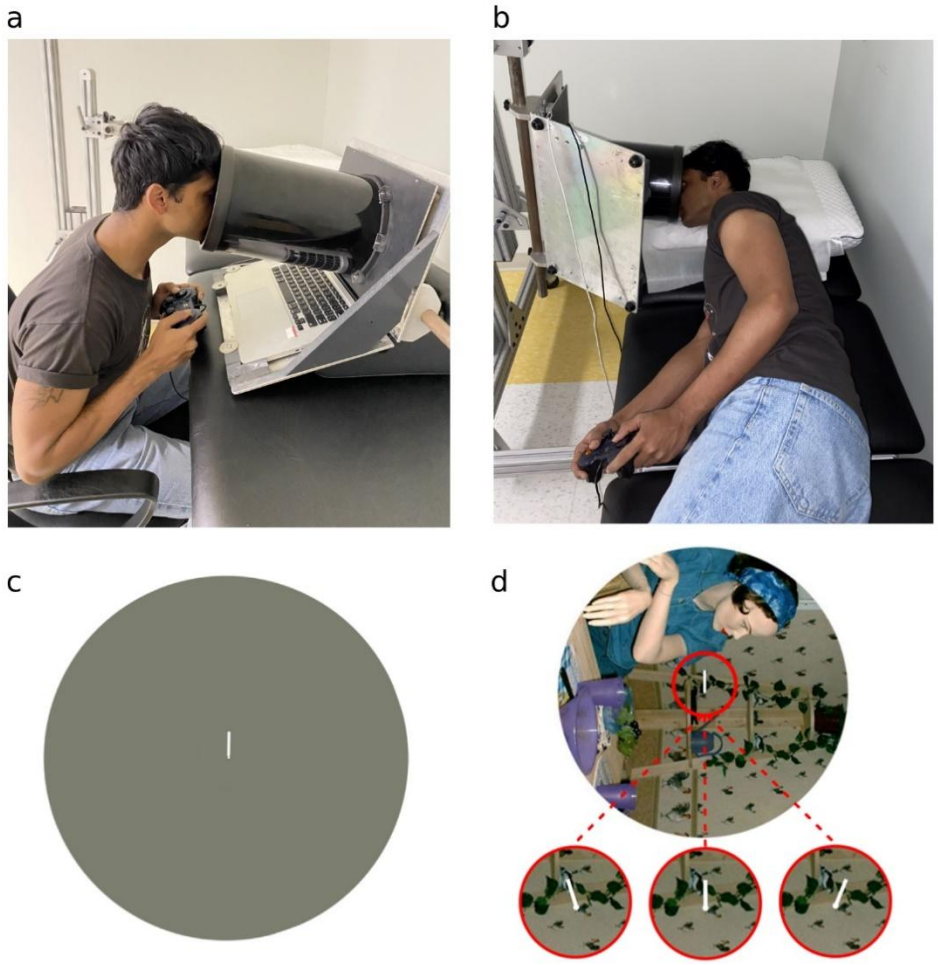


Figure 2.3: The two body postures used in these experiments: a) upright and b) right side down. Subjects viewed the display through a shroud to obscure all peripheral vision. An example of the Subjective Visual Vertical task on c) the grey background and d) on the visual background.

The task involved presenting 21-line orientations, varying from -50° to +50° in 5° increments, providing a comprehensive range for assessment (Figure *2.4*). To measure the influence of vision, 6 other backgrounds were added, alongside grey (Figure *2.5*). The background was a picture of a room (with clues all around the room that indicate which way is up). This background was presented in 6 different orientations (from 0 to 300 in 60° increments).

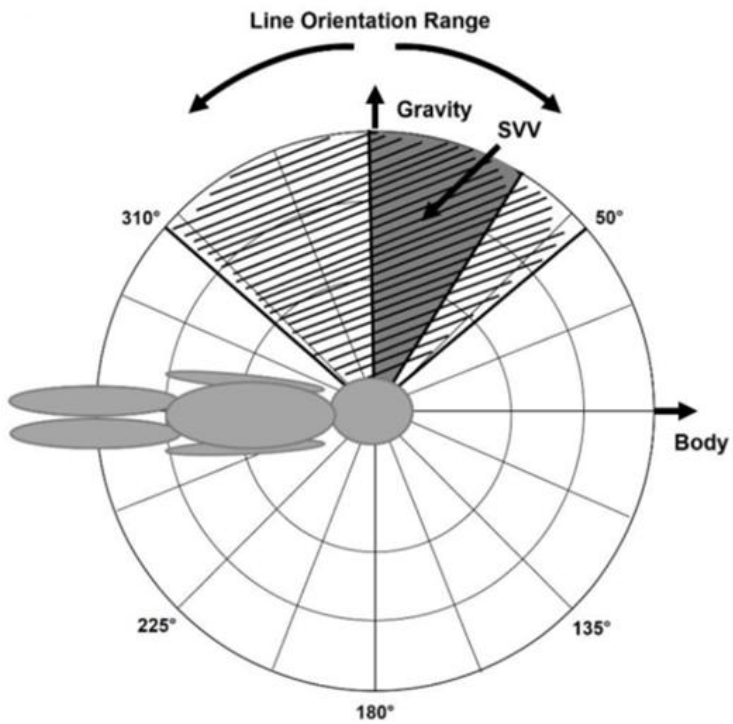


Figure 2.4: Polar illustration in earth coordinates relative to the right side down orientation. The hatched area delineates the spectrum of line orientations presented, displaying the extensive range considered in the study. In contrast, the shaded area illustrates the span of SVV estimates documented within the current investigation. It is important to note that the direction of upright is indicated by the vestibular system, primarily influenced by gravity (from Chung and Barnett-Cowan, 2023).

Each combination of line and background orientation was repeated eight times, totalling 1176 trials, typically completed within a 15-minute period. Participants utilized buttons on a Logitech Gamepad F310 to indicate whether they perceived the line as oriented clockwise or counterclockwise relative to their perceived vertical upright.

Figure 2.5: The visual backgrounds used in this experiment. The background was rotated anywhere from 0° to 300° in 60° increments.

### 2.3.4 Measures

Cybersickness was measured using the Fast Motion Sickness Scale (FMS), which measures sickness levels immediately following each VR session (Keshavarz & Hecht, 2011). Participants provided verbal ratings on a 20-point scale, with "0" indicating no sickness and "20" indicating severe or frank sickness.

### 2.3.5 Procedures

Participants completed a baseline SVV task, sat upright and lying down on their right-hand side on an examination table (using a foam pillow as a headrest to align their heads with a circular shroud attached to the laptop screen). They indicated whether a series of lines presented was oriented clockwise or counterclockwise from gravity ("the direction in which a ball would fall") via a 2-alternative forced choice task by button press on a gamepad.

Then, participants laid on their right side on an examination table to complete the SVV task. The laptop, set at a 90° clockwise angle, was mounted on a custom-built stand. This task involved a 2-alternative forced choice method, with participants indicating their responses via button presses on a gamepad. The completion of this task typically required approximately 15 minutes. Following the SVV task, participants were escorted to the VR play area, where they were equipped with the Oculus Quest system.

At the end of the VR session, participants' sickness levels were verbally assessed using the FMS. Immediately following this assessment, participants completed their final SVV task (only right side down), which took approximately 15 minutes (Figure *2.6*). We conducted the second SVV measurement post VR only in the right side down position to capture the immediate after-effects of the VR exposure before these effects decayed. However, omitting an upright SVV condition reduced the completeness of data feeding into the vector sum model, which typically requires both upright and right-side-down inputs to accurately estimate cue weightings, thereby compromising the model's accuracy in quantifying orientation contributions.

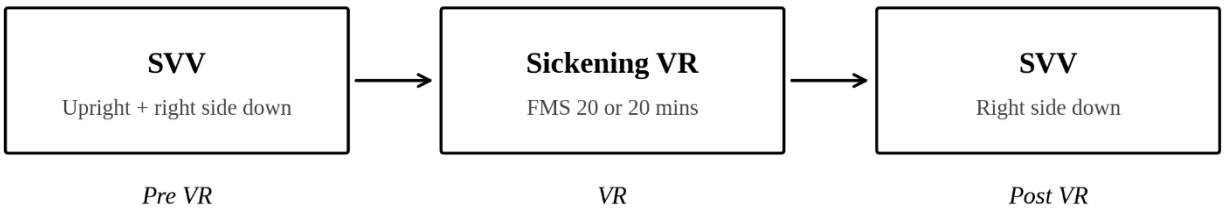


Figure 2.6: Schematic of experiment pipeline.

Post-exposure measurements of subjective visual vertical were obtained only in the right-side-down posture in order to capture immediate aftereffects of VR exposure while minimising task duration. Although this approach preserved sensitivity to postural effects, it constrained the ability to fully dissociate visual, gravitational, and body-based vector components following VR exposure. As a result, inferences regarding absolute cue weights post exposure should be interpreted with caution.

### 2.3.6 Data Analysis

Statistical analysis was conducted using SigmaPlot 12.5. To determine the SVV, a Gaussian function (Eq. 2.3) was applied to the data showing the percentage of times a line was judged as being tilted clockwise relative to gravity, plotted against the line's orientation. The SVV was identified as the line orientation ($x_0$) where there was an equal probability of the line being judged as tilted either clockwise or counterclockwise from the gravitational vertical.

(Eq 2.3) $$y = \frac{100}{1 + e^{\left(\frac{x - x_0}{b}\right)}}\%$$

The halfway (50%) inflection point of the sigmoid curve ($x_0$) represents the Point of Subjective Equality (PSE), which here represents the SVV estimate and the standard deviation of the function ($b$) as the just noticeable difference (JND) (Barnett-Cowan et al., 2010). A positive SVV value indicates that the line was perceived as vertically upright when it was oriented clockwise from true vertical, while a negative SVV score indicates that the line was perceived as vertically upright when it was oriented counterclockwise from true vertical. For each participant and condition, I assessed psychometric fit quality by inspecting convergence and by quantifying uncertainty in parameter estimates. Confidence intervals for PSE and JND were obtained using parametric bootstrapping of the fitted psychometric function, which provides robust uncertainty estimates for threshold parameters in binomial choice data (Wichmann & Hill, 2001).

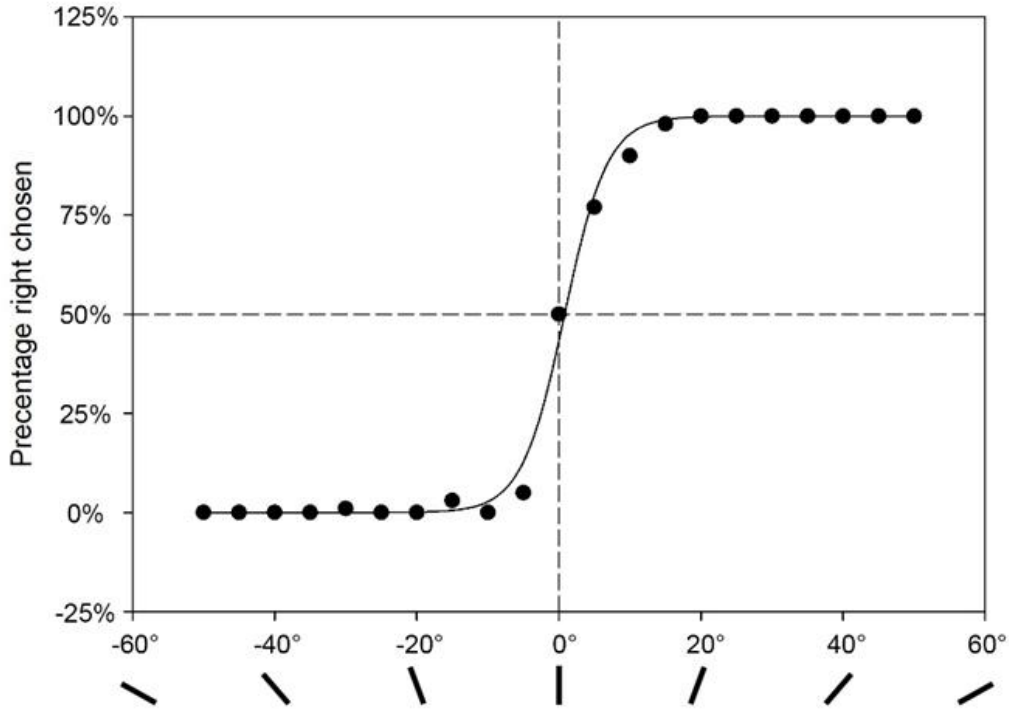


Figure 2.7: This figure illustrates how the SVV was measured. Participants chose whether the line was tilted left or right of vertical. Typical psychometric functions obtained from such responses are shown. The 50% point was taken as the SVV for the line judgments.

To assess the relationship between the SVV estimate and FMS scores, a three free-parameter Gaussian nonlinear regression model was fit to the FMS scores as a function of change in SVV from baseline after each VR exposure (Eq 2.2).

(Eq 2.4) $$y = ae\,\frac{100}{1 + e^{\left(\frac{x - x_0}{b}\right)}}$$

To evaluate whether VR exposure altered SVV across the full set of visual backgrounds, SVV PSE values from the right-side-down condition were analysed using a 2 (Condition: pre-VR, post-VR) × 7 (Background orientation: grey, 0, 60, 120, 180, 240, 300) repeated-measures ANOVA. Mauchly's test was used to assess sphericity for effects involving Background, and Greenhouse–Geisser corrected degrees of freedom are reported when sphericity was violated (Greenhouse & Geisser, 1959; Mauchly, 1940).

### 2.3.7 Vector Sum Model

To assess how each sensory cue contributed to SVV, a vector sum model (Mittelstaedt, 1983, 1988; Dyde et al., 2006) was used. In this model, every cue is represented by a vector of adjustable relative weight (Dyde et al., 2006). To capture the effects of various room orientations, the SVV was modelled by fitting each participant's data to Equation 2.3 via the Marquardt–Levenberg optimisation algorithm (Press, 1988). The relative contributions of vision, the body, and gravity are reflected in the lengths of their respective vectors that best fit the data. Importantly, the model assumes that each vector faithfully represents the actual direction indicated by its corresponding cue.

In this framework, vectors $\vec{v}$ and $\vec{b}$ correspond to the orientations of the visual cues and the body, respectively, while vector $\vec{g}$ indicates the direction of gravity. Only two free parameters are required, namely, the weights of two vectors relative to a third, which is arbitrarily set to 1. A schematic representation of the model is shown in Figure *2.8*.

(Eq 2.5) $$\overrightarrow{SVV} = \vec{v} + \vec{b} + \vec{g}$$

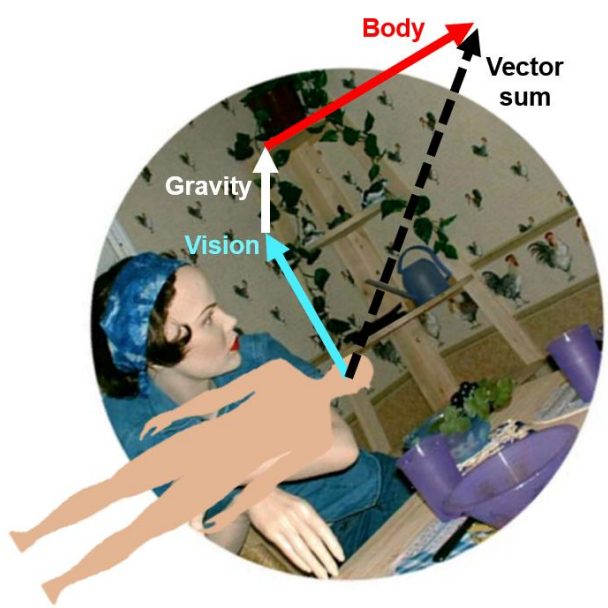


Figure 2.8: A weighted vector sum model, in which gravity, body orientation, and visual cues are summed to generate an estimate of vertical. The predicted outcome is depicted by the dotted black line, representing the vector sum.

(Eq 2.6) $$\boldsymbol{vy} = \text{vis} \cdot \cos(v)$$

$$\boldsymbol{v_x} = \text{vis} \cdot \sin(v)$$

$$\boldsymbol{y} = \text{b} + v_y + g \cdot \cos(\text{tilt})$$

$$x = v_x + g \cdot \sin(\text{tilt})$$

$$f = \tan^{-1}\left(\frac{x}{y}\right)$$

Where *vis* = visual background orientation, b = body vector along y-axis, *g* = represents the otolith-mediated gravitational signal which is a constant on Earth = 1, *tilt* = body orientation, *f* = perceptual upright (PU).

# 2.4 Results

## 2.4.1 Subjective Visual Vertical

The SVV measured against a grey background with the body-oriented RSD indicates how much gravity and body orientation influence perceptions of subjective vertical when visual information is absent. When RSD, responses closer to 0° indicate a bias towards the body's orientation, while responses closer to -90° indicate a bias towards gravity.

### 2.4.1.1 Grey Background

The results show that there was an SVV shift in a negative direction towards gravity for the Upright condition and a shift in a positive direction towards the body axis once lying RSD before and after intense VR exposure [Upright (M = -2.36, SE = 0.10); Pre $VR_{RSD}$ (M = 15.64, SE = 2.76); Post $VR_{RSD}$ (M = 14.48, SE = 2.79)] (Figure *2.9*a). To determine whether there was an effect of the VR exposure on the performance in the SVV data for the grey background, a one-way repeated measures ANOVA was conducted on the RSD SVV data. Assumptions of normality and homogeneity of variances were evaluated prior to conducting the repeated measures ANOVA. The Shapiro-Wilk test indicated that the data were normally distributed ($p > 0.05$). Mauchly's test of sphericity was violated for the SVV scores [$X^2(2) = 25.2, p < 0.001$] and the Greenhouse-Geisser corrected test revealed a significant main effect for the SVV scores across exposures [$F(2,38) = 35.6$, $p < 0.001$]. Post hoc tests with Bonferroni adjustments for multiple comparisons found no significant change in SVV scores following high intensity VR exposure relative to baseline ($t = 0.487, p = 1.00$). Significance was found between Pre $VR_{RSD}$ and the Upright condition ($t = 7.54, p < 0.001$), and Post $VR_{RSD}$ and the Upright condition ($t = 7.05, p < 0.001$) (Figure *2.9*a). For the JND scores, the Friedman test was performed and there was a significant

main effect across exposures $X^2(2) = 30.4, p < 0.001$] [Upright (M = 0.31, SE = 0.026); Pre $VR_{RSD}$ (M = 8.16, SE = 0.65); Post $VR_{RSD}$ (M = 7.29, SE = 0.57)] (Figure 2.9b).

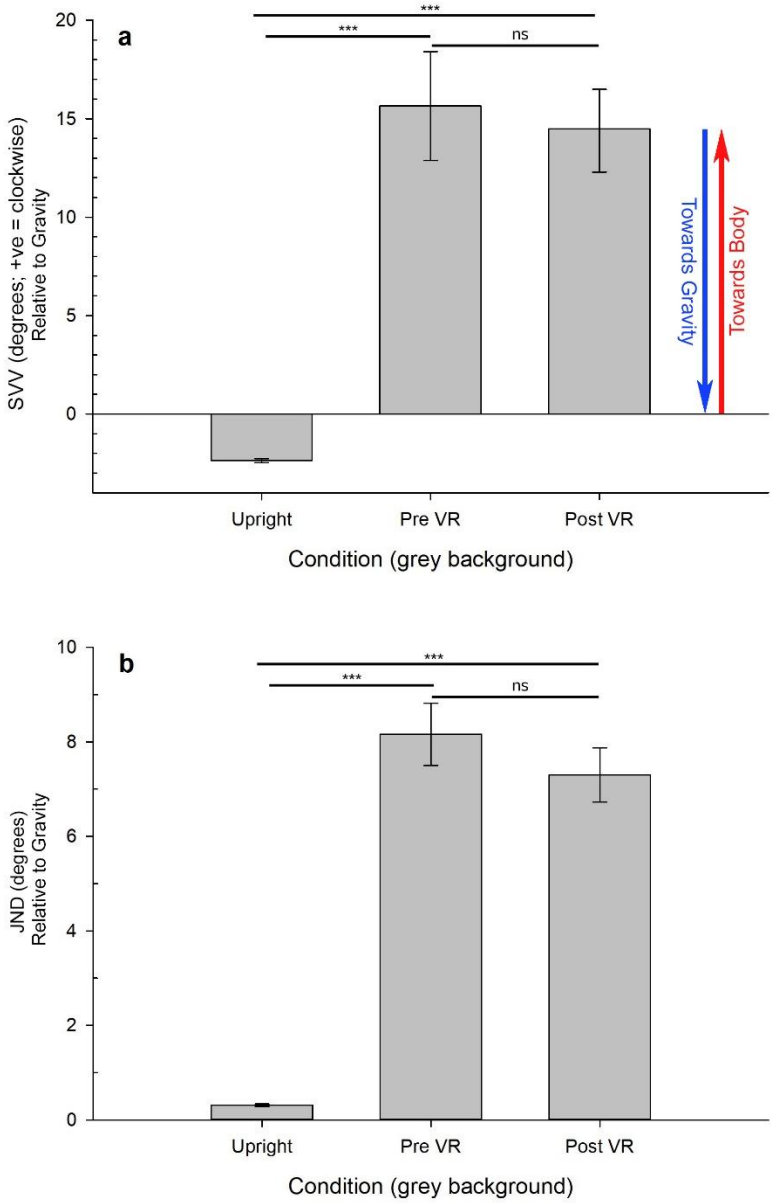


Figure 2.9: Summary of a) Subjective Visual Vertical (SVV) estimate relative to gravity and b) just-noticeable difference (JND) between upright, right side down pre and post intense VR exposures for grey background. Error bars are standard error.

### 2.4.1.2 All Visual Backgrounds

To determine whether there was an effect of the VR exposure on the performance in the SVV task for all visual backgrounds, the average SVV Pre $VR_{RSD}$ (Figure *2.10*a) and Post $VR_{RSD}$ (Figure *2.10*b) for all

visual backgrounds were plotted. The effect of visual backgrounds orientation on SVV can be seen in Pre $VR_{RSD}$ [Grey (M = 15.64, SE = 2.76); 0° (M = 25.42, SE = 3.49); 60° (M = 20.76, SE = 3.18); 120° (M = 15.33, SE = 2.37); 180° (M = 13.79, SE = 3.37); 240° (M = 8.86, SE = 3.61); 300° (M = 21.08, SE = 2.22)]. A shift in the negative direction towards gravity can be seen after intense VR exposure [Grey (M = 14.48, SE = 2.78); 0° (M = 19.06, SE = 3.70); 60° (M = 12.38, SE = 4.81); 120° (M = 12.77, SE = 2.47); 180° (M = 11.33, SE = 3.59); 240° (M = 4.55, SE = 3.98); 300° (M = 15.13, SE = 3.34)].

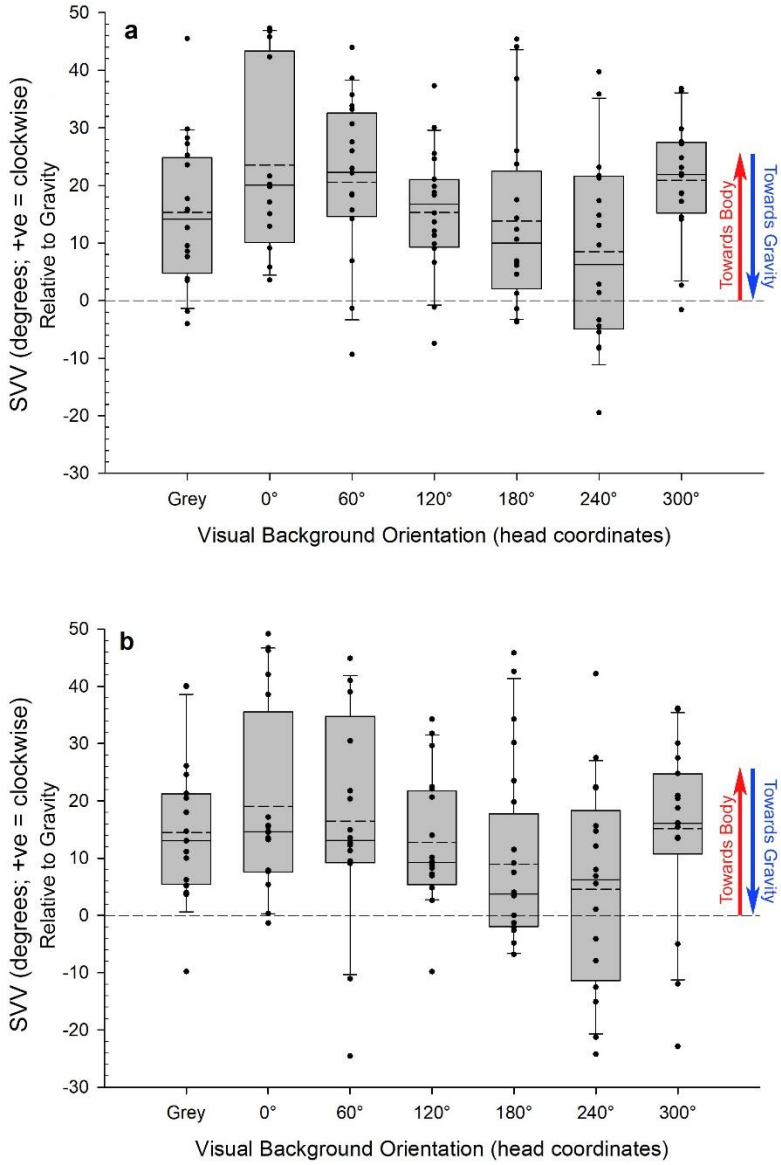


Figure 2.10: SVV estimate of a) RSD Pre VR and b) RSD Post VR of different visual background orientations relative to gravity. Solid lines within the box plot are the median and dashed lines are the mean. Error bars are standard error.

To see the change in SVV after VR, ΔSVV was plotted (Figure 2.11). The results show a shift in a negative direction towards gravity for all visual backgrounds after intense VR exposure [Grey (M = -1.16, SE = 1.11); 0° (M = -6.36, SE = 2.29); 60° (M = -8.38, SE = 4.07); 120° (M = -2.56, SE = 1.20); 180° (M = -2.47, SE = 1.95); 240° (M = -4.31, SE = 1.64); 300° (M = -5.95, SE = 2.13)].

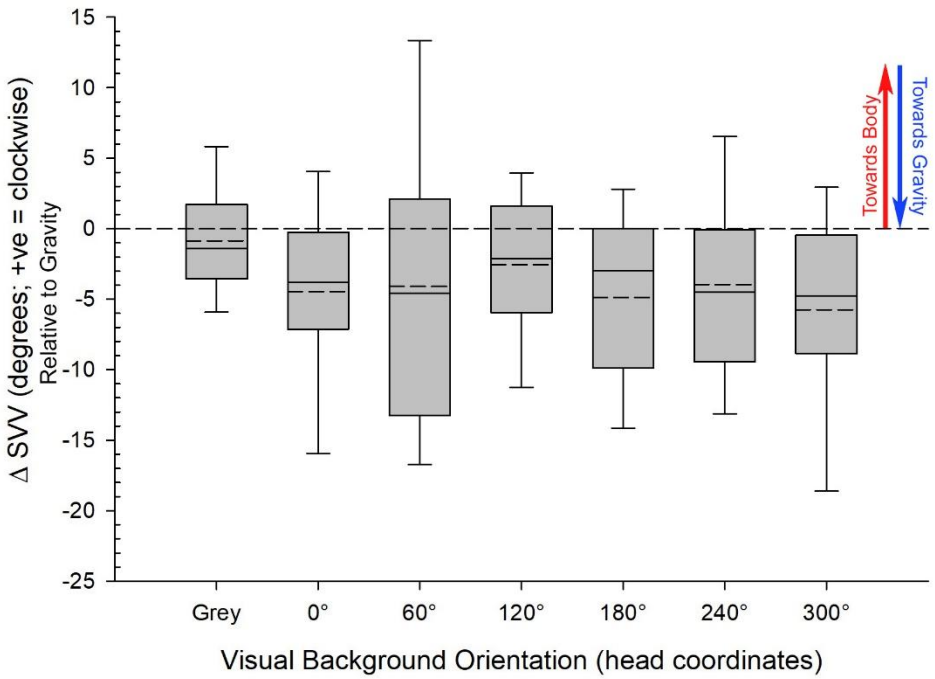


Figure 2.11: Change in SVV after intense VR exposure for different visual background orientations. Solid lines within the box plot are the median and dashed lines are the mean. Error bars are standard error.

| | | | | One sample t-tests | | |
|---|---|---|---|---|---|---|
| Room | Condition | SVV | s.e. | df | *t* | *p* |
| **Grey** | Upright | -2.36 | ±0.10 | 19 | -22.9 | **<0.001** |
| | Pre VR RSD | -74.7 | ±2.77 | 19 | -26.9 | **<0.001** |
| | Post VR RSD | -75.5 | ±2.78 | 19 | -27.1 | **<0.001** |
| **Room 0** | Upright | -2.50 | ±0.0003 | 19 | -8063.5 | **<0.001** |
| | Pre VR RSD | -66.5 | ±3.53 | 19 | -18.8 | **<0.001** |
| | Post VR RSD | -70.9 | ±3.70 | 19 | -19.2 | **<0.001** |
| | Upright | -2.45 | ±0.08 | 19 | -30.3 | **<0.001** |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Pre VR RSD | -69.4 | ±3.23 | 19 | -21.5 | **<0.001** |
| **Room 60** | Post VR RSD | -73.5 | ±4.16 | 19 | -17.7 | **<0.001** |
| | Upright | 2.41 | ±0.06 | 19 | 37.4 | **<0.001** |
| **Room 120** | Pre VR RSD | -74.7 | ±2.37 | 19 | -31.5 | **<0.001** |
| | Post VR RSD | -77.2 | ±2.47 | 19 | -31.2 | **<0.001** |
| | Upright | -2.60 | ±0.07 | 19 | -37.8 | **<0.001** |
| **Room 180** | Pre VR RSD | -76.2 | ±3.37 | 19 | -22.6 | **<0.001** |
| | Post VR RSD | -81.1 | ±3.58 | 19 | -22.7 | **<0.001** |
| | Upright | -2.56 | ±0.12 | 19 | -20.4 | **<0.001** |
| **Room 240** | Pre VR RSD | -81.5 | ±3.69 | 19 | -22.1 | **<0.001** |
| | Post VR RSD | -85.4 | ±3.98 | 19 | -21.5 | **<0.001** |
| | Upright | -2.76 | ±0.31 | 19 | -8.91 | **<0.001** |
| **Room 300** | Pre VR RSD | -69.1 | ±2.26 | 19 | -30.6 | **<0.001** |
| | Post VR RSD | -74.9 | ±3.33 | 19 | -22.5 | **<0.001** |

Table 2.1: Average SVV values were obtained from each combination of visual background (grey, room stimulus) and body orientation (up, RSD) before and after intense VR exposure. Positive values are clockwise relative to the observer. Standard errors represent the inter-subject variance of these values. One sample t-tests indicate whether the average SVV was significantly different from the orientation of gravity when upright (0°) and when RSD (-90°). Independent t-tests indicate whether the average SVV was significantly different across the participant groups. Significance ($p<0.05$) is highlighted in bold text.

| Room | Condition | Independent t-tests | | |
|---|---|---|---|---|
| | | *df* | *t* | *p* |
| Grey | Pre VR RSD | 38 | 0.22 | 0.830 |
| | Post VR RSD | | | |
| Room 0 | Pre VR RSD | 38 | 0.87 | 0.388 |
| | Post VR RSD | | | |

| | | | | |
|---|---|---|---|---|
| Room 60 | Pre VR RSD | 38 | 0.78 | 0.443 |
| | Post VR RSD | | | |
| Room 120 | Pre VR RSD | 38 | 0.74 | 0.463 |
| | Post VR RSD | | | |
| Room 180 | Pre VR RSD | 38 | 0.99 | 0.328 |
| | Post VR RSD | | | |
| Room 240 | Pre VR RSD | 38 | 0.73 | 0.470 |
| | Post VR RSD | | | |
| Room 300 | Pre VR RSD | 38 | 1.42 | 0.163 |
| | Post VR RSD | | | |

Table 2.2: Paired comparisons (paired t-tests) indicate whether the average SVV was significantly different across the conditions. Significance ($p<0.05$) is highlighted in bold text.

A two (Condition: pre-VR, post-VR) by seven (Background orientation) repeated-measures ANOVA was conducted on SVV PSE values. Mauchly's test indicated that sphericity was violated for the main effect of Background ($W = 0.006$, $\chi^2(20) = 71.48$, $p<0.001$) and the Condition × Background interaction ($W = 0.002$, $\chi^2(20) = 89.77$, $p<0.001$), so Greenhouse–Geisser corrected degrees of freedom are reported. There was a significant main effect of Condition, $F(1, 18) = 8.07$, $p = 0.012$, partial $\eta^2_p = 0.34$, indicating a gravity-ward shift in SVV following VR exposure averaged across backgrounds. There was also a significant main effect of Background, $F(2.52, 40.27) = 7.43$, $p = 0.001$, partial $\eta^2 = 0.32$, confirming that background orientation influenced SVV. The Condition × Background interaction was not significant, $F(2.34, 37.47) = 1.81$, $p = 0.173$, partial $\eta^2_p = 0.10$, although the medium effect size suggests that the magnitude of the shift may have varied across backgrounds.

Given departures from normality in several background-specific difference-score distributions, permutation-based ANOVA (5,000 permutations) was used as a robustness check. Permutation results converged with the parametric inference (Condition $p = 0.009$; Background $p<0.001$; Condition × Background $p = 0.095$), strengthening confidence in the main effects while indicating that evidence for differential reweighting across backgrounds did not reach conventional significance thresholds in this sample.

## 2.4.2 Subjective Visual Vertical and Cybersickness

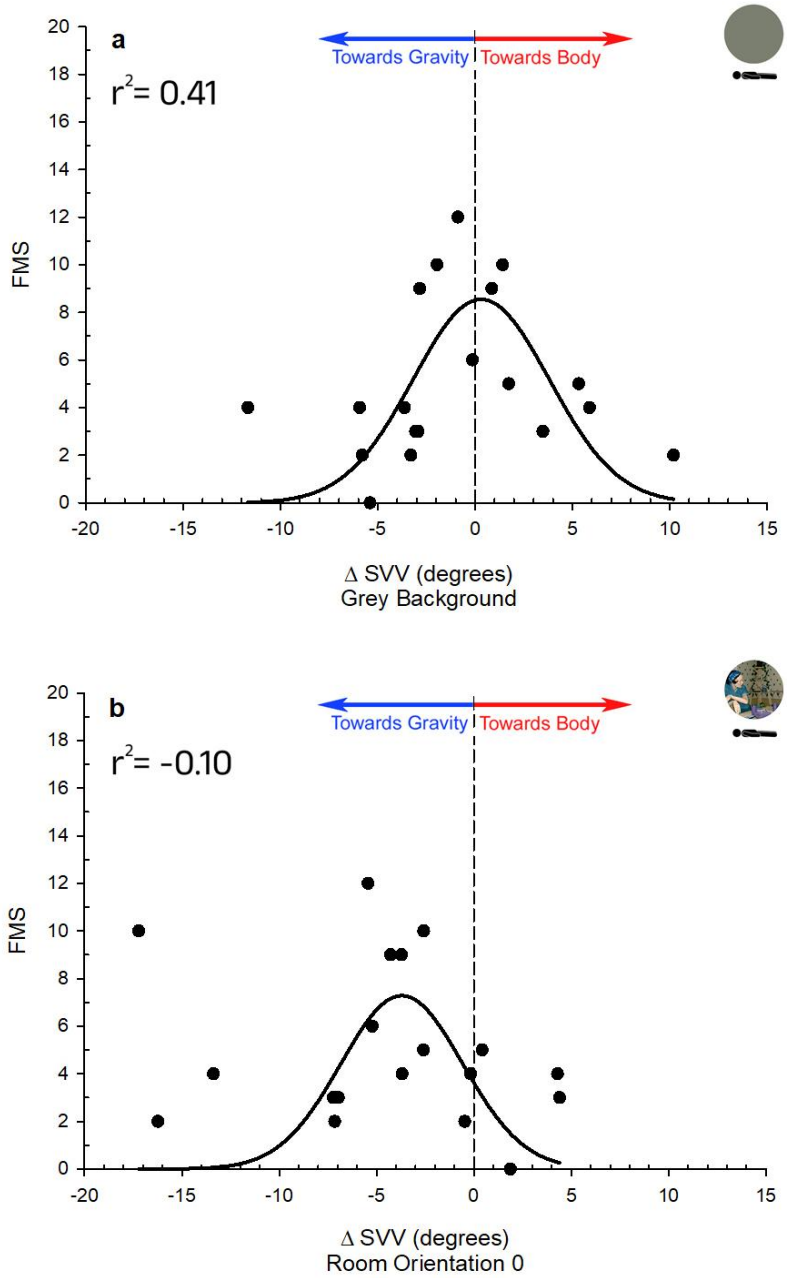


Figure 2.12: Gaussian model of FMS ratings relative to the right side down change in SVV for a) the grey background and b) for 0° room orientation after intense VR exposure. The dashed line indicates the upper specification of the regression of no difference in the SVV from baseline.

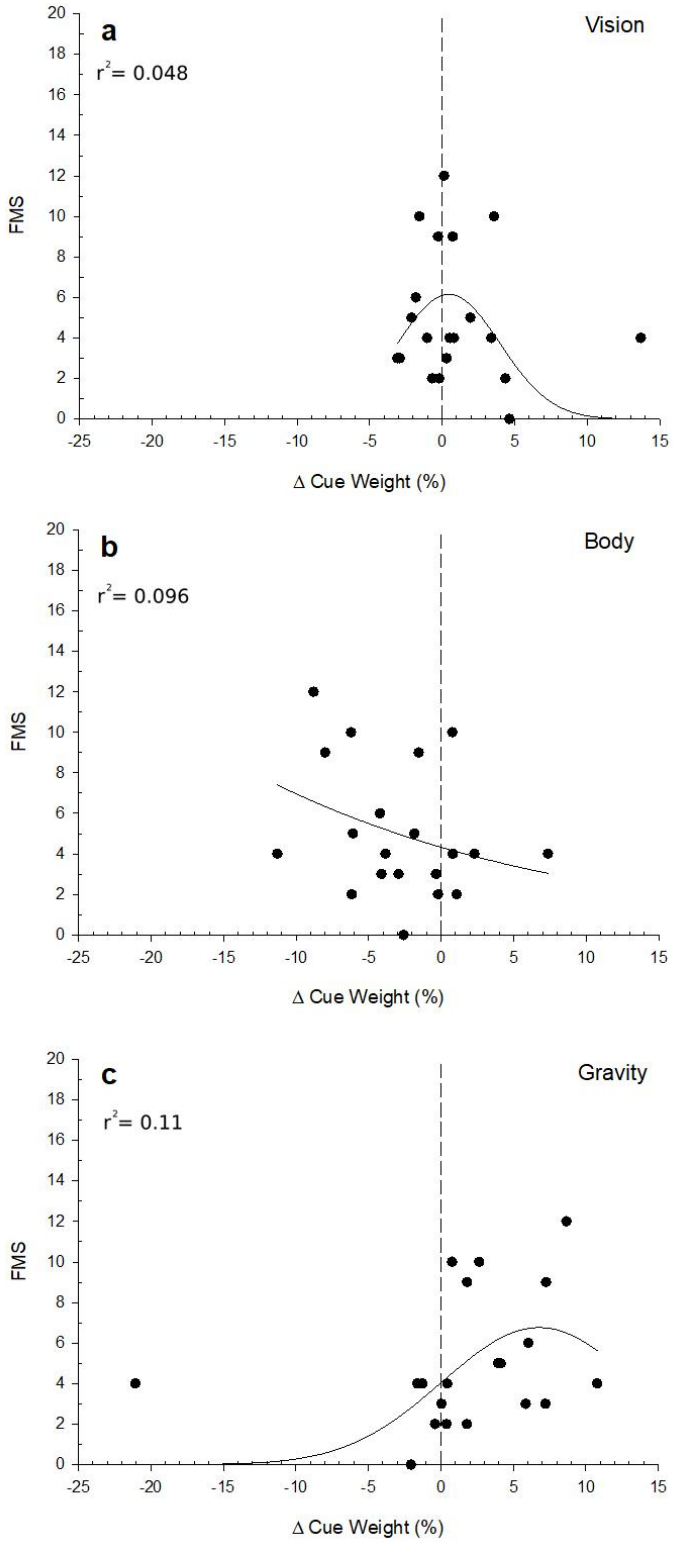


Figure 2.13: Gaussian model of FMS ratings relative to the change in cue weight per cents for a) vision and b) body, and c) gravity. The dashed line indicates the upper specification of the regression of no difference in the SVV from baseline.

| Participant | Δ Vision Cue Weight % | Δ Body Cue Weight % | Δ Gravity Cue Weight % |
|---|---|---|---|
| **P1** | -1.81 | -4.22 | 6.03 |
| **P2** | 13.70 | 7.37 | -21.07 |
| **P3** | 4.36 | -6.16 | 1.80 |
| **P4** | -1.04 | 2.31 | -1.26 |
| **P5** | -0.68 | 1.07 | -0.39 |
| **P6** | -1.56 | 0.78 | 0.78 |
| **P7** | 0.31 | -0.35 | 0.04 |
| **P8** | -0.27 | -1.54 | 1.81 |
| **P9** | -2.11 | -1.84 | 3.95 |
| **P10** | 0.13 | -8.80 | 8.67 |
| **P11** | -0.20 | -0.20 | 0.41 |
| **P12** | 1.95 | -6.09 | 4.14 |
| **P13** | 4.64 | -2.58 | -2.06 |
| **P14** | -3.09 | -4.12 | 7.21 |
| **P15** | 3.40 | -3.85 | 0.45 |
| **P16** | 0.81 | 0.81 | -1.62 |
| **P17** | 0.74 | -8.00 | 7.26 |
| **P18** | -2.92 | -2.93 | 5.85 |
| **P19** | 0.52 | -11.30 | 10.79 |
| **P20** | 3.57 | -6.22 | 2.64 |
| **Average** | **1.02** | **-2.79** | **1.77** |

Table 2.3: Output of the weighted vector sum model for vision and body relative to gravity (set to 1) from the best fits to the SVV before and after VR expressed as percentage (e.g. vision% = $\vec{v}/(\vec{v}+\vec{b}+\vec{g})$).

A two-way repeated-measures ANOVA examined the effects of Condition (pre vs. post) and Cue Weights (vision, body, gravity), treating Participant as a repeated factor. Although the normality and equal variance tests failed ($p<0.05$), the analysis proceeded with caution. Results revealed a significant main effect of Cue Weights, $F(2,38) = 842.72$, $p < 0.001$, $\eta^2_p = 0.63$, and a significant Condition × Cue Weights interaction, $F(2,38) = 17.43$, $p < 0.001$, $\eta^2_p = 0.37$. The main effect of Condition alone was not significant, $F(1,19) = 1.44$, $p = 0.246$, $\eta^2_p = 0.07$. These findings indicate that cue weights differed across cue types and that the effect of condition depended on cue type, but overall, there was no main effect of condition at the group level.

A two-way ANOVA was conducted on Δ Cue Weight %, computed as ((post – pre) × 100), with Participant and Cue Weight (vision, body, gravity) as factors. Although the normality test failed ($p<0.05$), the equal variance test passed ($p = 1.000$). The Δ Cue Weight % ($p = 0.053$) did not reach significance,

meaning there was no statistically reliable difference among participants' overall percentage change in cue weights, nor among the three cue types, after controlling for each other.

| Cue | Mean Difference | 95% CI Lower | 95% CI Upper |
|---|---|---|---|
| **Body** | -0.24 | -0.35 | -0.14 |
| **Gravity** | 0.00 | 0.00 | 0.00 |
| **Vision** | 0.21 | 0.11 | 0.33 |

Table 2.4: Bootstrap 95% confidence intervals for paired mean differences (Post – Pre) in cue weights (vision, body, gravity). Mean differences and confidence intervals were estimated using 10,000 bootstrap resamples.

### 2.4.3 Cluster Analysis

To further characterize individual differences in post VR reweighting, I conducted an exploratory k-means cluster analysis using each participant's change in cue weights from pre to post VR (ΔVision, ΔBody, ΔGravity; percentage points). The aim was to determine whether participants naturally separated into distinct reweighting profiles based on the direction and magnitude of redistribution across cues. Because cue weights are compositional (increases in one cue necessarily imply decreases in one or more others), the clustering is interpreted as differences in relative redistribution among the three cues rather than fully independent changes.

To determine the most appropriate number of clusters, k-means solutions were evaluated across a range of cluster counts (k) using the elbow method, which quantifies how within-cluster dispersion decreases as additional clusters are added. Specifically, I computed the within-cluster sum of squares (inertia) across candidate values of k and identified the elbow point, where adding more clusters yields only marginal reductions in inertia, to guide the selection of k.

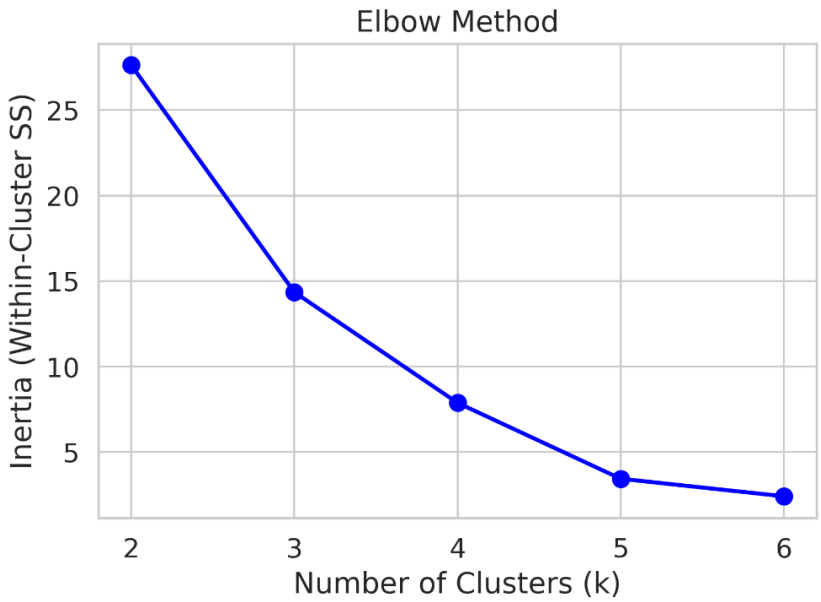


Figure 2.14: Elbow method for selecting the number of clusters (k) in cue-weight reweighting profiles.

As a complementary criterion, I assessed each candidate k using silhouette analysis, which quantifies how well each participant fits within their assigned cluster relative to neighbouring clusters. Higher mean silhouette values indicate stronger separation between clusters and better within-cluster cohesion. Together, the elbow and silhouette approaches provide convergent, data-driven guidance for selecting k before describing the resulting cue-weight profiles.

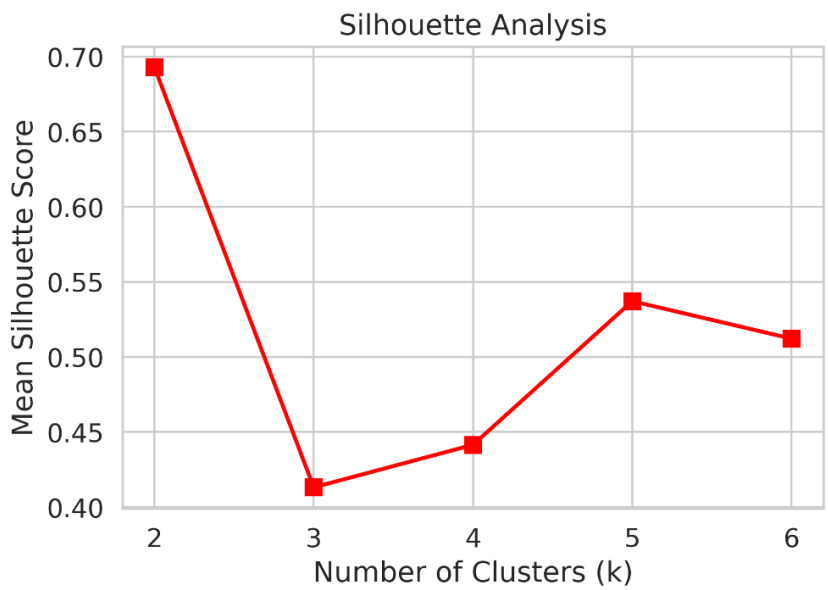

Figure 2.15: Silhouette analysis for selecting the number of clusters (k) in cue-weight reweighting profiles.

## 2.5 Discussion

The purpose of this study was to examine the impact of prolonged VR exposure on sensory reweighting and to explore its potential association with the severity of cybersickness. Sensory reweighting was evaluated through the systematic bias observed in the subjective visual vertical task, characterized by its inclination towards the body axis when performed in a 90° clockwise roll tilt orientation (Alberts et al., 2015, 2016; Clemens et al., 2011; De Vrijer et al., 2009b). Our findings indicated that a 20-minute exposure to an intense VR environment resulted in an alteration in SVV bias. Specifically, there was a notable shift away from the body axis and towards the gravitational upright, as evidenced by a decrease in the PSE. This finding suggests that the VR experience may have led to an increased reliance on vestibular cues, a decreased dependence on the idiotropic body cue, or a combination of both, influencing the perception of subjective vertical (Chung & Barnett-Cowan, 2023; Mittelstaedt, 1983). This effect was also observed by Chung and Barnett-Cowan (2023), where they saw a significant shift in the bias of the SVV away from the body axis towards the gravitational upright.

### 2.5.1 Sensory Reweighting and Cybersickness Susceptibility

A further aim of this study was to explore how changes in SVV relate to cybersickness. Our modelling suggests a relationship between the reweighting of sensory cues (vision, body, and gravity) and the degree of motion sickness symptoms. A similar Gaussian relationship for the grey background to that reported by Chung and Barnett-Cowan (2023) was observed in Figure *2.12*a, where individuals reporting a higher sickness level (higher FMS) have a lower change in SVV after VR. This experiment aimed to add visual backgrounds to assess the effects of vision, and as seen in Figure *2.12*b, in the presence of vision, most participants shift towards gravity compared to the grey background. Then, from this new position of the SVV, some people who have more change from this offset get less sick. These findings align with a growing body of literature indicating that VR-induced sensory mismatches (particularly involving vision and gravity) are a key driver of cybersickness (Ang & Quarles, 2023; Caserman et al., 2021; Kim et al., 2022; Weech et al., 2020).

The Gaussian modelling of Fast Motion Sickness ratings relative to changes in SVV and cue weights suggests that the relationship between reweighting and cybersickness is not necessarily linear. However, the cue-weight results should be interpreted with caution. Descriptively, the model output was consistent

with reduced body weighting after VR and with small average increases in vision and gravity-related weighting. The clearest and most consistent change was the reduction in body weighting, whereas evidence for gravity up-weighting was weaker and should be treated as descriptive rather than definitive. Accordingly, Figure *2.13*a illustrates the Gaussian relationship between FMS and Δ vision cue weight, indicating that the visual weighting, on average, remained unchanged after VR exposure. In contrast, Figure *2.13*b shows an almost inverse pattern for FMS and Δ body cue weight, where the negative shift suggests that body cues were downweighed post VR. Finally, Figure *2.13*c shows the Gaussian relationship between FMS and Δ gravity cue weight, revealing a complementary upweighting of gravitational cues.

When the functional form of the FMS relationships was tested directly, the strongest support for a non-linear association emerged for ΔSVV in the grey condition. This is also the cleanest SVV context in the chapter because it minimises directional visual orientation cues and therefore provides a more direct readout of post-VR changes in verticality perception. The fitted peak near minimal ΔSVV supports the interpretation that higher sickness is associated with little to no perceptual updating in SVV, which is consistent with the idea that failure to adapt verticality perception under conflict can be maladaptive. At the same time, the absence of a detectable association for the 0° background indicates that not all background-specific ΔSVV measures are equally informative about sickness severity, likely due to increased variability and the additional influence of the visual scene in those conditions.

For cue weights, the group-level pattern was small on average and differed by cue type, but the individual-differences links to FMS were weaker and less consistent than the grey ΔSVV relationship. The most consistent cue-weight signal was an association between Δ gravity cue weight and FMS, suggesting that participants who showed larger increases in gravity weighting tended to report higher sickness. This pattern can be interpreted cautiously in two ways that are not mutually exclusive: it may reflect compensatory reweighting in the most symptomatic participants, or it may reflect overcorrection that accompanies discomfort.

To probe how participants reweight sensory cues, cue weights were compared before and after VR exposure (Table *2.3*). The results indicate that cue weights differed by cue type and that the effect of VR exposure depended on cue type, consistent with redistribution among cues rather than a uniform shift across all cues. When the size of these changes was examined across participants, the overall pattern was in the expected direction but did not reach conventional significance for Δ cue weight percentage. A likely contributor is the modest sample size, which limits power to detect small to moderate individual-differences effects in model-derived measures.

Several assumption checks also indicated departures from normality in the present analyses. Although repeated-measures ANOVA is often robust to moderate deviations from normality at similar sample sizes, key inferences were interpreted cautiously and corroborated using robustness checks, including permutation-based and rank-based approaches where appropriate. Under this framework, marginal or non-significant findings are treated as inconclusive rather than as evidence of absence.

Effect size analyses indicated that the main effect of cue type was large (partial $\eta^2 = 0.63$), while the interaction between condition and cue type was also substantial (partial $\eta^2 = 0.37$). In contrast, the main effect of condition alone was small (partial $\eta^2 = 0.07$). This suggests that differences in cue weighting are primarily driven by the type of sensory cue and by the interaction with VR exposure, rather than by VR exposure alone at the group level.

Recent research supports the idea that individuals who are more visually dependent (i.e., those who rely heavily on visual cues for orientation) are more prone to cybersickness, as they struggle to reweight cues when visual-vestibular conflicts arise in VR (Chung & Barnett-Cowan, 2023; McManus & Harris, 2023). Conversely, those who adapt their sensory weighting more effectively may experience less discomfort, reinforcing the importance of individual differences in sensory integration.

A key limitation is that SVV was assessed in a recumbent posture post-exposure, which may have altered the availability and reliability of body-related cues relative to upright stance. As a result, fitted "weights" should be interpreted as model-based indices of cue influence within this specific testing context, rather than as direct estimates of stable sensory gains that generalize across postures (Dyde et al., 2006; Peterka, 2002). Future work could strengthen causal interpretation by collecting SVV in both upright and recumbent postures pre and post exposure, and by pairing perceptual measures with independent markers of multisensory balance control dynamics, which are known to show time-varying reweighting (Assländer & Peterka, 2016). More broadly, systematic reviews emphasize substantial heterogeneity in cybersickness measurement and susceptibility, so identifying robust mechanistic markers will likely require convergent evidence across perceptual, postural, and physiological outcomes (Biswas et al., 2024).

### 2.5.2 Individual Variability in Cybersickness Susceptibility

Individual variability in cybersickness susceptibility was substantial, and the present data suggest that the most consistent individual-differences signal was linked to changes in graviceptive weighting rather than to changes in visual or body weighting alone. While cue weights are interdependent by definition, formal tests did not provide strong evidence that Δ vision or Δ body cue weights reliably predicted FMS in this sample. In contrast, Δ gravity cue weight showed a monotonic association with sickness severity,

suggesting that participants who increased gravity weighting more tended to report higher FMS. This relationship should be interpreted cautiously because the direction could reflect compensatory reweighting in the most symptomatic participants, overcorrection under conflict, or both. Given the modest sample size and the additional estimation noise introduced by model-based cue-weight fitting, null associations for Δ vision and Δ body should be treated as inconclusive rather than as evidence of absence.

However, individual variability in cybersickness responses must also be considered. Recent studies indicate that factors such as age, gender, and prior VR experience significantly influence adaptation to sensory conflicts (Kourtesis et al., 2024). For example, women have been found to report more severe cybersickness symptoms than men, possibly due to differences in field-of-view sensitivity or hormonal influences on vestibular function (Abu Selo et al., 2025; Grassini & Laumann, 2020; Stanney, Fidopiastis, et al., 2020; Stanney, Lawson, et al., 2020). Additionally, VR-naive users may exhibit greater susceptibility due to a lack of prior exposure, whereas experienced users show signs of neural adaptation that reduce symptom severity over time (Adhanom et al., 2022b). These factors highlight the need for individualized approaches to cybersickness mitigation, as different user groups may require tailored training or VR adjustments to optimise comfort. In the present study, these participant-level factors were not the primary focus and the sample size limits stratified analyses, but they remain plausible contributors to the observed heterogeneity and should be recorded and modelled explicitly in future work.

Cluster analyses (Figure *2.14* and Figure 2.15) initially indicated that a k = 2 solution was driven by a single participant (P2) whose cue-weight changes were markedly larger than the rest of the sample. Under k-means, this observation consistently formed a singleton cluster, indicating that the dominant structure in the dataset was an outlying reweighting response rather than stable subgroup separation. Because singleton clusters are not interpretable as a meaningful subtype in this context, P2 was excluded for the purpose of clustering only to evaluate whether any coherent multi-participant profiles were present among the remaining participants.

With P2 excluded, the two-cluster solution remained the most interpretable, yielding two groups that differed in the direction and magnitude of post-VR sensory reweighting. The first group, termed Flexible Adapters (N = 8; P1, P9, P10, P12, P14, P17, P18, P19), was characterized by a substantial reduction in body cue reliance accompanied by a corresponding increase in gravity weighting (ΔBody = -5.92%, ΔGravity = +6.74%), with minimal change in visual weighting (ΔVision = -0.82%). The second group, termed Rigid Perceivers (N = 11; P3, P4, P5, P6, P7, P8, P11, P13, P15, P16, P20), showed negligible changes across all three cue types following VR exposure (ΔVision = +1.21%, ΔBody = -1.45%,

ΔGravity = +0.24%), suggesting little post-VR sensory recalibration. The overall cluster quality was modest (silhouette score = 0.375), consistent with the exploratory nature of this analysis.

Flexible Adapters reported somewhat higher FMS scores on average (M = 5.88, SD = 3.14, Mdn = 5.0) compared to Rigid Perceivers (M = 4.55, SD = 3.50, Mdn = 4.0); however, this difference did not reach statistical significance (Mann-Whitney U = 59.0, p = 0.227, rank-biserial r = -0.341; Welch's t = 0.868, p = 0.398, Cohen's d = 0.400). The non-parametric test was preferred given evidence of non-normality in the Rigid Perceivers group (Shapiro-Wilk W = 0.846, p = 0.038). These results suggest a small-to-medium effect in the expected direction, with greater sensory reweighting trending toward higher cybersickness, though the sample size was insufficient to detect this effect reliably.

To assess whether P2's exclusion altered the group-level correlations, a sensitivity analysis was conducted by repeating the Spearman correlations between each Δ cue weight and FMS score with and without P2. The correlation between Δ gravity cue weight and FMS was maintained and marginally strengthened following P2's exclusion (full sample: ρ = 0.46, p = 0.041; without P2, N = 19: ρ = 0.49, p = 0.032), indicating that the association was not driven by this outlying observation. Associations for Δ vision and Δ body cue weights remained negligible in both analyses, consistent with the null findings reported for the full sample.

The present study hypothesized that individuals who reweight sensory cues more substantially following VR exposure would report lower cybersickness, on the basis that flexible multisensory adaptation reflects a functional response to the conflict imposed by the virtual environment. The cluster-level comparison does not support this hypothesis. Although the difference was not statistically significant, Flexible Adapters, who showed the largest shifts in body and gravity cue weighting, tended to report higher FMS scores than Rigid Perceivers, a pattern in the opposite direction to what was predicted. The direction of the effect, where those who reweighted most actively also tended to report higher sickness, runs counter to the idea that flexible sensory adaptation should be protective. One possible explanation is that the reweighting observed in Flexible Adapters reflects a reactive rather than a pre-emptive process: rather than smoothly recalibrating during VR exposure, these individuals may have experienced greater perceptual disruption, which both drove their sickness and forced a larger post-exposure recalibration of cue weights. Rigid Perceivers, by contrast, may have maintained a more stable internal model throughout the session, resulting in less perceptual conflict and lower symptom burden. This interpretation may suggest that individuals with more rigid multisensory priors are paradoxically buffered against acute cybersickness because they suppress conflicting signals rather than attempting to integrate them.

### 2.5.3 Neural Mechanisms and Implications for Cybersickness Mitigation

Our findings complement earlier work by Weech, Moon, and Troje (2018), who demonstrated that adding mild vestibular noise stimulation can reduce the severity of visually induced motion sickness, presumably by improving the consistency of visual-vestibular integration (Weech et al., 2018). It is possible that participants who showed larger post-VR shifts in SVV engaged more effective updating of internal orientation priors under conflict, whereas participants with minimal ΔSVV may have maintained a more rigid estimate, consistent with sustained prediction error signalling.

Neurophysiological data indicate that individuals who struggle to recalibrate sensory expectations during VR exposure exhibit increased activity in brain regions associated with sensory prediction errors, such as the posterior insula and anterior cingulate cortex (Chang et al., 2023). These findings suggest that cybersickness may, in part, stem from a failure of the brain's predictive coding mechanisms, leading to persistent sensory conflicts. Within predictive coding frameworks, insular and cingulate responses can be interpreted as signalling persistent visuo-vestibular prediction errors that should drive updating of internal models and, downstream, reallocation of cue weights. In this view, individuals who show sustained prediction error signalling may be those who fail to downweight conflicting visual motion evidence or to upweight stable graviceptive cues efficiently, thereby prolonging conflict and symptom expression (Ernst & Bülthoff, 2004; Seth & Friston, 2016).

Beyond passive monitoring, recent evidence points to a top-down route for symptom control. Applying low-frequency (0.08 Hz) transcranial alternating-current stimulation to the dorsolateral prefrontal cortex (dlPFC) nearly abolished the sway illusions and nausea produced by sinusoidal galvanic vestibular stimulation, indicating that dlPFC can exert top-down inhibition of vestibular inputs (McCarthy et al., 2023). Although these results are promising, translating neuromodulation approaches to VR cybersickness mitigation will require replication across VR paradigms and validation that changes in symptoms are accompanied by measurable changes in multisensory weighting.

Additionally, electroencephaloghy-based studies suggest that cybersickness may be associated with altered brain oscillations in sensory integration regions, including decreased alpha power in the parietal cortex and increased theta activity in vestibular processing areas (Chang et al., 2023). These findings open the door for real-time physiological monitoring as a potential strategy for mitigating cybersickness by dynamically adjusting VR parameters based on neural feedback. One interpretation is that reduced parietal alpha reflects increased sensory sampling and attentional engagement during conflict resolution, whereas increased theta reflects control and error-monitoring processes recruited as the system attempts

to stabilize orientation under mismatch. More broadly, these spectral shifts are consistent with transient reconfiguration of a frontoparietal control network interacting with vestibular-insular cortex, which would predict changes not only in local power but also in functional connectivity across these regions. This framing generates a testable prediction for future work, namely that individuals with lower cybersickness will show faster normalization of alpha-theta dynamics as cue weights stabilize (Cavanagh & Frank, 2014; Lopez et al., 2012).

Complementary evidence from Functional Near-Infrared Spectroscopy (fNIRS) further supports the involvement of multisensory cortical networks. Recent research has found that cybersickness during VR was associated with bilateral deactivation of the angular gyrus, interpreted as suppression of vestibular cortical processing during visual-vestibular conflict. This pattern is consistent with the sensory reweighting account proposed here: if the nervous system resolves visuo-vestibular conflict by downweighting vestibular input, reduced activation in vestibular-associated cortical regions would be the expected neural correlate (Yeo et al., 2024).

### 2.5.4 Alternative Measures of Sensory Reweighting

Although the present findings shed light on how SVV shifts and cue reweighting occur following a single intense VR session, it is worth noting that the SVV task is not the most sensitive method for detecting the influence of vision on vertical perception. Future research could consider using alternative tasks such as the OCHART, which is known to provide more nuanced and sensitive measures of visually driven orientation illusions (Dyde et al., 2006).

McManus & Harris (2023) demonstrated that OCHART is significantly more sensitive than SVV in detecting visual cue dominance, making it a more effective tool for quantifying individual differences in multisensory integration. By adopting the more sensitive OCHART paradigm, we aim to capture subtler changes in visual cue weighting. In addition, investigating interindividual differences in susceptibility to cybersickness (e.g., by examining gender, age, or prior VR experience) could provide insight into why some participants exhibit stronger or weaker reweighting responses.

Overall, the data support VR-induced changes in verticality perception and suggest that individual differences in how graviceptive information is reweighted are linked to cybersickness severity, motivating more sensitive visual-dependence measures in Chapter 3.

## 2.6 Conclusion

In this study, the contributions of graviceptive and body-related cues to verticality perception were examined before and after exposure to an intense VR experience. Across conditions, VR exposure was associated with a robust gravity-ward shift in subjective visual vertical estimates. Model-derived cue weights further suggested that VR exposure influenced the redistribution of cue contributions in a cue-dependent manner, consistent with sensory reweighting rather than a uniform shift across cues. This reweighting pattern aligns with previous work showing that VR-induced sensory conflicts can alter the way individuals integrate multiple cues to maintain a stable sense of upright.

Individual differences analyses suggest that higher sickness was most consistently associated with minimal change in ΔSVV in the grey condition and with larger increases in gravity cue weighting, whereas associations with changes in visual and body cue weights were weaker and less consistent in this sample. These observations reinforce the idea that cybersickness is deeply intertwined with the processes governing multisensory integration and how the nervous system balances or reweighs conflicting inputs.

Prior research has suggested that sensory reweighting mechanisms, particularly the ability to adjust visual-vestibular cue weighting, may underlie individual differences in motion sickness susceptibility (Chung & Barnett-Cowan, 2023). Although SVV captured robust post-VR changes in verticality perception, it may not be the most sensitive assay for isolating visually driven individual differences in orientation processing that contribute to cybersickness. This aligns with previous research suggesting that SVV is more strongly influenced by vestibular inputs than visual cues (McManus & Harris, 2023), meaning it may not fully reflect how visual dependence contributes to cybersickness. In contrast, studies utilizing Oriented CHAracter Recognition Tests have demonstrated that OCHART is more sensitive to visual influences on orientation perception, making it a potentially stronger predictor of cybersickness (Dyde et al., 2006).

Given these limitations, our next study (Chapter 3) will incorporate OCHART as an alternative. If OCHART proves to be a more sensitive measure of visual influences, this could offer further support for the role of visual dependence in cybersickness susceptibility and provide insights into who is most at risk in VR environments. A deeper understanding of these processes will aid in designing VR experiences that minimise motion sickness and optimise user comfort.

# Chapter 3
# Examining Sensory Reweighting and Oriented CHAracter Recognition Tests (OCHART) Shifts: Insights into Cybersickness in Virtual Reality

## 3.1 Abstract

Virtual reality (VR) environments present conflicting sensory cues that can disrupt spatial orientation and provoke cybersickness. This study used the Oriented CHAracter Recognition Test (OCHART) to quantify the perceptual upright (PU) before and after high-intensity VR exposure and to estimate how visual, bodily, and gravitational cues contribute to upright perception. Participants completed OCHART tasks across various visual background orientations and in both upright and right-side-down postures. While PU judgements showed strong visual attraction across background orientations ($p < 0.001$), VR exposure did not produce a reliable group-level shift in PU, with no main effect of condition ($p = 0.618$) and no condition-by-background interaction ($p = 0.622$). Vector sum modelling suggested modest post-VR cue reweighting in the pooled sample, characterised by a reduced reliance on body-referenced cues and an increased reliance on gravity. A significant individual difference result was the positive association between the change in visual cue weight and cybersickness severity; specifically, participants who increased their visual weighting following VR reported higher Fast Motion Sickness (FMS) scores (Spearman $\rho = 0.37$, $p = 0.027$). Model comparison indicated that this relationship was best described by a linear function ($R^2 = 0.120$, $p = 0.039$), providing no support for a curvilinear pattern. These results show that OCHART is sensitive to visual influences in VR and suggest that an increased reliance on visual cues post-exposure represents a maladaptive response that exacerbates visuo-vestibular conflict. Overall, this work suggests that cybersickness is driven by a failure to effectively down-weight unreliable visual signals in favour of stable gravitational cues, highlighting that individual differences in sensory plasticity are the primary determinants of user tolerance in immersive environments.

## 3.2 Introduction

As discussed in Chapter 2, Virtual Reality (VR) offers a unique platform for investigating sensory integration and perception because it can decouple normally congruent sensory cues. While VR technology has progressed over the past decade (Jang et al., 2022), cybersickness remains a primary obstacle to its widespread adoption (Caserman et al., 2021). Understanding why some individuals experience higher levels of discomfort than others requires a closer examination of how multiple sensory cues (vestibular, visual, and body) are combined and reweighted. In Chapter 2, we saw that measures such as the Subjective Visual Vertical (SVV) can detect changes in the relative contributions of vestibular and body cues after VR exposure. For instance, exposure to high-intensity VR experiences can alter the weighting of vestibular cues, as evidenced by shifts in SVV estimates, potentially correlating with cybersickness susceptibility (Chung & Barnett-Cowan, 2023).

Cybersickness is commonly attributed to sensory cue conflicts between visual motion signals and vestibular/proprioceptive feedback (Oman, 1990; Reason & Brand, 1975). Traditional models posit that when the brain receives incongruent information, such as seeing movement in VR while physically stationary, this mismatch triggers discomfort. However, such models do not fully explain why some individuals are more affected than others, prompting exploration of how people integrate and reweight sensory inputs when cues conflict (Assländer & Peterka, 2014; Bles et al., 1998).

While early orientation studies assumed that sensory cues are always fused, recent research shows the brain can switch to a causal inference mode and actively discard an outlier cue under large conflicts, consistent with causal inference models of multisensory upright (Dokka et al., 2019; Fetsch et al., 2013). These models suggest that under large conflicts, the brain downweighs the less reliable or inconsistent cue, leading to perceptual recalibration. The idea that the brain can infer whether sensory cues come from a common or separate source is foundational to Causal Inference models of multisensory perception (Körding et al., 2007; Rideaux et al., 2021). This mechanism may explain adaptive behaviours in VR, such as shifts in perceived upright, and offers a promising framework for predicting individual differences in cybersickness.

Recent research highlights that the brain's ability to flexibly reweight sensory inputs is fundamentally supported by neuroplasticity (Gangemi et al., 2023). Exposure to novel or conflicting sensory environments, such as those encountered in VR, can induce rapid synaptic modifications and functional reorganization within multisensory cortical networks. These neuroplastic mechanisms allow individuals to recalibrate perceptual strategies, maintain orientation, and adapt to ongoing sensory conflict. Critically,

such plasticity occurs not only in early sensory areas but also in higher-order multisensory integration centres such as the posterior parietal cortex, temporoparietal junction, and superior temporal sulcus, which are key for integrating visual, vestibular, and somatosensory signals and updating internal spatial representations (Drigas & Sideraki, 2024; Gangemi et al., 2023).

Alongside these neural mechanisms, recent studies increasingly emphasize the importance of individual differences in cybersickness susceptibility and sensory adaptation (Cheng et al., 2022; Kourtesis et al., 2024). Motion sickness susceptibility during adulthood has emerged as the most prominent predictor of cybersickness, with gaming experience as the second most significant factor (Kourtesis et al., 2024). Gender-related differences have also been documented, with females experiencing approximately 1.7 times higher susceptibility than males, attributed to factors such as interpupillary distance, field of view sensitivity, and hormonal fluctuations (Stanney et al., 2020).

The SVV provides a direct measure of perceived vertical alignment; however, the technique is not without its limitations. As highlighted in the previous chapter, the SVV requires observers to make explicit judgments about line orientation relative to gravity, a task that demands insight and potentially invokes cognitive strategies. Further, since VR is predominantly a vision-based medium, understanding how visual cues affect perceived orientation becomes critical. Although we know from Chapter 2 that downweighting vestibular cues and reweighting other sensory inputs can mitigate conflicts, we have yet to explore more sensitive methods that can better isolate the influence of vision on perceived orientation.

To address these considerations, this chapter focuses on the Oriented CHAracter Recognition Test (OCHART), introduced by Dyde et al. (2006) as an indirect measure of the perceptual upright (PU). The PU is the orientation in which objects, especially letters, are most easily recognised (Jolicoeur, 1985). By having participants discriminate between rotated versions of the letters ‘p’ and ‘d,’ OCHART circumvents many of the biases that can arise in explicit alignment tasks like the SVV. In OCHART, the average of the p-to-d and d-to-p transition orientations defines the PU, providing a stable index of how one integrates visual, body, and gravitational cues to determine what “upright” really is. OCHART offers distinct advantages over SVV for VR research. While SVV relies heavily on gravity and can be cognitively biased, OCHART bypasses explicit alignment judgments by exploiting orientation-sensitive character recognition (Dyde et al., 2006). This makes it ideal for quantifying the PU in environments where gravity cues may be unreliable or overridden by immersive visuals, such as in VR.

This measure is especially relevant in a VR context. We know from previous work that when body orientation is altered (e.g., lying on one’s side), the PU shifts its alignment from gravity to a greater

reliance on body orientation (Barnett-Cowan et al., 2012; Dyde et al., 2006). Moreover, when the visual surround is tilted relative to the observer, the PU is more affected than the SVV (Dyde et al., 2006). This makes OCHART uniquely sensitive to the influence of visual orientation cues and therefore ideal for testing the hypotheses that emerged from our discussion of cybersickness and sensory reweighting in Chapter 2. Specifically, if certain populations or individuals (e.g., those who experience heightened cybersickness) are more visually dependent, we should detect greater shifts in their PU compared to controls or those less susceptible to cybersickness. Large inter-individual variability characterizes visual dependency. Josupeit (2024) recently reevaluated the classic rod-and-frame task as a cybersickness screening instrument and showed that visual dependency alone did not predict cybersickness (Josupeit, 2024).

In addition, OCHART allows for the calculation of individual sensory weights using vector decomposition of PU. As discussed in Chapter 2, sensory reweighting is a dynamic process in which the central nervous system adjusts the relative importance of each sensory modality based on contextual demands and signal reliability (Maurer et al., 2006; Peterka, 2002). The vector sum model provides a framework to predict the PU by assigning vector lengths (weights) to gravity, body, and visual orientation cues (Barnett-Cowan et al., 2012; Dyde et al., 2006). While vector summation models have previously been used to predict navigational behaviour (Cheng & Spetch, 1995), Dyde et al. (2006) extended this idea to orientation perception tasks. Thus, the OCHART measure, together with the vector sum model, offers a powerful means to quantify how and why orientation perception differs across individuals and contexts – an understanding that may shed light on why some are more susceptible to VR-induced sensory disturbances.

Building on these considerations, Chapter 3 tests three linked predictions using OCHART and vector sum modelling. First, PU should vary systematically with visual background orientation, replicating the strong visual modulation reported in prior PU work (Dyde et al., 2006). Second, if VR exposure induces sensory reweighting, cue weights should change differentially from pre- to post-VR, which will be evaluated as a Condition × Cue Type interaction (vision, body, gravity) on cue weight percentages. Third, individual differences in reweighting should relate to cybersickness severity, such that participants who show less adaptive redistribution of cue weights, or who increase reliance on vision under conflict, will report higher sickness (Chung & Barnett-Cowan, 2023).

This chapter, therefore, has two main objectives: (1) to measure the influence of vision on orientation perception using OCHART (a more sensitive measure than traditional line alignment tasks) under upright and right side down (RSD) body orientations and tilted visual surrounds, and (2) to apply the vector sum

model to quantify how gravity, body, and visual cues combine to predict the PU. By comparing these results to predictions derived from the vector sum model, and by examining symmetrical conditions such as upright versus RSD, we can determine if the PU changes symmetrically and if the model's assumptions hold true (Dyde et al., 2006).

In summary, Chapter 3 builds upon the multisensory integration concepts introduced in Chapter 2 and employs the OCHART to evaluate how orientation perception adapts when vision, gravity, and body cues are misaligned, before and after exposure to intense VR stimuli. Chung and Barnett-Cowan (2023) found that individuals who exhibited greater changes in upright perception after VR reported less sickness. This supports the hypothesis that flexible reweighting, or the ability to adapt one's internal model of upright, may protect against cybersickness. This chapter builds on that insight by applying OCHART as a sensitive probe of such adaptation. By indirectly measuring the PU in these different conditions and applying the vector sum model, we seek not only to understand how the weights assigned to each sensory cue evolve, but also to determine whether individuals who exhibit greater flexibility in shifting their PU are less susceptible to experiencing cybersickness. These insights will advance our understanding of the mechanisms underlying VR-induced discomfort and guide the development of strategies to mitigate it.

## 3.3 Methods

### 3.3.1 Participants

Data were collected from twenty participants (8 males and 12 females; mean age = 22.9, SD = ±4.5) recruited from the University of Waterloo. Fourteen participated on a voluntary basis, and the remaining participants were compensated at a rate of $10 per hour. All participants had normal or corrected-to-normal vision and no history of vestibular or neurological disorders. The study was approved by the University Research Ethics Committee and informed consent was obtained in accordance with the Declaration of Helsinki.

Because Echo VR servers were discontinued before the target sample size could be reached, the primary analyses for this chapter are based on these 20 participants (of whom 18 produced reliable vector sum model fits). To address the resulting power limitation, a supplementary pooled analysis was also conducted. This pooled sample combined the 18 Experiment 2 participants with 18 participants from Experiment 3 (Chapter 4) who completed the no-EVS condition, yielding a total sample of N = 36. The pooled analysis is justified on methodological grounds: both samples completed the same OCHART protocol, viewed the same visual backgrounds at the same orientations, were tested in the same body

postures (upright and right-side-down), used the same VR hardware (Oculus Quest), and were exposed to the same VR game (Echo VR) for the same duration (20 minutes). Both samples were drawn from the same University of Waterloo student population. Critically, the Experiment 3 no-EVS condition involved no electrical stimulation, making it procedurally equivalent to Experiment 2. FMS did not differ significantly between Experiment 2 and Experiment 3 (Mann–Whitney $U = 101.0$, $p = 0.055$). However, baseline body cue weights differed (Experiment 3 > Experiment 2; Mann–Whitney $U = 74.0$, $p = 0.006$), indicating greater body dominance at baseline in the Experiment 3 subsample. This pooled sample size exceeds the $N = 34$ threshold required for 80% power as determined by the a priori power analysis.

#### 3.3.1.1 A-priori Power Analysis

An a priori power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2007) to estimate the minimum sample size required to detect within-factor effects in a repeated-measures ANOVA The analysis assumed a medium effect size ($f = 0.25$), $\alpha = .05$, desired power of 0.95, a correlation among repeated measures of $r = 0.50$, and a nonsphericity correction $\varepsilon = 0.75$. The resulting calculations yielded a noncentrality parameter of $\lambda = 15.19$, a critical $F(1.50, 78) = 3.44$, and a required total sample size of $N = 54$, providing an actual power of approximately 0.954. Recruitment, however, had to be halted at $N = 20$ because the Echo VR game servers were unexpectedly discontinued, preventing further data collection. As a result, the study retains sufficient power to detect medium-to-large effects but may be underpowered for small-to-medium effects. The a priori analysis showed that achieving 90% power would require a total sample of N=44, whereas 80 % power would be attained with N=34.

### 3.3.2 Apparatus

The VR content for this experiment was delivered using the Oculus Quest system. The Oculus Quest has a dual OLED display with a resolution of 1600x1440 (per eye) at 72Hz refresh rate. It is a completely wireless system featuring 6 degrees-of-freedom (6DOF) inside out tracking allowing the user to freely move around. Here we allowed participants to move within a 2.4x2.4m designated play area.

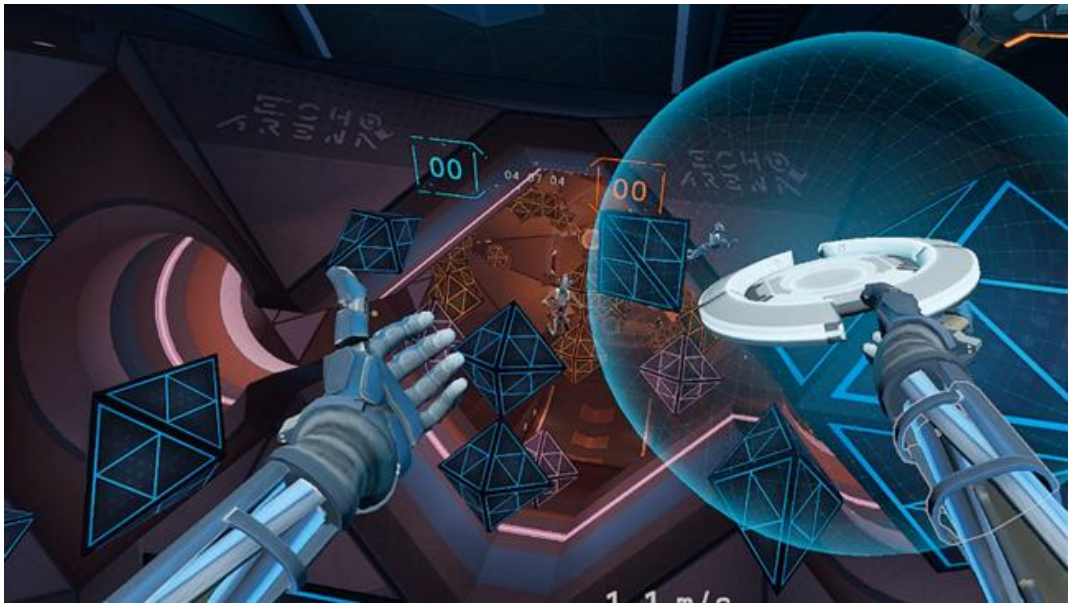


Figure 3.1: Screen capture of VR environment of the game used in the VR exposure, Echo VR.

The VR exposure consisted of a 20-minute session of Echo VR (Figure *3.1*), a zero-gravity arena game rated as "intense" on the Oculus Quest store and known to produce strong visuo-vestibular conflicts. The goal of the game was stealing the discs and scoring them into the opponent's (computer bots) net. Immediately before and after VR exposure, participants completed the Fast Motion Sickness Scale (FMS) to quantify cybersickness severity.

### 3.3.3 Stimulus

To assess participants' PU, the OCHART was employed. In this task, participants discriminate between the letters 'p' and 'd,' which, when rotated 180°, transform into each other. By identifying the angles at which the perceived letter transitions from 'p' to 'd' and vice versa, the PU can be indirectly determined. A polarized room image with clear orientation cues was used as the background, presented in six orientations (0° to 300° in 60° increments), each combined with multiple letter orientations. This yielded 1344 total trials, performed on a MacBook Pro laptop with a shroud installed to limit peripheral distractions, making the screen appear as a 35° diameter circle at a distance of 25 cm.

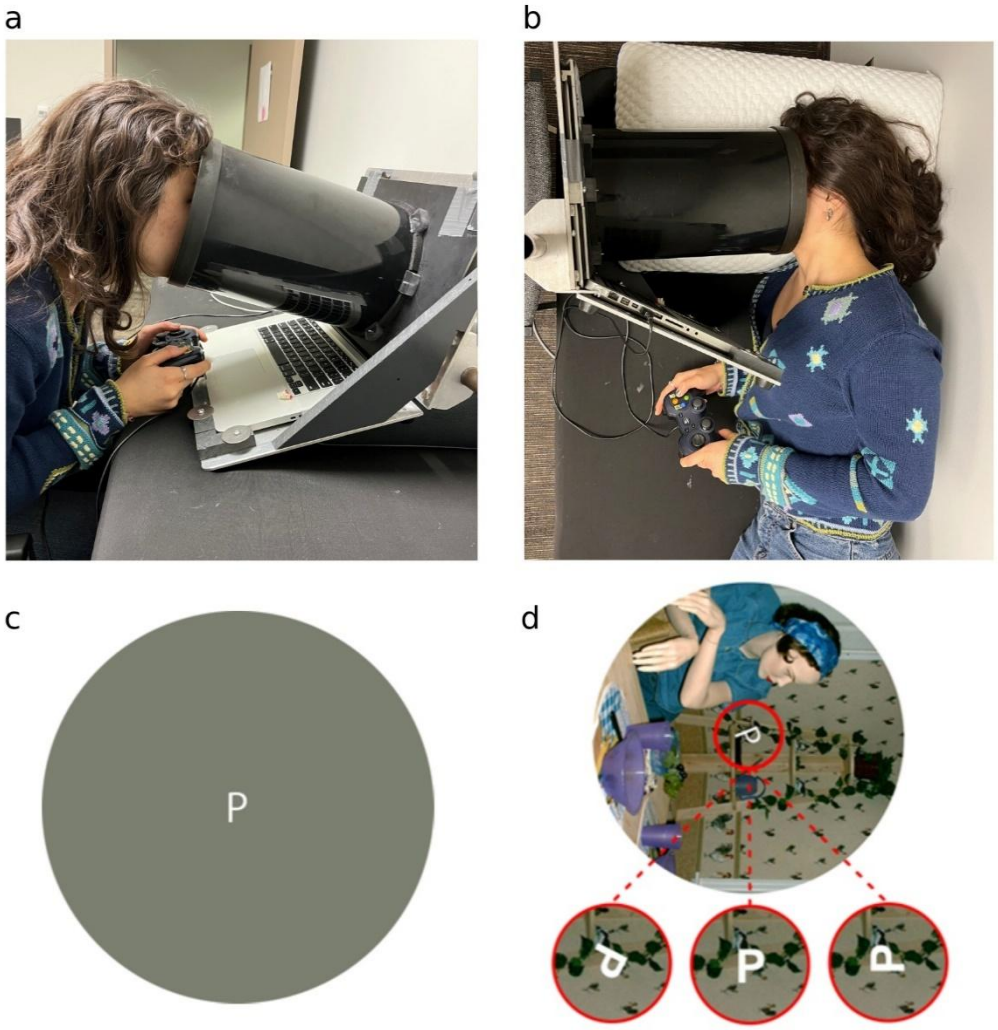


Figure 3.2: The two body postures used in these experiments: a) upright and b) right side down. Subjects viewed the display through a shroud to obscure all peripheral vision. An example of the OCHART task on c) the grey background and d) on the visual background. Participants were asked to determine whether they were presented with the letter ‘p’ or ‘d.’ Each combination of the letter orientation and background orientation was repeated eight times and that brought the total amount of trials to 1344.

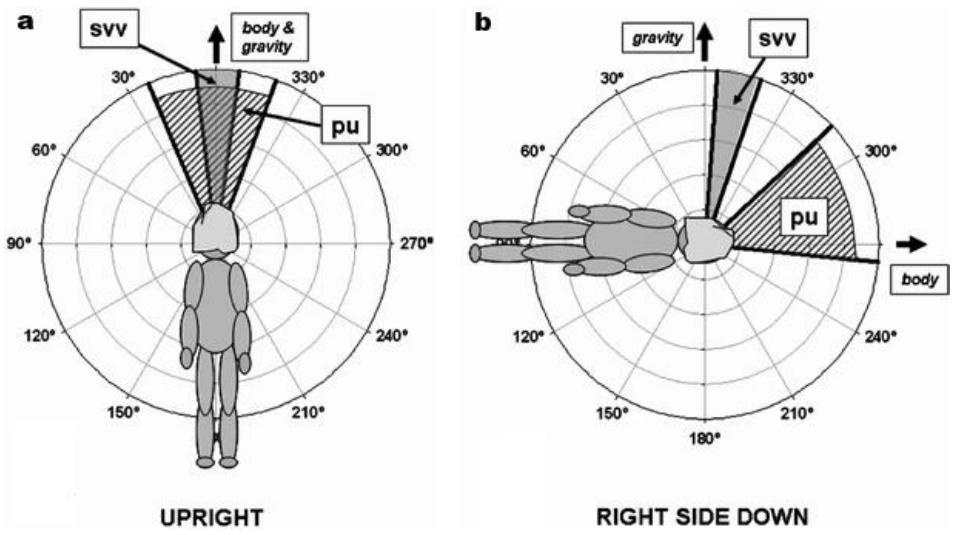


Figure 3.3: A polar summary in earth coordinates of the extent of the effect of the visual background for the SVV (solid grey area) and PU (hatched area) for a) upright and b) right-side-down body postures. The shaded segments indicate the full extent of effect that the orientation of the visual background had on the respective measures (from Dyde et al., 2006).

Each letter orientation and background orientation was repeated eight times, totalling 1344 trials, typically completed within a 15-minute period. Participants utilized buttons on a Logitech Gamepad F310 to indicate whether they perceived the letter 'p' or the letter 'd.'

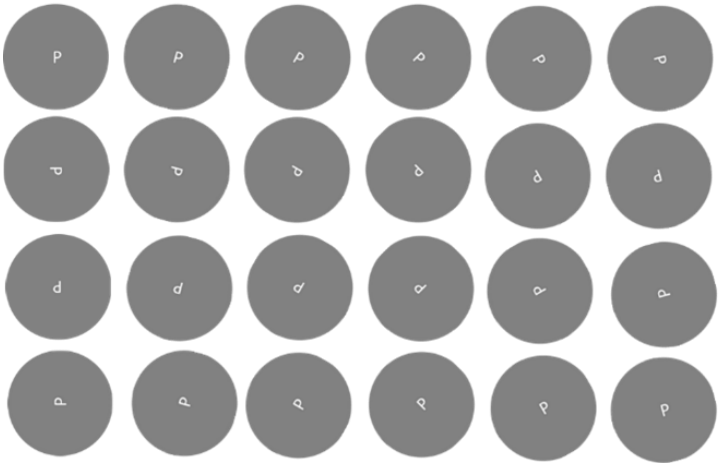

Figure 3.4: Twenty-four orientations of letter probe displayed against a grey background.

Figure 3.5: Experimental visual backgrounds, rotated in 60° increments from 0° to 300°.

### 3.3.4 Measures

Cybersickness was measured using the FMS, which measures sickness levels immediately following each VR session (Keshavarz & Hecht, 2011). Participants provided verbal ratings on a 20-point scale, with '0' indicating no sickness and '20' indicating severe or frank sickness.

### 3.3.5 Procedures

Participants first completed a baseline OCHART task both in an upright seated position and while lying on their right side on an examination table (Figure *3.2*). A foam pillow served as a headrest to ensure proper alignment of the head with the circular shroud attached to the laptop screen. During this initial task, participants used a two-alternative forced choice paradigm, pressing buttons on a gamepad to indicate whether the letter was a 'd' or a 'p.'

For the upright OCHART, the laptop was placed on a table and the participants sat on a chair with their face as close as possible to the circular shroud. For the RSD assessment, the laptop was rotated 90° clockwise and mounted on a custom-built stand. Again, participants provided their responses through a two-alternative forced choice paradigm button press format. Each OCHART session typically required approximately 15 minutes to complete. Following this, participants were escorted to the VR play area and equipped with the Oculus Quest system.

At the conclusion of the VR session, participants' sickness levels were assessed verbally using the FMS. Immediately after this evaluation, participants performed the final OCHART task (see Figure *3.6*).

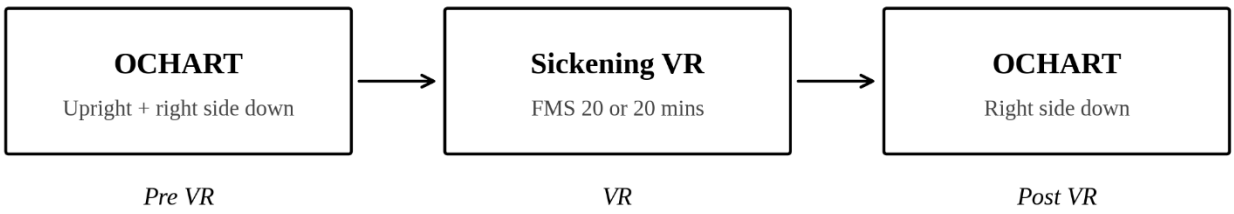

Figure 3.6: Flowchart of the experiment procedure.

### 3.3.6 Data Analysis

Statistical analysis was conducted using SigmaPlot 12.5. To determine the perceptual upright (PU), sigmoidal psychometric functions were fitted to the proportion of trials in which each letter orientation was identified as "p" rather than "d," plotted as a function of letter orientation angle. Following the original OCHART procedure, two separate sigmoid fits were used to estimate the two category transition points: the orientation at which responses shifted from "p" to "d," and the orientation at which responses shifted from "d" to "p" (Dyde et al., 2006).

(3.1) $$y = \frac{100}{1 + e^{\left(\frac{x - x_0}{b}\right)}}\%$$

In Eq. 3.1, $x$ denotes the letter orientation angle, $x_0$ denotes the 50% point of the sigmoid, and $b$ is a spread parameter reflecting the steepness of the transition curve. Smaller $b$ values indicate a steeper transition and therefore more precise discrimination between "p" and "d."

The two fitted 50% points, $x_{01}$ and $x_{02}$, were then averaged to estimate the participant's perceptual upright:

(3.2) $$PU = \frac{(x_{01} + x_{02})}{2} - 180$$

Here, $x_{01}$ and $x_{02}$ represent the two category boundaries separated by approximately 180°. Averaging these values gives the midpoint between the two ambiguous orientations, corresponding to the observer's PU. Subtracting 180° re-centres the estimate so that 0° corresponds to the canonical upright orientation of the character. For consistency with prior OCHART reporting, discrimination precision was indexed using the fitted slope parameter $b$ b, which is referred to here as the JND estimate.

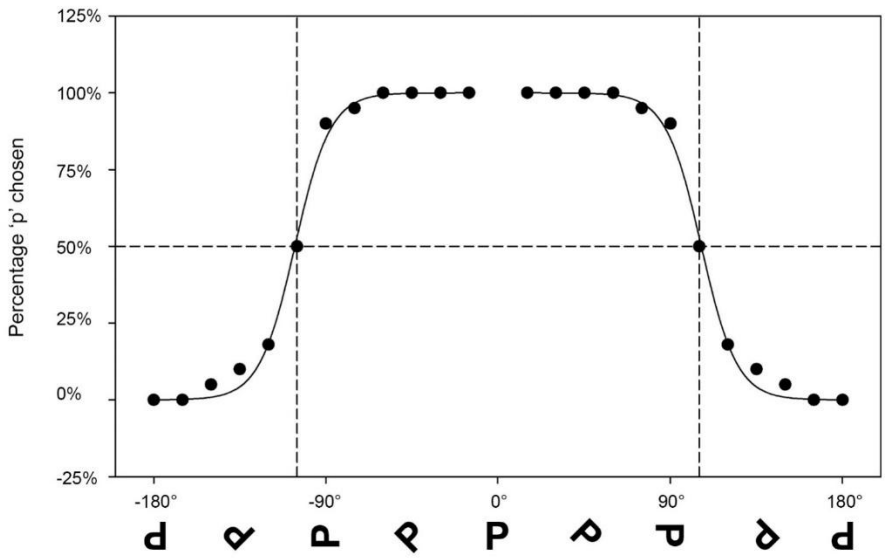


Figure 3.7: Illustration of how the PU were measured. Participants chose whether they thought the symbol was a 'p' or 'd.' Typical psychometric functions obtained from such responses are shown. The two 50% points were averaged to give the PU.

In addition to computing PU from psychometric fits, cue weights for vision, gravity, and body orientation were extracted using a vector sum model (Dyde et al., 2006). Statistical analyses included two (Condition: Pre-VR RSD, Post-VR RSD) × seven (Background Orientation) within-subjects repeated-measures ANOVA, with Greenhouse-Geisser corrections when sphericity was violated. Given non-normality in several cells and in the Pre-to-Post change scores, permutation-based ANOVA and nonparametric follow-ups were used to verify robustness of inference. Associations between PU change and cybersickness were assessed using rank-based correlations.

### 3.3.7 Vector Sum Model

To determine how visual, bodily, and gravitational cues contribute to the perception of upright, we applied a vector sum model (Dyde et al., 2006; Mittelstaedt, 1983; Mittelstaedt & Fricke, 1988). In this model, each cue is represented as a vector pointing in its respective direction, with a weight indicating its relative influence. By fitting participants' responses to this model using the Marquardt-Levenberg optimisation algorithm, we quantified the contribution of each cue to the OCHART (Press et al., 1992). The relative contributions of vision, the body, and gravity are reflected in the lengths of their respective vectors that best fit the data.

In this model, vectors $\vec{v}$ and $\vec{b}$ represent the orientations signalled by visual and body cues, respectively, while $\vec{g}$ denotes the gravitational direction. We fixed the length of the gravity vector at 1 and adjusted the

lengths of the visual and body vectors to best fit the data. This approach assumes that each vector accurately reflects the direction indicated by its corresponding sensory input.

A schematic representation of this model is provided in Figure *3.8*, illustrating how the combined influence of these cues determines the perceived vertical.

(3.3) $$\overrightarrow{PU} = \vec{v} + \vec{b} + \vec{g}$$

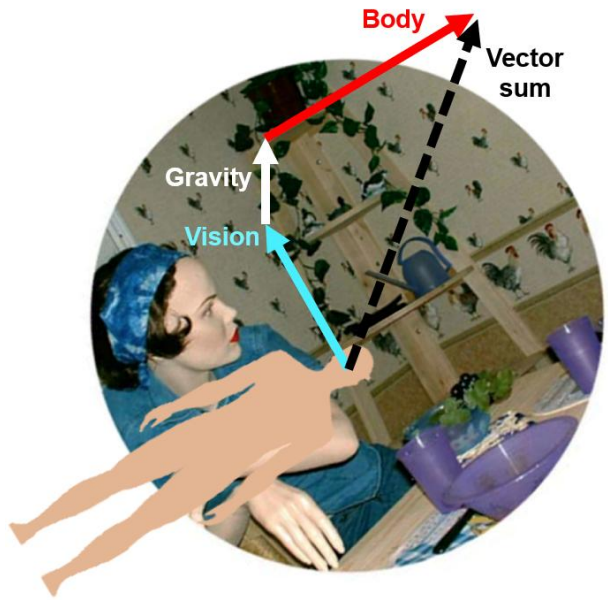


Figure 3.8: A weighted vector sum model, in which gravity, body orientation, and visual cues are summed to generate an estimate of vertical. The predicted outcome is depicted by the dotted black line, representing the vector sum.

This model conceptualizes each sensory input (vision, body, gravity) as a directional vector with a corresponding weight, and the resultant PU is the vector sum of these influences. Gravity is typically held constant (normalised to 1.0), while the relative weights of vision and body cues are adjusted to fit participant responses. This approach enables a direct comparison of sensory contributions and allows for pre vs. post VR change detection on an individual level.

(Eq 3.4) $$\boldsymbol{vy} = \text{vis} \cdot \cos(v)$$

$$\boldsymbol{v_x} = \text{vis} \cdot \sin(v)$$

$$\boldsymbol{y} = \text{b} + v_y + g \cdot \cos(\text{tilt})$$

$$x = v_x + g \cdot \sin(\text{tilt})$$

$$f = \tan^{-1}\left(\frac{x}{y}\right)$$

Where *vis* = visual background orientation, b = body vector along y-axis, *g* = represents the otolith-mediated gravitational signal which is a constant on Earth = 1, *tilt* = body orientation, *f* = perceptual upright (PU).

### 3.3.8 Cluster Analysis

To explore whether participants showed separable profiles of sensory redistribution following VR exposure, an exploratory cluster analysis was conducted on the pooled PU cue-weight dataset (Experiment 2 combined with Experiment 3 no-EVS participants). Clustering was performed on pre-to-post change scores for cue weight percentages (Δ vision, Δ body, Δ gravity). Prior to clustering, multivariate outliers were screened using both univariate z-score thresholds ($|z| > 3.0$) and Mahalanobis distance ($p < .001$). Because the pooled sample combined participants from two studies, Kruskal-Wallis tests were used to check for study-related distribution differences in cue weight percentages and FMS prior to treating the dataset as a single mixed sample.

## 3.4 Results

### 3.4.1 Perceptual Upright

The PU measured against a grey background with the body-oriented RSD indicates how much gravity and body orientation influence perceptions of upright when visual information is absent. Responses in the positive direction indicate a bias towards the body's orientation, while responses towards the negative indicate a bias towards gravity.

#### 3.4.1.1 Grey Background

The PU from the OCHART was biased in a negative direction toward gravity in the right-side-down posture both before and after VR exposure. Mean PU values for the grey background were Upright (M = -4.36, SE = 1.84), Pre-VR RSD (M = -41.06, SE = 3.43), and Post-VR RSD (M = -41.62, SE = 3.75) (Figure *3.9*a). The Pre-to-Post VR difference in the right-side-down posture was small (ΔM = 0.56°) and not statistically reliable, indicating that VR exposure did not produce a systematic shift in grey-background PU.

To test the effect of posture and VR exposure on grey-background PU, a one-way repeated-measures ANOVA (Upright, Pre-VR RSD, Post-VR RSD; N = 18) was conducted. Assumptions were met (Shapiro–Wilk: all $p > 0.80$; Mauchly's test: $p = 0.512$). There was a significant main effect of condition, $F(2, 34) = 68.25$, $p < 0.001$, $\eta^2_p = 0.80$. Holm-corrected pairwise comparisons indicated that Upright differed from both Pre-VR RSD and Post-VR RSD (both $p < 0.001$), while Pre-VR RSD and Post-VR RSD did not differ ($p = 0.894$). Permutation testing (5,000 permutations) and a Friedman test converged on the same conclusion (both $p < 0.001$), confirming that the effect reflects the body-tilt manipulation rather than VR exposure.

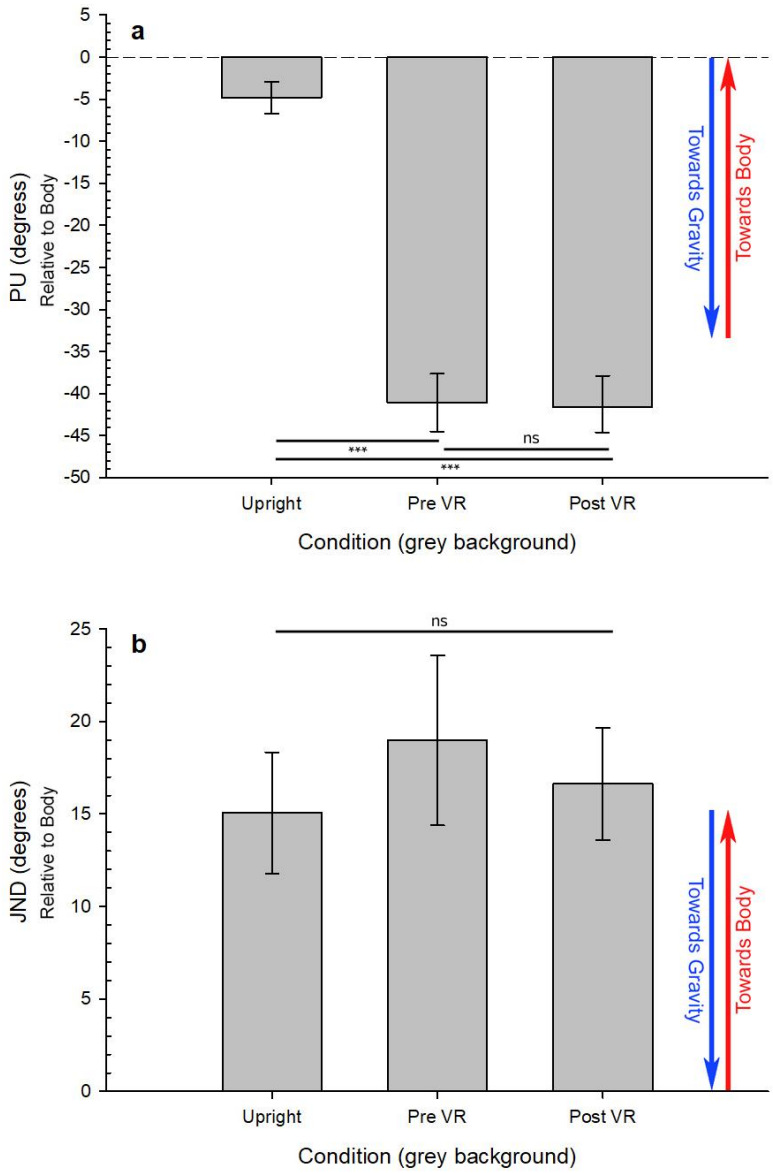

Figure 3.9: Summary of a) Perceptual Upright (PU) estimate and b) just-noticeable difference (JND) between upright, right side down pre and post intense VR exposures for grey background. Error bars are standard error.

A one-way repeated-measures ANOVA was conducted on JND for the grey background (Upright, Pre-VR RSD, Post-VR RSD). JND values were right-skewed and violated normality in all three conditions (all $p \leq 0.001$), although sphericity was met (Mauchly's test: $p = 0.242$). The condition effect was not significant, $F(2, 32) = 1.38$, $p = 0.266$, $\eta^2_p = 0.08$. Given the severe normality violations, a Friedman test was used as the primary robustness check and was also not significant, $\chi^2(2) = 3.08$, $p = 0.215$. Holm-corrected pairwise comparisons were non-significant (all $p > 0.44$), consistent with no reliable change in discrimination precision across conditions. The power of the test performed was low (0.049), suggesting these results should be interpreted with caution.

#### 3.4.1.2 All Visual Backgrounds

To determine whether there was an effect of VR exposure on performance in the OCHART task across all visual backgrounds, the average PU RSD Pre VR (Figure *3.10*a) and Post VR (Figure *3.10*b) for each visual background were plotted. The effect of visual background orientation on PU can be seen in Pre VR [Grey (M = -41.06, SE = 3.43); 0° (M = -27.97, SE = 5.27); 60° (M = -20.25, SE = 5.03); 120° (M = -38.52, SE = 3.91); 180° (M = -55.16, SE = 6.29); 240° (M = -61.82, SE = 4.49); 300° (M = -49.52, SE = 3.26)]. Following VR exposure, a similar pattern was observed in Post VR [Grey (M = -41.62, SE = 3.74); 0° (M = -26.84, SE = 6.19); 60° (M = -24.92, SE = 4.67); 120° (M = -44.38, SE = 6.09); 180° (M = -57.29, SE = 6.53); 240° (M = -60.63, SE = 5.79); 300° (M = -51.80, SE = 5.05)].

A two (Condition: Pre-VR RSD, Post-VR RSD) × seven (Background Orientation) within-subjects repeated-measures ANOVA was conducted on PU. Mauchly's test indicated that sphericity was violated for the Background main effect and the Condition × Background interaction, so Greenhouse-Geisser corrected results are reported. There was a significant main effect of Background, indicating that PU varied robustly with visual background orientation, $F(GG)(1.46, 24.88) = 19.59$, $p < 0.001$, $\eta^2_p = 0.54$. In contrast, there was no main effect of Condition, $F(1, 17) = 0.26$, $p = 0.618$, $\eta^2_p = 0.02$, and no Condition × Background interaction, $F(GG)(1.44, 24.47) = 0.38$, $p = 0.622$, $\eta^2_p = 0.02$. Permutation-based tests (5,000 permutations) converged on the same conclusions, confirming that the Background effect was robust while VR exposure did not produce a systematic group-level change in PU.

Because the Condition main effect and the Condition × Background interaction were not significant, follow-up comparisons focused on the main effect of Background, collapsing across pre- and post-VR conditions. Holm-corrected paired comparisons indicated that several background orientations differed

reliably in PU. Specifically, 60° differed from 120°, 180°, 240°, and 300°; 0° differed from 120°, 180°, 240°, and 300°; grey differed from 0°, 60°, 120°, 240°, and 300°; and 300° differed from 180° and 240° (all p_Holm < 0.05). In contrast, several pairs previously reported as significant in the manuscript were no longer significant following correction of the error term, including 120° vs 180°, grey vs 180°, and 180° vs 240°. Overall, the pattern of significant contrasts was consistent with the expected visual modulation of perceptual upright, with the strongest differences occurring between backgrounds associated with more body-aligned PU (0°, 60°) and those associated with more gravity-aligned PU (180°, 240°, 300°), while grey and 120° tended to fall between these ranges.

| Background | Mean Difference | 95% CI Lower | 95% CI Upper |
|---|---|---|---|
| **Grey** | 1.50 | -9.10 | 11.3 |
| **0 Degrees** | 4.13 | -13.3 | 24.2 |
| **60 Degrees** | -2.67 | -15.4 | 10.2 |
| **120 Degrees** | -5.17 | -17.6 | 7.17 |
| **180 Degrees** | 4.09 | -10.2 | 18.2 |
| **240 Degrees** | 4.80 | -6.03 | 15.6 |
| **300 Degrees** | -0.38 | -11.2 | 9.49 |

Table 3.1: Bootstrap 95% confidence intervals for paired mean differences (Post – Pre) in perceptual upright across visual backgrounds. Mean differences and 95% confidence intervals were estimated using 10,000 bootstrap resamples.

One-sample tests on ΔPU (Post minus Pre) confirmed that no background showed a statistically reliable shift after VR exposure using either a one-sample t test (all $p \geq 0.349$) or a Wilcoxon signed-rank test (all $p \geq 0.182$), and all 95% confidence intervals included zero.

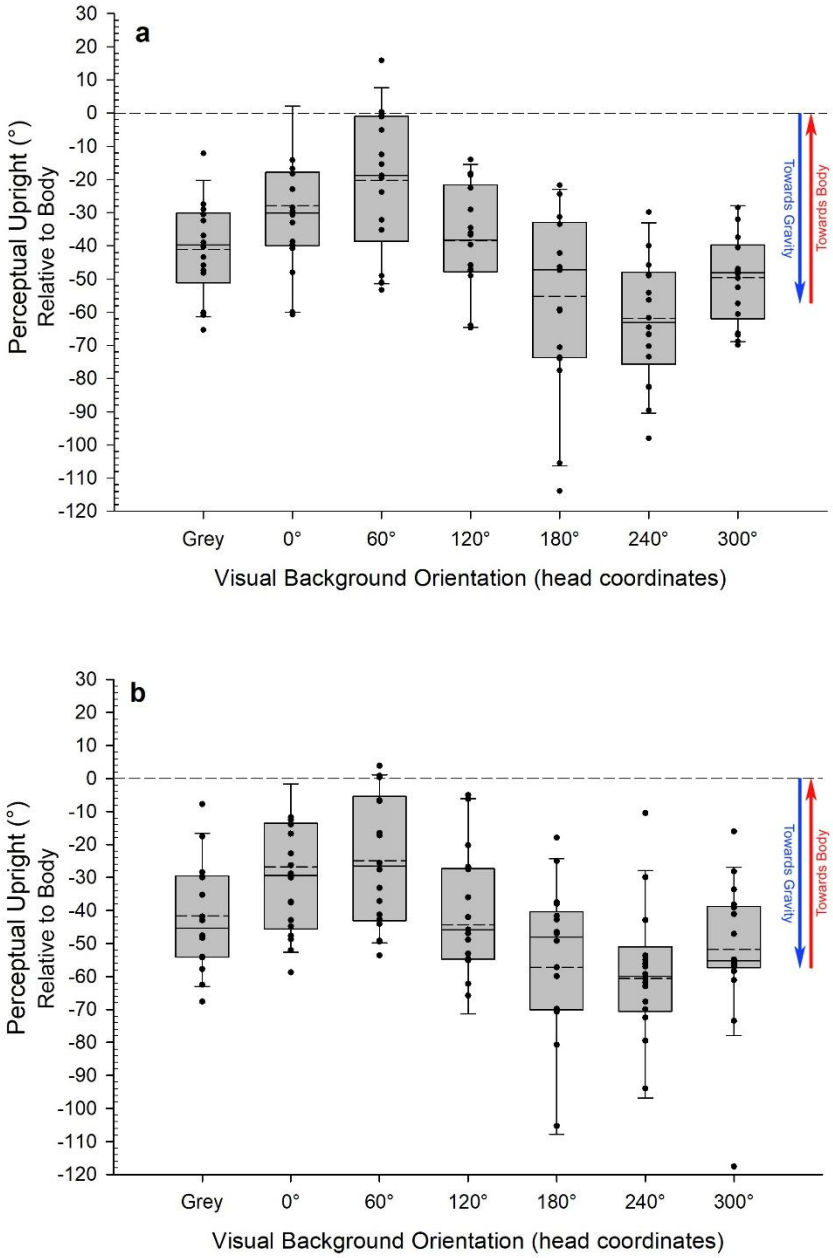


Figure 3.10: Perceptual Upright estimate of right side down a) Pre VR and b) post VR of different visual background orientations. Solid lines within the box plot are the median and dashed lines are the mean. Error bars are standard error.

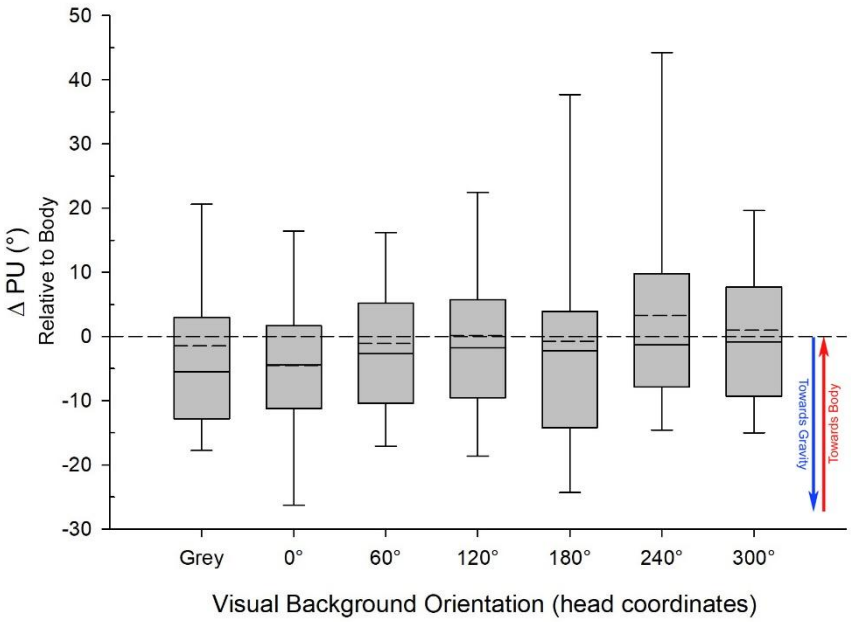


Figure 3.11: Change in Perceptual Upright after intense VR exposure for different visual background orientations lying right side down. Solid lines within the box plot are the median and dashed lines are the mean. Error bars are standard error.

### 3.4.2 Perceptual Upright and Cybersickness

Associations between PU change and cybersickness severity were assessed by correlating ΔPU with post-VR FMS for each background. Because ΔPU distributions violated normality across all backgrounds, rank-based correlations were emphasised. No background showed a statistically significant association between ΔPU and FMS using Spearman's rho or Kendall's tau (all $p \geq 0.176$). Although Pearson correlations reached nominal significance for the 60°, 120°, and 300° backgrounds, these effects did not replicate with rank-based methods and are therefore interpreted as sensitivity to extreme observations rather than robust associations.

The hypothesized Gaussian relationship between cybersickness and PU change was tested directly for the grey and 0° backgrounds (Figure *3.12*). For the grey background, the Gaussian model did not provide evidence of an association. For the 0° background, the Gaussian model performed worse than an intercept-only model (negative $R^2$), indicating that the Gaussian curve does not describe a reliable relationship in this dataset.

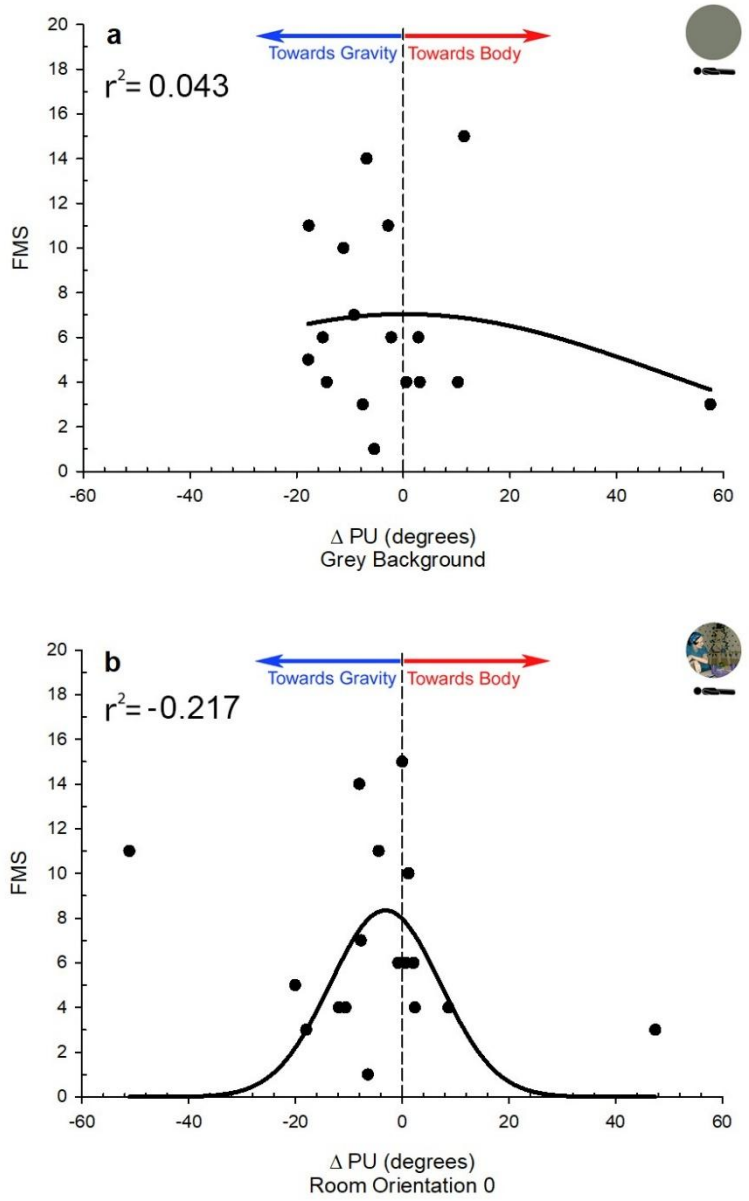


Figure 3.12: Gaussian model of FMS ratings relative to the right side down change in PU for a) the grey background and b) for 0° room orientation after intense VR exposure. The dashed line indicates the upper specification of the regression of no difference in the PU from baseline.

### 3.4.3 Cue Weights Before and After VR

The vector sum model (Eq. 3.4) was fitted to each participant's PU values across the 12 condition cells (6 visual backgrounds × 2 body orientations), separately for the pre-VR and post-VR sessions. Of the 20 participants, 18 produced reliable model fits ($R^2 \geq 0.50$) and did not show extreme body dominance (body weight ≤ 20), and were retained for cue weight analyses. Two participants were excluded due to poor model convergence, consistent with the extreme body dominance pattern described in Chapter 4.

#### 3.4.3.1 Descriptive statistics

At the group level, visual cue weighting decreased slightly following VR (Δ cue weight = −0.617%), whereas body cue weighting increased modestly (+0.419%) and gravity cue weighting increased marginally (+0.197%). Overall, these shifts were small in magnitude. Notably, the direction of change in body weighting was opposite to the hypothesised pattern, and contrasts with Chapter 2, where SVV-derived cue weights showed body down-weighting and gravity up-weighting following VR exposure.

| Cue | Pre VR Weight | Post VR Weight | ΔPU | Δ Cue weight (%) |
|---|---|---|---|---|
| **Vision** | 0.510 | 0.474 | -0.036 | -0.617 |
| **Body** | 1.268 | 1.323 | 0.055 | 0.419 |
| **Gravity** | 1 | 1 | 0 | 0.197 |
| **Vector Sum** | 2.778 | 2.797 | | |

Table 3.2: Mean vector sum model parameters and cue weight percentage changes for Experiment 2 participants (N = 18).

To improve statistical power and assess the robustness of cue weight changes, the 18 Experiment 2 participants were pooled with 18 participants from Experiment 3 who completed the no-EVS condition (see Section 3.3.1 for pooling justification). Table *3.3* summarizes the pooled cue weight results.

| Cue | Pre VR Weight | Post VR Weight | ΔPU | Δ Cue weight (%) |
|---|---|---|---|---|
| **Vision** | 0.552 | 0.503 | -0.049 | -0.218 |
| **Body** | 1.753 | 1.570 | -0.183 | -1.205 |
| **Gravity** | 1 | 1 | 0 | 1.423 |
| **Vector Sum** | 3.305 | 3.072 | | |

Table 3.3: Mean vector sum model parameters and cue weight percentage changes for the pooled sample (N = 36).

In the pooled sample, body cue weighting decreased following VR exposure (Δ cue weight = −1.21%), whereas gravity cue weighting increased (+1.423%). Vision cue weighting continued to decrease (−0.218%). Relative to the Experiment 2-only analysis, the pooled sample showed a larger increase in gravity weighting (Experiment 2: +0.197%) and a reversal in the direction of body weighting, from a modest increase in Experiment 2 to a modest decrease in the pooled sample. The overall vector sum also

decreased from 3.305 to 3.072, consistent with a reduction in the overall magnitude of cue weights following VR exposure. This pooled pattern of body down-weighting and gravity up-weighting aligns with the direction of cue weight changes observed in Chapter 2 using SVV-derived estimates.

The Experiment 3 no-EVS participants were, on average, more body-dominant at baseline (mean body weight = 2.24) than the Experiment 2 participants (mean body weight = 1.27), which contributes to the higher pooled baseline body weight (mean = 1.75). Descriptively, the no-EVS group also showed stronger body down-weighting following VR exposure. This pattern is consistent with a regression-to-the-mean account, or alternatively, with the interpretation that individuals with more extreme baseline weighting profiles have greater scope for adaptive adjustment under visuo-vestibular conflict.

#### 3.4.3.2 Repeated-measures ANOVA on cue weight percentages

To test whether VR exposure produced differential reweighting across sensory cues, a 2 (Condition: pre-VR, post-VR) × 3 (Cue Type: vision, body, gravity) repeated-measures ANOVA was conducted on cue weight percentages, with Greenhouse–Geisser corrected degrees of freedom reported where sphericity was violated.

For Experiment 2 ($n = 18$), there was a significant main effect of Cue Type, $F(1.42, 24.12) = 19.60$, $p < 0.001$, $\eta p^2 = 0.54$, confirming that the three cues contributed unequally to PU. However, the Condition × Cue Type interaction was not significant, $F(1.02, 17.42) = 0.05$, $p = 0.834$, $\eta p^2 = 0.003$, indicating that cue weight percentages did not change differentially following VR exposure at the group level. For the pooled sample ($n = 36$), the main effect of Cue Type remained large, $F(1.29, 45.27) = 59.31$, $p < 0.001$, $\eta p^2 = 0.63$. The Condition × Cue Type interaction was again not significant, $F(1.06, 37.03) = 0.51$, $p = 0.490$, $\eta p^2 = 0.014$.

Overall, the null Condition × Cue Type interaction in both analyses indicates that VR exposure did not produce statistically reliable differential reweighting at the group level. This null result should be interpreted in light of substantial inter-individual variability, as the standard deviations on Δ cue weight percentages (approximately 10% to 14%) were an order of magnitude larger than the corresponding mean changes (approximately 0.2% to 1.4%).

#### 3.4.3.3 Paired comparisons

In the Experiment 2 data alone, no individual cue weight percentage change reached significance: Δ vision, $t(17) = -1.22$, $p = 0.240$; Δ body, $t(17) = 0.09$, $p = 0.928$; Δ gravity, $t(17) = 0.17$, $p = 0.870$.

Wilcoxon signed-rank tests converged on the same conclusion (all $p \geq 0.18$). In the pooled sample, no cue reached conventional significance: Δ vision, $t(35) = -0.42$, $p = 0.675$; Δ body, $t(35) = -0.66$, $p = 0.514$; Δ gravity, $t(35) = 0.78$, $p = 0.441$. Wilcoxon signed-rank tests for body ($p = 0.108$) and gravity ($p = 0.115$) approached marginal levels, with directions consistent with the reweighting hypothesis.

#### 3.4.3.4 Cue weight changes and cybersickness

To test whether individual differences in cue reweighting predicted cybersickness severity, Δ cue weight percentages were correlated with post-VR FMS scores. Because several Δ cue weight distributions violated normality, both parametric and rank-based correlations are reported, with Spearman rank correlations emphasised as the primary test.

Δ vision cue weight percentage was significantly and positively associated with FMS in the pooled sample (Pearson $r = 0.35$, $p = 0.039$; Spearman $\rho = 0.37$, $p = 0.027$; Kendall $\tau = 0.27$, $p = 0.026$; Figure *3.13*). Participants who increased their visual cue weighting following VR exposure tended to report higher cybersickness. A linear model provided the best fit for this relationship ($R^2 = 0.120$, $F(1, 34) = 4.64$, $p = 0.039$). In contrast, neither Δ body cue weight percentage (Spearman $\rho = -0.10$, $p = 0.557$) nor Δ gravity cue weight percentage (Spearman $\rho = -0.03$, $p = 0.850$) were significantly associated with FMS. For both body and gravity, the intercept-only model provided the best fit by AIC, and neither linear, quadratic, nor Gaussian models significantly improved upon it (all F-test $p > 0.20$). Figure *3.13* presents the scatterplots for all three cue types.

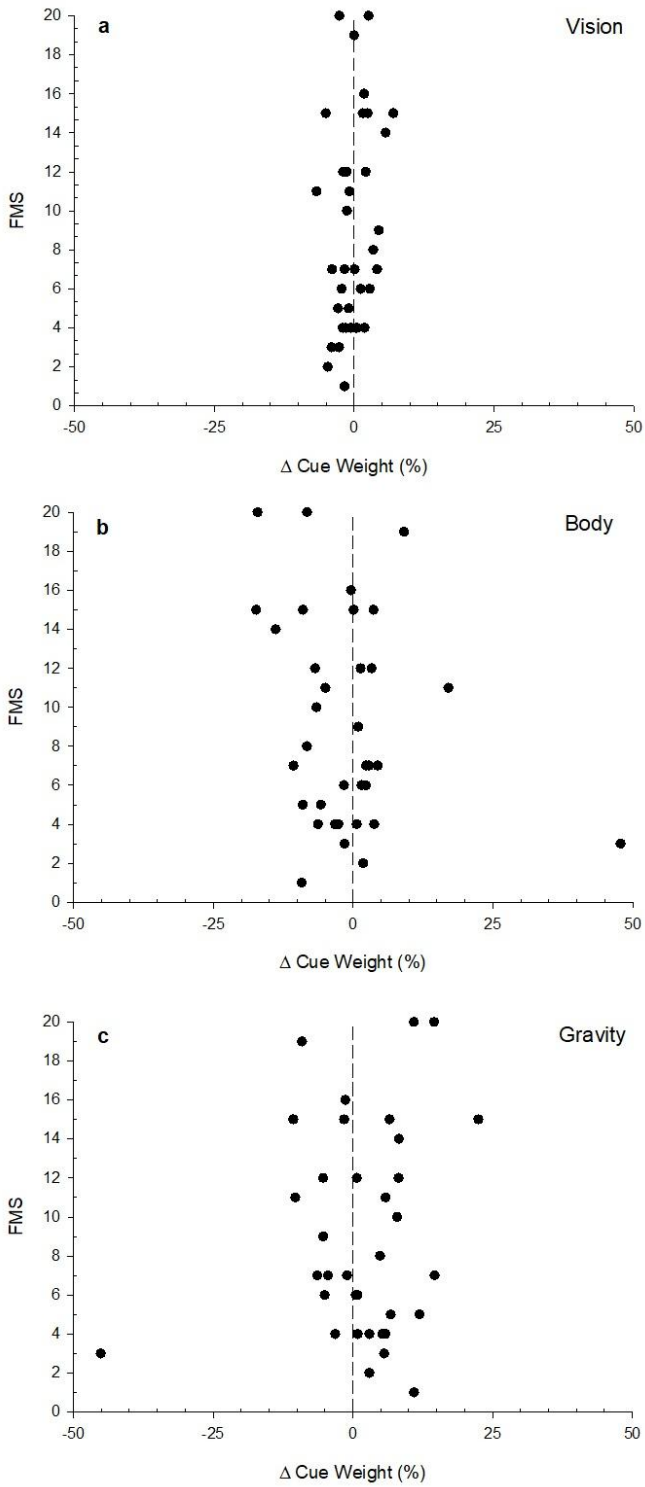


Figure 3.13: Scatter plots of FMS ratings relative to the change in cue weight per cents for a) vision and b) body, and c) gravity in the pooled sample (N = 36). The dashed line indicates the upper specification of the regression of no difference in the PU from baseline.

### 3.4.4 Cluster Analysis

Outlier screening in the pooled cue-weight change space identified one multivariate outlier (P12, Experiment 2). P12 showed extreme cue-weight changes (Δ vision = -2.74%, Δ body = 47.85%, Δ gravity = -45.11%), exceeding the univariate threshold (max |z| = 4.54) and meeting the Mahalanobis criterion ($D^2 = 20.13$, $p = 0.000160$). Because this observation disproportionately influenced unsupervised partitions, P12 was excluded from the primary clustering analysis (primary N = 35). Before clustering, Experiment 2 and Experiment 3 participants were compared to assess whether pooling introduced systematic differences. Kruskal-Wallis tests indicated no significant study differences in cue weight percentages (vision: $H = 1.047$, $p = 0.306$; body: $H = 0.039$, $p = 0.843$; gravity: $H = 0.279$, $p = 0.597$). FMS also did not differ significantly between studies ($H = 3.088$, $p = 0.0788$). These checks support treating the pooled dataset as a mixed sample for exploratory individual-differences analyses.

K-means clustering indicated that a two-cluster solution provided the best separation by silhouette score (k = 2, silhouette = 0.381). The resulting clusters were labelled Flexible Adapters (N = 16) and Rigid Perceivers (N = 19), and cluster membership was balanced across study origin (Flexible: Experiment 2 = 8, Experiment 3 = 8; Rigid: Experiment 2 = 9, Experiment 3 = 10). The Flexible Adapters profile was characterized by larger cue redistribution, particularly reduced body weighting and increased gravity weighting (Δ body: M = -8.63%, SD = 4.43; Δ gravity: M = 9.41%, SD = 4.71; Δ vision: M = -0.78%, SD = 2.97). In contrast, Rigid Perceivers showed smaller and, on average, opposite-signed redistribution for body and gravity (Δ body: M = 2.47%, SD = 4.49; Δ gravity: M = -2.86%, SD = 4.26; Δ vision: M = 0.39%, SD = 3.23).

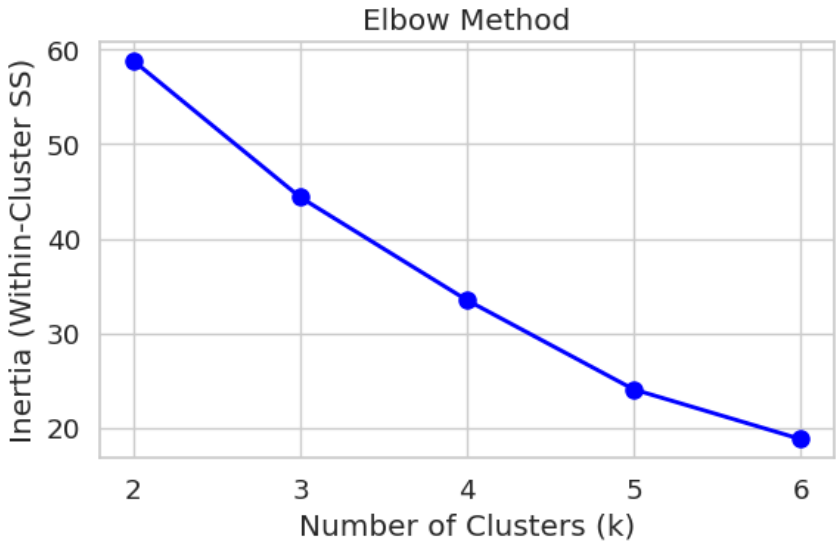

Figure 3.14: Elbow method for selecting the number of clusters (k) in cue-weight reweighting profiles.

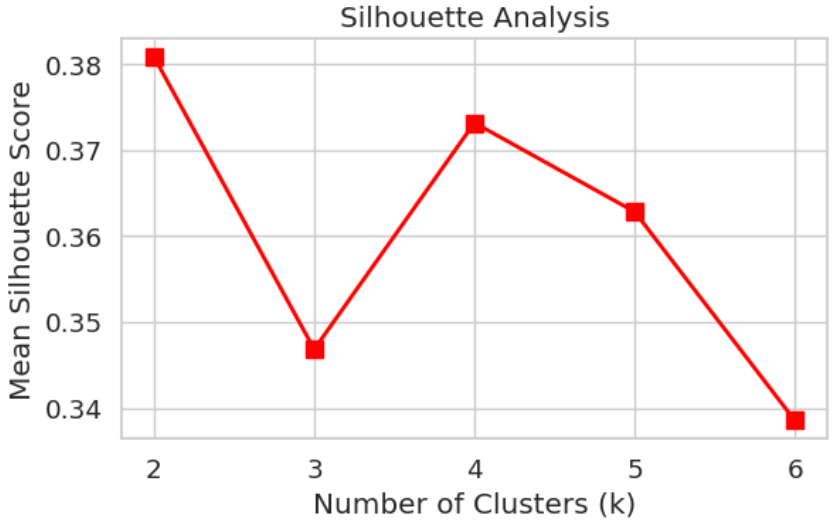


Figure 3.15: Silhouette analysis for selecting the number of clusters (k) in cue-weight reweighting profiles.

Despite these differences in cue redistribution, cybersickness severity did not differ reliably between clusters. Flexible Adapters reported FMS scores of M = 9.62 (SD = 5.97; Mdn = 9.0; range = 1 to 20), while Rigid Perceivers reported M = 8.74 (SD = 4.89; Mdn = 7.0; range = 2 to 19). Both cluster distributions were consistent with normality (Shapiro-Wilk $p \geq 0.10$), and variances did not differ (Levene $p = 0.300$). A Mann-Whitney U test found no between-cluster difference ($U = 161.0$, $p = 0.777$; rank-biserial $r = -0.059$), and Welch's t-test converged on the same conclusion ($t = 0.476$, $p = 0.638$; Cohen's $d = 0.163$).

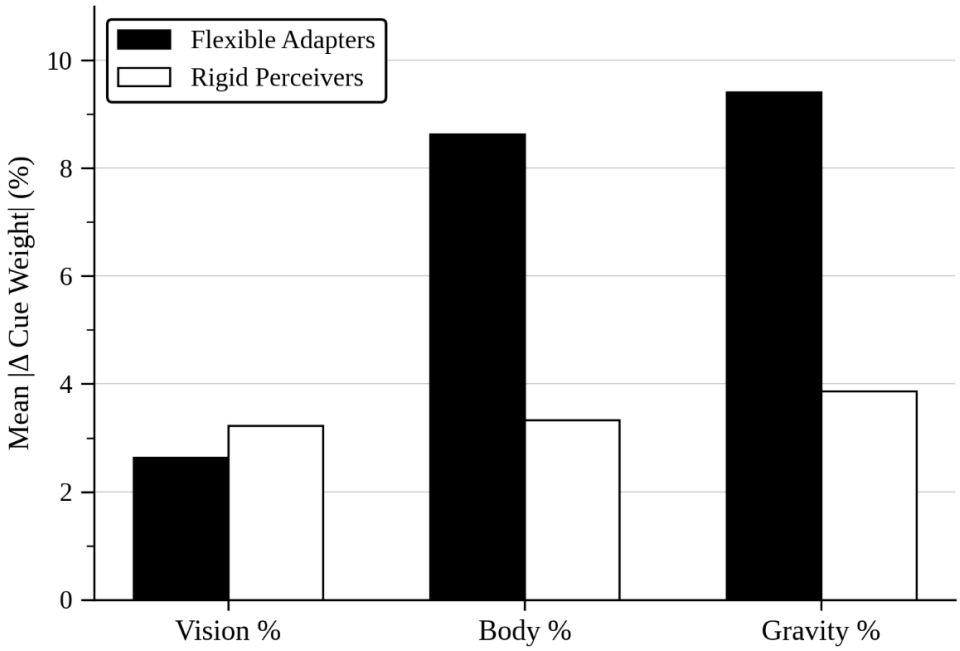


Figure 3.16: Mean Δ cue weight profiles for Flexible Adapters and Rigid Perceivers. This figure shows the absolute reweighting magnitude.

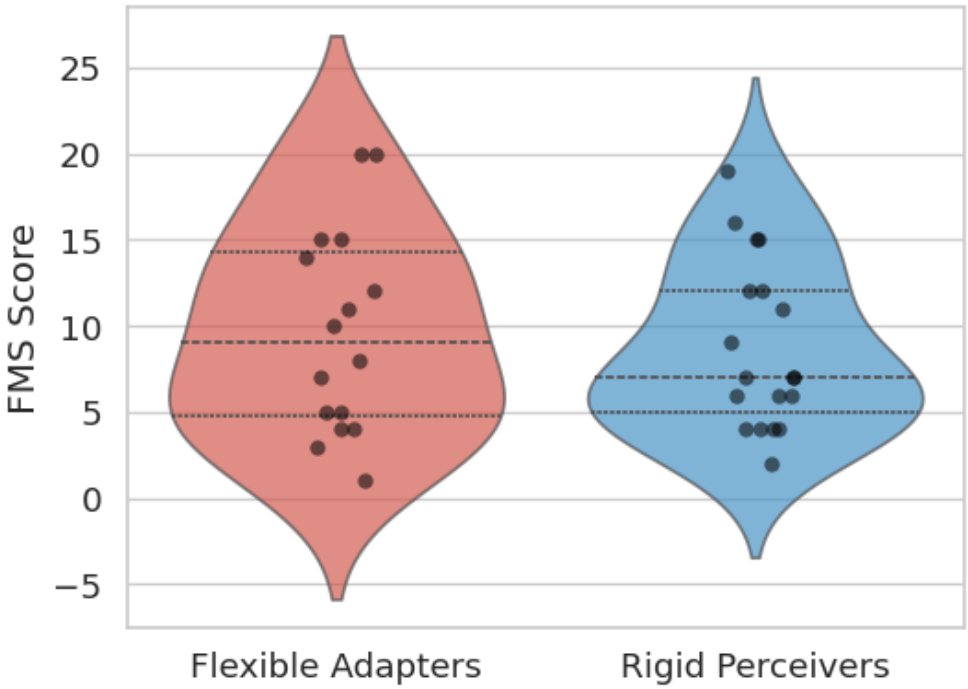


Figure 3.17: FMS score distributions by cluster shown as violin plots with individual data points overlaid. Dashed lines within each violin indicate quartiles. The Mann-Whitney test indicated no significant between-cluster difference (U = 161.0, p = 0.777).

## 3.5 Discussion

The present study expands upon prior work (Chapter 2) by employing OCHART to measure the PU as a function of multisensory cue integration. OCHART is particularly well-suited for examining cue conflict in immersive environments due to its high sensitivity to shifts in PU across multiple sensory frames. Unlike tasks such as the rod-and-frame or SVV, which primarily emphasize gravity-vision conflicts, the OCHART enables simultaneous manipulation of visual, gravitational, and body-based cues with fine angular resolution (Dyde et al., 2006). The task's design permits modelling perceptual outcomes through vector summation, allowing researchers to estimate the relative weights of each sensory input on a per-participant basis. This makes OCHART a powerful tool not only for identifying group-level changes in sensory integration but also for characterizing individual differences in multisensory flexibility, which have been linked to cybersickness resilience (Chung & Barnett-Cowan, 2023).

Furthermore, OCHART is advantageous in pre-post intervention designs, such as VR exposure, because it permits quantification of reweighting without relying solely on subjective reports. The integration of body orientation trials (e.g., upright vs. right-side down) also enables disambiguation between body and gravitational contributions (Jenkin et al., 2004). Given these strengths, the use of OCHART in this study allows for a more comprehensive understanding of how sensory cue weights adapt following neuromodulatory or immersive stimuli and how these adaptations relate to subjective states like cybersickness. The present findings show that PU is strongly modulated by visual background orientation, but that intense VR exposure did not produce a systematic group-level shift in PU in the right-side-down condition.

### 3.5.1 Interpretation of PU Shifts and VR Exposure Effects

In the grey background, PU in the right-side-down posture was consistently biased toward gravity both pre and post VR, but the Pre-to-Post difference was small and not statistically reliable. This indicates that, under these conditions, any VR-related changes in PU were not expressed as a consistent group-level shift, even though SVV showed clearer evidence of a gravity-ward shift in Chapter 2. This finding echoes the SVV-based trends reported previous work and supports the broader hypothesis that intense VR can induce a reorganization of sensory weights, potentially as an adaptive response to visuo-vestibular conflict (Chung & Barnett-Cowan, 2023). Notably, although the shift toward gravity in the RSD condition post VR was more pronounced than pre-VR, it did not reach statistical significance, highlighting the need for additional statistical modelling and within-subject comparisons across cue conditions.

Effect size analyses indicated that the orientation of the visual background explained a large proportion of variance in PU ($\eta^2_p = 0.54$), whereas the effects of VR exposure and its interaction with background orientation were small ($\eta^2_p = 0.02$ for each). This pattern underscores the dominant role of visual context in shaping perceptual upright, while indicating that VR exposure did not reliably shift PU at the group level in this dataset. However, it is likely that group-level effects are obscured by substantial inter-individual variability in sensory reweighting following VR exposure.

In conditions with visual backgrounds, PU was strongly influenced by the orientation of the visual environment. These results replicate and extend findings from Dyde et al. (2006), reinforcing that visual context has an influence on PU in scenarios where gravitational cues are inconsistent or ambiguous. Although the general direction of PU shifts aligns with visual cue orientation, the degree of shift appears to vary between individuals, hinting at significant inter-individual differences in visual dependency, a factor known to correlate with cybersickness susceptibility (Josupeit, 2024; McManus & Harris, 2023).

### 3.5.2 Perceptual Upright and Cybersickness

The relationship between changes in PU and cybersickness ratings was explored by plotting the shift in PU against FMS scores following intense VR exposure. As shown in Figure *3.12*a, for the grey background, there was no reliable association between the magnitude or direction of PU shifts and FMS ratings when the relationship was tested directly. Rank-based correlations were non-significant, and formal tests of the hypothesized Gaussian relationship did not support a non-linear association. This indicates that, in the absence of strong visual orientation cues, changes in PU alone were not predictive of cybersickness severity.

In contrast, when the room background was at 0° relative to the head (Figure *3.12*b), the scatterplot can appear suggestive of curvature, but this pattern was not supported by inferential tests. Rank-based correlations again did not identify a reliable association between ΔPU and FMS, and the Gaussian model performed worse than an intercept-only model for the 0° background. Although the scatterplots suggest curvature in some background conditions, these PU–FMS relationships should be treated as exploratory. Given the non-normal distribution of PU change scores, rank-based associations provide a more appropriate summary of monotonic relationships, and any apparent Gaussian pattern should be interpreted cautiously until supported by formal nonlinear model comparisons and cross-validated stability. At the current sample size, PU change alone explains only a modest and inconsistent proportion of variance in cybersickness ratings, suggesting that PU shifts are, at best, an incomplete marker of the mechanisms

linking sensory conflict to symptom severity. This underscores the need to look beyond PU shifts alone and instead consider model-derived sensory weights as a more direct index of multisensory reweighting.

These findings align with current Bayesian Causal Inference models of multisensory integration, where the brain dynamically weighs the likelihood that signals originate from a common source (Fetsch et al., 2013). Under conditions of intense visual-vestibular conflict present in VR, the central nervous system may favour cue segregation over integration, leading to individual differences in adaptive strategies and potential transitions between "forced fusion" and causal inference modes (de Winkel et al., 2018). This theoretical perspective may account for the observed variability in how participants shift their perceptual upright and their susceptibility to cybersickness.

### 3.5.3 Vector Sum Model and Cue Weight Estimation

The vector sum model quantified the relative contributions of vision, body, and gravity cues to PU before and after VR exposure. In the Experiment 2-only analysis (N = 18), post-VR cue weight changes were small, and the direction of the body weight change was unexpected. Specifically, body weighting increased by 0.419%, while gravity weighting increased by only 0.197%. This pattern does not align with the body down-weighting and gravity up-weighting observed using SVV in Chapter 2, nor with prior work showing that SVV can exhibit systematic changes following VR exposure that are interpretable as sensory reweighting (Chung & Barnett-Cowan, 2023). One plausible explanation is limited precision in the Experiment 2-only estimates. With N = 18, the analysis may have been underpowered to detect small but consistent shifts, such that sampling variability could plausibly account for the direction inconsistency.

The pooled analysis (N = 36) is consistent with this interpretation. When Experiment 3 no-EVS participants were included, the direction of body cue weighting reversed (from 0.419% to -0.121%), and gravity up-weighting increased substantially (from 0.197% to 1.423%). This pooled pattern, in which body cues are down-weighted and gravity cues are up-weighted following VR exposure, is convergent with Chapter 2's SVV findings and with the reweighting direction reported by Chung and Barnett-Cowan (2023). Taken together, the convergence across two perceptual measures (SVV and OCHART) and a larger combined sample supports the inference that VR exposure is associated with a shift away from body-referenced orientation cues and towards gravitational cues, at least under the current stimulus conditions and modelling assumptions.

The Experiment 3 no-EVS participants contributed disproportionately to this pooled pattern because they were more body-dominant at baseline (mean body weight = 2.24 vs. 1.27 in Experiment 2). Descriptively,

they also showed stronger post-VR body down-weighting. This observation is consistent with reliability-based accounts of multisensory cue integration, in which cue weights depend on relative reliability and can adjust when cue reliability changes (Ernst & Banks, 2002; Fetsch et al., 2012). Under this framework, individuals with higher initial reliance on a given cue could show larger absolute shifts if that cue becomes less reliable in the relevant context. However, regression to the mean cannot be excluded, and future work should explicitly test whether baseline body dominance prospectively predicts the magnitude of post-VR reweighting.

Notably, the OCHART-derived baseline cue weight distribution differed from the SVV-derived distribution. In the pooled OCHART sample, body cues accounted for approximately 53% of total weighting at baseline, compared to approximately 15% in the Chapter 2 SVV data. This contrast is consistent with prior work showing that SVV is typically gravity-dominant, whereas PU assessed with OCHART yields a larger contribution from body cues and a more distributed influence across cues (Dyde et al., 2006; Barnett-Cowan et al., 2013). Practically, this implies that OCHART may provide greater dynamic range for detecting changes in body and visual weighting, whereas SVV may be more sensitive to changes that primarily involve the balance between body and gravity. Considered together, the complementary sensitivities of these measures strengthen the reweighting interpretation when both point in the same direction, as observed in the pooled analysis.

The strongest individual differences in the pooled analysis was a positive association between Δ vision cue weight and FMS (Spearman $\rho = 0.37$, $p = 0.027$). Participants who increased their visual weighting following VR reported higher cybersickness. Model comparison confirmed that this relationship was best described by a linear model ($R^2 = 0.120$, $F(1, 34) = 4.64$, $p = 0.039$) rather than a Gaussian function; the Gaussian did not converge to a bell-shaped curve and instead approximated a monotonic increase, yielding no improvement over the linear fit. This is notable because Chapter 2 hypothesized a Gaussian (inverted-U) relationship between cue weight change and cybersickness, but the pooled data did not support a curvilinear relationship for vision. In the pooled analysis, the linear pattern is consistent with a directional effect: increasing reliance on visual cues in the face of unreliable VR visual input represents a maladaptive weighting strategy that exacerbates the visuo-vestibular conflict rather than resolving it.

To evaluate whether this association was robust to sample composition, study group (Experiment 2 vs Experiment 3 no-EVS) was included as a covariate in a linear regression predicting FMS from Δ vision cue weight. The overall model did not reach significance ($R^2 = 0.132$, $F(2, 33) = 2.52$, $p = 0.096$), and the Δ vision term was not significant after controlling for study group ($\beta = 0.389$, $t(33) = 1.37$, $p = 0.182$). Study group also did not reach significance ($\beta = 2.592$, $t(33) = 1.48$, $p = 0.148$). A within-group centred

analysis produced a similarly non-significant association (Spearman $\rho = 0.222$, $p = 0.193$). When examined separately, the vision–FMS association was numerically present in Experiment 2 (Spearman $\rho = 0.461$, $p = 0.054$, $N = 18$) but absent in Experiment 3 no-EVS (Spearman $\rho = 0.024$, $p = 0.925$, $N = 18$), suggesting that the pooled effect was driven primarily by Experiment 2. Accordingly, the vision–FMS relationship is treated as exploratory and should be replicated in a larger and more uniform sample before firm conclusions are drawn.

This finding complements the Chapter 2 observation that higher FMS was associated with minimal change in SVV for the grey background condition. Both results converge on the conclusion that inflexible or maladaptive reweighting is associated with greater sickness. The key difference is that the OCHART-derived vision cue weight isolates the visual contribution specifically, whereas SVV conflates visual and gravitational influences. The specificity of the vision-FMS association reported here, combined with the null associations for body and gravity, suggests that it is the visual component of the reweighting response that most directly predicts cybersickness outcomes. This is consistent with sensory conflict theory (Oman, 1990; Reason, 1978), which posits that cybersickness arises from mismatch between visual and vestibular signals: an individual who responds to this mismatch by further increasing visual reliance would amplify rather than reduce the conflict signal.

The absence of significant associations for body and gravity cue weight changes does not mean these cues are irrelevant to cybersickness. The large variability in $\Delta$ body and $\Delta$ gravity percentages (SDs of 14.3 and 11.5, respectively) relative to the range of FMS scores may have obscured any underlying relationship. Additionally, the compositional constraint of the vector sum model, whereby vision, body, and gravity percentages must sum to 100%, means that changes in one cue necessarily affect the others, making it difficult to isolate independent contributions.

### 3.5.4 Individual Differences and Cluster Analysis

The cluster analysis revealed two separable post-VR reweighting profiles in the pooled sample. Flexible Adapters showed a pronounced shift away from body-based cues and toward gravity, while Rigid Perceivers showed the opposite pattern of smaller, gravity-downweighting changes. These groupings were stable across methods (83% k-means/hierarchical agreement) and balanced across study of origin, confirming that the solution was not an artifact of pooling two experiments. The Flexible Adapter pattern (body down-weighting and gravity up-weighting) aligns with the adaptive response to visuo-vestibular conflict described in Chapter 2, and with prior work linking sensory flexibility to reduced cybersickness susceptibility (Chung & Barnett-Cowan, 2023).

However, the cluster solution did not predict cybersickness severity. FMS scores were nearly identical between groups and the effect size was negligible (Cohen's d = 0.163). This null result stands in contrast to the significant positive association between Δ vision cue weight and FMS (reported in Section 3.4.3.4) and suggests that the direction of body and gravity redistribution was not the relevant factor for symptom severity in this dataset. Rather, it was the degree to which visual weighting increased (irrespective of the body/gravity pattern) that showed the more meaningful relationship with FMS. Both Flexible Adapters and Rigid Perceivers spanned the full range of FMS scores, which may explain why cluster membership did not stratify sickness outcomes.

While two statistically separable groups exist in the cue weight space, the practical separation between them is limited, and some participants in opposing clusters had overlapping profiles. This cluster analysis should therefore be interpreted as a characterization of the distribution of individual reweighting strategies in this pooled sample, rather than as evidence of discrete perceptual subtypes. Future work with larger samples could test whether these profiles are stable across sessions, predictive of symptom severity under stronger conflict, or modified by the vestibular neuromodulation examined in Chapter 4.

### 3.5.5 Limitations and Future Directions

The clearest effects in this chapter were driven by the task manipulations (especially visual background), whereas VR-related pre to post changes were small and occurred alongside large inter-individual variability. This is not surprising in VR sickness work, where symptom severity and adaptation are heterogeneous even in relatively controlled HMD studies (Saredakis et al., 2020). In that context, null group-level effects (for example, the non-significant Condition × Cue Type interaction) should be treated as evidence that there was no reliable mean reweighting signal, not evidence that reweighting did not occur for a subset of participants.

The main individual-differences result, the positive association between Δ vision cue weight and post-VR FMS, is statistically reliable but modest in explanatory strength ($R^2 \approx 0.12$). Practically, this is better framed as a mechanistically interpretable vulnerability marker than as a stand-alone predictor of cybersickness, since most variance in sickness remains unexplained by any single behavioural index (Saredakis et al., 2020).

FMS is efficient and well validated for capturing sickness quickly (Keshavarz & Hecht, 2011), but a single post-exposure rating cannot distinguish early onset from late escalation or quantify symptom growth over time. Future work should include brief repeated FMS probes across exposure blocks, and

where feasible, a multidimensional questionnaire to separate nausea, oculomotor strain, and disorientation components that may relate differently to reweighting.

A further consideration is that the present design captures only a single VR exposure per participant. Recent work on Cybersickness Abatement from Repeated Exposure (CARE) demonstrates that cybersickness can be substantially reduced across four to five sessions, with benefits that generalise to novel VR experiences not encountered during training (Kelly et al., 2026). Critically, Doty and colleagues (2025) showed that CARE can be achieved while minimising discomfort during the training phase itself, using field-of-view restriction during early exposures that is removed once tolerance develops. This suggests that the reweighting processes measured here may represent only the initial response to conflict, whereas the more functionally relevant adaptation may unfold over repeated sessions. If so, the modest $R^2$ values observed for the vision-FMS association may partly reflect the fact that single-session reweighting is only one component of a broader multi-session adaptation trajectory. A practical implication is that individual differences in first-session reweighting capacity may predict not just acute susceptibility, but also the rate of adaptation across sessions, a hypothesis that future repeated-exposure designs could test directly. Importantly, Palmisano and Constable (2022) found that sickness reduction over two days was specific to the same game and did not generalise, suggesting that the number of training sessions and the protocol design (e.g., FOV restriction) may be critical factors determining whether CARE transfers across contexts.

There are additional limits imposed by the modelling framework. Cue weights expressed as percentages are compositional, and changes in one cue necessarily imply compensatory changes in the others, which complicates interpretation of "unique" cue contributions. More importantly, the vector sum model assumes obligatory integration with fixed weights within a condition. Under strong conflict, behaviour may be better characterised as switching between integration and segregation depending on inferred common cause (Körding et al., 2007). If some participants intermittently segregated cues across trials or backgrounds, fixed-weight estimates would be expected to become less stable and relationships with cybersickness would be attenuated. Finally, OCHART is intentionally sensitive to visual orientation context, which is a strength for quantifying visual contributions, but it can also make small post-VR shifts harder to detect when visual background effects dominate the upright estimate (Dyde et al., 2006). This likely contributes to the pattern observed here: robust Background effects, but weak and inconsistent pre to post effects at the group level.

These points motivate a straightforward next step. Chapter 3 identifies a plausible maladaptive pattern (visual up-weighting tracking higher sickness), but it cannot test causality because VR exposure is a

complex perturbation rather than a targeted reliability manipulation. This is the rationale for Chapter 4, which moves from correlational individual differences to a within-subject vestibular manipulation using stochastic EVS, motivated by the idea that altering vestibular reliability during VR can shift cue weighting and, in some contexts, change cybersickness outcomes (Weech et al., 2020). Chapter 4 therefore tests this hypothesis directly by manipulating vestibular reliability during VR, while recognizing that the effect of vestibular noise may be parameter-sensitive and could either reduce or exacerbate cybersickness depending on how it interacts with the ongoing sensory conflict.

## 3.6 Conclusion

In summary, OCHART proved to be a sensitive tool for probing visual influences on perceptual upright, with PU varying strongly as a function of visual background orientation. This confirms that the task robustly captures visual contextual effects on upright perception and is well suited for quantifying how vision, body, and gravity cues combine under conflict. However, VR exposure did not produce a reliable group-level shift in PU, nor did cue weight percentages show significant differential reweighting in the Experiment 2 sample alone. This null group-level result likely reflects the combination of high inter-individual variability in reweighting direction and the relatively modest sample size, rather than evidence that reweighting did not occur for any participants.

The pooled analysis (N = 36) told a more coherent story. The direction of cue weight change converged with Chapter 2, with body cues down-weighted and gravity cues up-weighted following VR. This convergence across SVV and OCHART, two measures with fundamentally different baseline cue weight distributions, provides the strongest evidence in this dissertation that VR-induced conflict triggers adaptive cue redistribution. Importantly, OCHART's greater sensitivity to visual cues and more evenly distributed baseline weight profile complements the gravity-dominated SVV, meaning the two tasks together constrain the reweighting interpretation more tightly than either could alone.

The clearest individual-differences finding was that increased visual cue weighting after VR was associated with higher cybersickness, suggesting that maladaptive visual up-weighting is a more informative marker of susceptibility than the direction of body or gravity redistribution. The exploratory cluster analysis reinforced this conclusion: although two separable reweighting profiles emerged (Flexible Adapters and Rigid Perceivers), cluster membership did not predict sickness severity, indicating that how much visual weighting increased mattered more than which body-gravity strategy a participant adopted. These results motivate a shift from group-level perceptual shifts to individual-level cue weight dynamics as the primary unit of analysis for understanding cybersickness vulnerability.

A key limitation of both Chapters 2 and 3 is that VR exposure is a complex perturbation that simultaneously alters visual, vestibular, and proprioceptive statistics in ways that cannot be independently controlled. The cue weight changes observed here emerge from the full sensory disruption of VR, making it difficult to attribute reweighting patterns to any single modality. To isolate the contribution of vestibular reliability specifically, what is needed is a manipulation that targets one cue's precision independently of the VR content itself. Chapter 4 addresses this by applying stochastic electrical vestibular stimulation during VR to selectively degrade vestibular input, testing whether altering the reliability of a single modality produces the kind of adaptive reweighting that Chapters 2 and 3 suggest is protective against cybersickness.

# Chapter 4

# Investigating the Effects of Electrical Vestibular Stimulation on Multisensory Integration and Cybersickness in Virtual Reality

## 4.1 Abstract

This chapter examined whether stochastic electrical vestibular stimulation (EVS) alters multisensory integration and reduces cybersickness during immersive virtual reality (VR). Twenty-eight participants completed two VR sessions in a counterbalanced within-subject design, one with bilateral stochastic EVS and one without stimulation (no-EVS), while immersed in an intense zero-gravity VR environment. Perceptual upright (PU) was measured before and after each session using the Oriented CHAracter Recognition Test (OCHART) in upright and right-side-down postures. Cybersickness severity was assessed with the Fast Motion Sickness Scale (FMS), and task duration was used as a behavioural tolerance measure. Contrary to the primary hypothesis, EVS significantly increased cybersickness severity and reduced VR tolerance. Participants reported higher FMS scores in the EVS condition than in the no-EVS condition, and more participants terminated the task early under EVS. Despite this behavioural effect, EVS did not produce a reliable group-level change in perceptual upright across visual backgrounds, nor did it produce a significant overall change in model-derived cue weighting in the subset of participants with stable vector sum model fits. Model fitting was unstable for a substantial subset of the sample, particularly when PU showed minimal sensitivity to body tilt. These findings suggest that, under the present stimulation parameters, stochastic EVS acted as an additional perturbation rather than a countermeasure, and they support a parameter-sensitive interpretation in which vestibular noise can either attenuate or exacerbate cybersickness depending on the specific stimulation and VR context.

## 4.2 Introduction

Virtual reality (VR) head-mounted displays now deliver photorealistic scenes and six-degree-of-freedom motion, but they still provoke cybersickness in 20-60% of users, especially when visual self-motion cues conflict with vestibular and proprioceptive inputs (Reason & Brand, 1975; Weech et al., 2019). The consensus is that large visuo-vestibular discrepancies drive autonomic reactions such as nausea, dizziness, and oculomotor strain that can limit VR exposure times and compromise research or training outcomes (Stanney, Fidopiastis, et al., 2020). One promising countermeasure is to directly modulate vestibular afferents so that their moment-to-moment output becomes less reliable, encouraging the brain to down-weight vestibular cues and thereby reduce the perceived conflict (Weech et al., 2020). Electrical Vestibular Stimulation (EVS) offers such a route: a safe, non-invasive technique in which small current (< 4 mA peak-to-peak) is delivered between electrodes placed over the left and right mastoid processes, producing precise modulation of the firing rates of vestibular nerve afferents (Fitzpatrick & Day, 2004).

Recent research highlights that individual susceptibility to cybersickness varies considerably and can impact the effectiveness of vestibular interventions in VR. Factors such as a person's prior motion sickness history, sex, and gaming experience are now recognised as significant predictors of both cybersickness severity and adaptation capacity (Chang et al., 2021; Kourtesis et al., 2024). Considering these sources of heterogeneity is essential when evaluating vestibular modulation techniques like EVS, as individual differences in sensory integration may shape outcomes.

### 4.2.1 Vestibular Stimulation

Stochastic or “noisy” EVS is typically implemented as a zero-mean Gaussian current (0.3–25 Hz) with an rms amplitude of 0.5–1 mA delivered bilaterally (Mulavara et al., 2011, 2015). EVS directly alters the firing of vestibular afferents by polarizing the vestibular nerve: cathodal current increases firing probability, whereas anodal current decreases it (Fitzpatrick & Day, 2004). Importantly, EVS is peripheral, meaning that it modulates the otolith-canal nerve bundle before central processing, allowing experimenters to probe or perturb vestibular contributions without introducing mechanical head motion. Classical studies used Galvanic Vestibular Stimulation (GVS), quasi-DC, or low-frequency step currents to evoke coherent sensations of roll and corresponding sway reflexes (Wardman et al., 2003), but such galvanic patterns can themselves be nauseogenic and thus are poorly suited for cybersickness mitigation.

Previous work confirms that stimulation can perturb the vestibular network. Bilateral mastoid stimulation activates the parieto-insular vestibular cortex (PIVC), cerebellar nodulus, and hippocampal formation, all

regions implicated in spatial orientation and motion perception (Indovina et al., 2020; Utz et al., 2010). Functional MRI with alternating-current EVS (0.1–5 Hz sinusoidal, ±2.5 mA) demonstrates frequency-dependent activation of the posterolateral thalamus, posterior insula/operculum, supramarginal gyrus at the temporo-parietal junction, supplementary motor area, and cerebellar vermis, together with pronounced deactivation of primary and extrastriate visual cortex (Stephan et al., 2005). These results delineate a thalamo-insular vestibular pathway and underscore the reciprocal suppression of visual areas during strong vestibular drive. Together these findings suggest that noisy EVS can be leveraged as a selective reliability cue by degrading vestibular precision during VR, the central nervous system may place more emphasis on vision and proprioception, thereby diminishing cue conflict and the resultant discomfort.

### 4.2.2 EVS Application in VR Research

Empirical support for the application of EVS and GVS in VR is growing. In the key noisy-GVS cybersickness paper, Weech et al. (2020) applied bilateral noisy GVS at ±1750 μA during VR and observed reduced cybersickness on a verbal sickness scale during intensely nauseogenic content, but not during moderate content, and the effect did not generalize to questionnaire-based measures and extinguished within approximately 3-6 minutes of further VR exposure. Groth and colleagues (2022) applied omnidirectional GVS to simulate realistic vestibular cues during omnidirectional locomotion in VR, finding reduced cybersickness and greater spatial coherence in user navigation. These improvements suggest that adding artificial vestibular signals can either disrupt or enhance the brain's natural balance signals to help visual and motion cues feel more consistent (Groth et al., 2022).

In parallel, Imaoka et al. (2022) reported that EVS in aging populations improved balance control and attentional focus under immersive VR conditions, supporting a broader vestibulo-cognitive interaction hypothesis (Imaoka et al., 2022). Meanwhile, Gim et al. (2025) and Sra et al. (2019) have begun integrating GVS with haptic and contextual motion cues in vehicular VR, demonstrating that even brief pulses of EVS can restore spatial updating and reduce vection lag in confined virtual environments (Gim et al., 2025; Sra et al., 2019). Critically, several studies also show that GVS can exacerbate symptoms or slow adaptation when the induced vestibular signal increases, rather than reduces, the effective sensory conflict. Using model-designed GVS waveforms, Allred et al. (2025) demonstrated that they can produce opposite sickness outcomes, including substantial increases in motion sickness severity under detrimental waveforms, highlighting strong parameter and waveform dependence. In an immersive 'virtual ship motion' paradigm, Qi et al. (2021) similarly reported that GVS aggravated balance disturbance and delayed cybersickness habituation.

Two mechanisms have been proposed to explain these benefits. The first mechanism is conflict attenuation, whereby the reduction in vestibular signal reliability decreases the discrepancy between visual and inertial cues, and the second mechanism is sensory reweighting, in which the central nervous system adaptively downweighs unreliable vestibular input based on Bayesian inference (Fetsch et al., 2013). Both mechanisms can be quantified behaviourally via orientation tasks such as the Subjective Visual Vertical (SVV) or the Oriented CHAracter Recognition Test (OCHART), which index the relative weighting of gravity, body, and visual cues (Chung & Barnett-Cowan, 2023; Dyde et al., 2006). Collectively, these results indicate that EVS, when carefully tuned, serves not merely to mask vestibular input, but to sculpt sensory integration in a manner that promotes realism, immersion, and comfort in virtual environments.

The main objectives of this chapter are to: (1) determine whether EVS alters the weighting of sensory cues underlying perceived upright in VR; (2) assess whether these changes predict cybersickness outcomes; and (3) quantify inter-individual variability in adaptation to EVS during immersive VR exposure. This chapter extends the OCHART approach from Chapter 3 to investigate how degrading vestibular reliability with EVS influences multisensory orientation perception and cybersickness in challenging VR. Specifically, we hypothesize that applying stochastic EVS during intense VR will reduce the effective weighting of vestibular cues, resulting in a PU that is more strongly influenced by visual information, as captured by OCHART. Consistent with our previous findings, we expect that individuals who show greater flexibility in shifting their PU in response to sensory conflict, indicating better adaptive reweighting, will report lower levels of cybersickness following VR with EVS. We predict that cue weights calculated using the vector sum model will shift such that gravitational cues will be down weighted in response to the unreliable input induced by EVS, resulting in a compensatory upweighting of bodily and visual cues.

## 4.3 Methods

### 4.3.1 Participants

Data were collected from 28 participants (16 males and 12 females; mean age = 21.6, SD = ±2.59) recruited from the University of Waterloo. All participants reported having no auditory or vestibular disorders and had normal or corrected-to-normal vision. This experiment was reviewed and approved by the University of Waterloo Research Ethics Committee, and all participants provided informed written consent in accordance with the committee's guidelines.

#### 4.3.1.1 A-priori Power Analysis

An a priori power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2007) to estimate the required sample size for detecting within-factor effects in a repeated-measures ANOVA. For this calculation, I used the effect size observed in the previous study (Chapter 2), which employed a similar two-way repeated-measures design. Specifically, the post-hoc analysis of Chapter 2 yielded a Cohen's f of 0.28. The analysis assumed this effect size ($f = 0.28$), an alpha of 0.05, desired power of 0.80, and a nonsphericity correction $\varepsilon = 1.0$. These parameters indicated a minimum total sample size of $N = 28$.

### 4.3.2 Apparatus

The VR content for this experiment was delivered using the Oculus Quest system. The Oculus Quest has a dual OLED display with a resolution of 1600x1440 (per eye) at 72Hz refresh rate. It is a completely wireless system featuring 6 degrees-of-freedom (6DOF) inside out tracking allowing the user to freely move around. Here we allowed participants to move within a 2.4x2.4m designated play area.

One commercially available VR game was selected for the VR content, ADR1FT (Figure *4.1*). This game was chosen based on user reviews on the Oculus Quest store, as it was rated as "intense." The game places the participants inside a zero-gravity suit, where head turns update the view naturally, but all translation (thrust, roll, pitch, yaw) is driven by an Xbox controller. Thus, rich 6DOF optic flow is generated without matching vestibular input, creating the "intense" visuo-vestibular conflict sought for this study. Controls were demonstrated and practiced briefly before the trial.

Figure 4.1: Screen capture of neurogenic VR environment of the game used for the VR exposure, ADR1FT.

### 4.3.3 Stimulus

To assess participants' PU, the OCHART was employed. In this task, participants discriminate between the letters 'p' and 'd,' which, when rotated 180°, transform into each other. By identifying the angles at which the perceived letter transitions from 'p' to 'd' and vice versa, the PU can be indirectly determined.

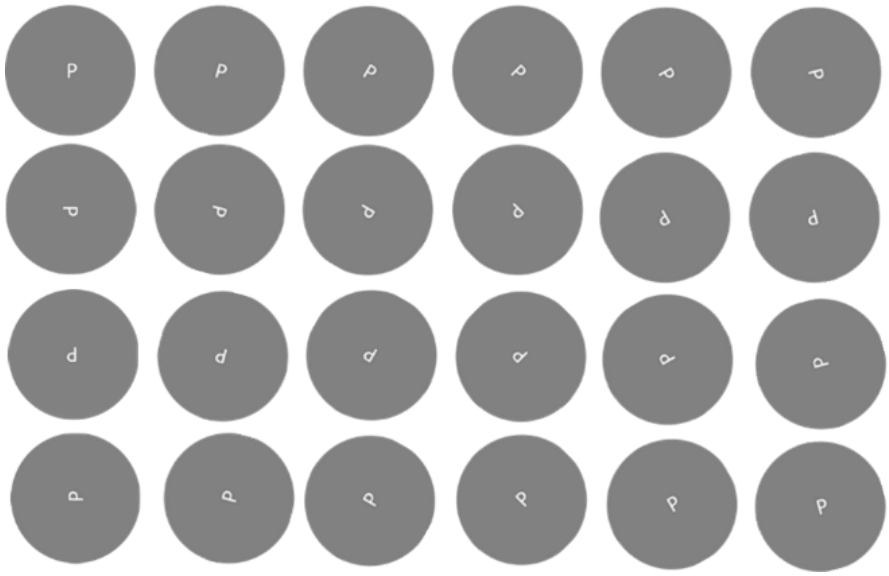

Figure 4.2: Twenty-four orientations of letter probe displayed against a grey background.

A polarized room image with clear orientation cues was used as the background, presented in six orientations (0° to 300° in 60° increments), each combined with multiple letter orientations. This yielded 1344 total trials, performed on a MacBook Pro laptop with a shroud installed to limit peripheral distractions, making the screen appear as a 35° diameter circle at 25 cm. Throughout the task, participants indicated whether the displayed character appeared as a 'p' or a 'd' using a Logitech Gamepad F310.

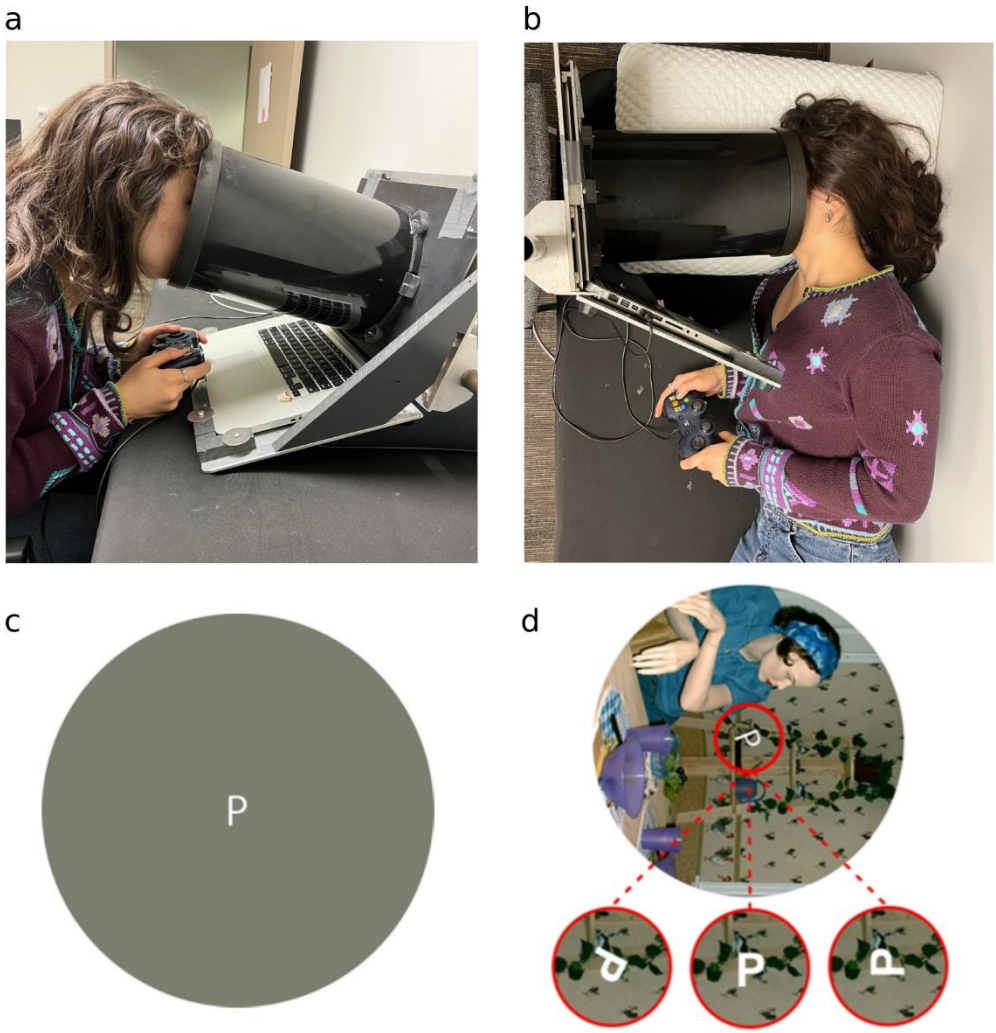


Figure 4.3: The two body postures used in these experiments: a) upright and b) right side down. Subjects viewed the display through a shroud to obscure all peripheral vision. An example of the OCHART task on c) the grey background and d) on the visual background. Participants were asked to determine whether they were presented with the letter 'p' or 'd.' Each combination of the letter orientation and background orientation was repeated eight times and that brought the total amount of trials to 1344.

### 4.3.4 Procedures

The procedure for this chapter closely followed the design described in Chapter 3, with the addition of two experimental sessions per participant, one with EVS and one with no-EVS, assigned in a counterbalanced order across participants. Each session took place on a separate day.

At the start of each session, participants completed baseline OCHART tasks in two positions: upright seated and lying on their right side, following the same setup and instructions as in Chapter 3. For the upright assessment, participants sat with their face close to the circular shroud attached to the laptop

screen; for the right side down (RSD) assessment, the laptop was rotated 90° and mounted on a custom stand, and participants lay on an examination table with a foam pillow for head support. In both cases, participants indicated whether the letter stimulus appeared as a ‘p’ or a ‘d’ using a two-alternative forced choice on a gamepad.

For the EVS condition, after baseline OCHART tasks, participants were prepped with electrodes for vestibular stimulation and then played ADR1FT in VR for up to 20 minutes, or until they reported a score of 20 on the Fast Motion Sickness Scale (FMS). For the no-EVS condition, participants completed the same VR gameplay without any electrode preparation or EVS, replicating the protocol from Chapter 3.

Immediately after the VR session in both conditions, participants’ sickness levels were verbally assessed using the FMS. They then completed the OCHART task again, this time only in the right-side-down position. All equipment setup, task instructions, and response recording were identical to those described previously.

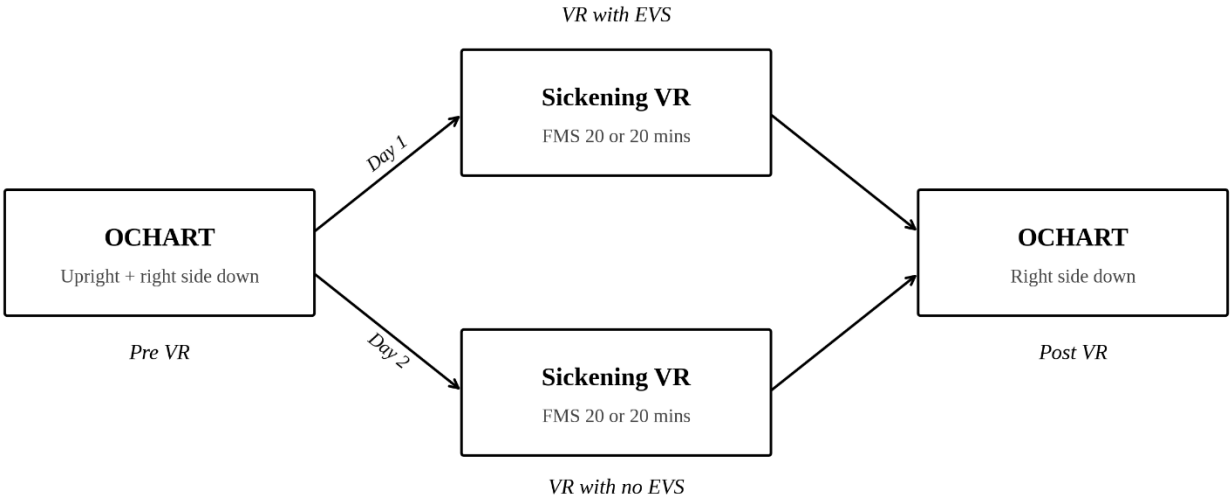


Figure 4.4: Flowchart of the experiment procedure.

### 4.3.5 Sickness Measures

Cybersickness was measured using the FMS, which measures sickness levels immediately following each VR session (Keshavarz & Hecht, 2011). Participants provided verbal ratings on a 20-point scale, with ‘0’ indicating no sickness and ‘20’ indicating severe or frank sickness.

### 4.3.6 neuroConn DC-STIMULATOR PLUS

Continuous stochastic bilateral EVS was delivered using the neuroConn DC-STIMULATOR PLUS with reusable 35 $cm^2$ silicone electrodes placed over the left and right mastoid processes (Fitzpatrick & Day, 2004). The stimulation consisted of a zero-mean, Gaussian white noise current, sampled at 1280 Hz, with an amplitude range of ±2500 μA. This protocol is well-established for modulating vestibular afferent activity in a non-invasive and safe manner (Serhan et al., 2018; Silva et al., 2024).

The amplitude of ±2500 μA was selected to exceed the perceptual threshold for vestibular stimulation, with the intent of producing a detectable perturbation of vestibular signals during VR exposure. This amplitude is within the safe range for transcutaneous EVS (Fitzpatrick & Day, 2004) and is comparable to or below amplitudes used in prior VR-cybersickness GVS studies.

### 4.3.7 EVS Preparation

Parker Spectra 360 Electrode Gel was used to ensure low-impedance contact, and 3M clear medical tape secured the electrodes and wires in place. The stimulator and battery pack were carried in a small backpack worn by participants during VR exposure, allowing for unrestricted movement within the play area. Active stimulation ramped up to the target amplitude over 8 seconds, was delivered continuously throughout the ~20-minute VR session and then ramped down over 5 seconds at the end.

In the no-EVS condition, participants underwent the same VR procedure without any vestibular stimulation; the device was not activated, and no current was delivered during the VR session. This allowed for direct comparison between standard VR exposure and VR with concurrent vestibular stimulation.

Figure 4.5: Example of electrode placement for bilateral EVS. The participant is shown with a 35 cm² silicone electrode positioned over both mastoid processes and prepped for stimulation.

### 4.3.8 Data Analysis

Statistical analysis was conducted using SigmaPlot 12.5. To determine the PU, sigmoidal psychometric functions were fitted to the proportion of trials in which each letter orientation was identified as "p" rather than "d," plotted as a function of letter orientation angle. Following the original OCHART procedure, two separate sigmoid fits were used to estimate the two category transition points: the orientation at which responses shifted from "p" to "d," and the orientation at which responses shifted from "d" to "p" (Dyde et al., 2006).

(4.1) $$y = \frac{100}{1 + e^{\left(\frac{x - x_0}{b}\right)}}\%$$

The two fitted 50% points, $x_{01}$ and $x_{02}$, were then averaged to estimate the participant's perceptual upright:

(4.2) $$PU = \frac{(x_{01} + x_{02})}{2} - 180$$

Here, $x_{01}$ and $x_{02}$ represent the two category boundaries separated by approximately 180°. Averaging these values gives the midpoint between the two ambiguous orientations, corresponding to the observer's PU. Subtracting 180° re-centres the estimate so that 0° corresponds to the canonical upright orientation of the character. For consistency with prior OCHART reporting, discrimination precision was indexed using the fitted slope parameter $b$ b, which is referred to here as the JND estimate.

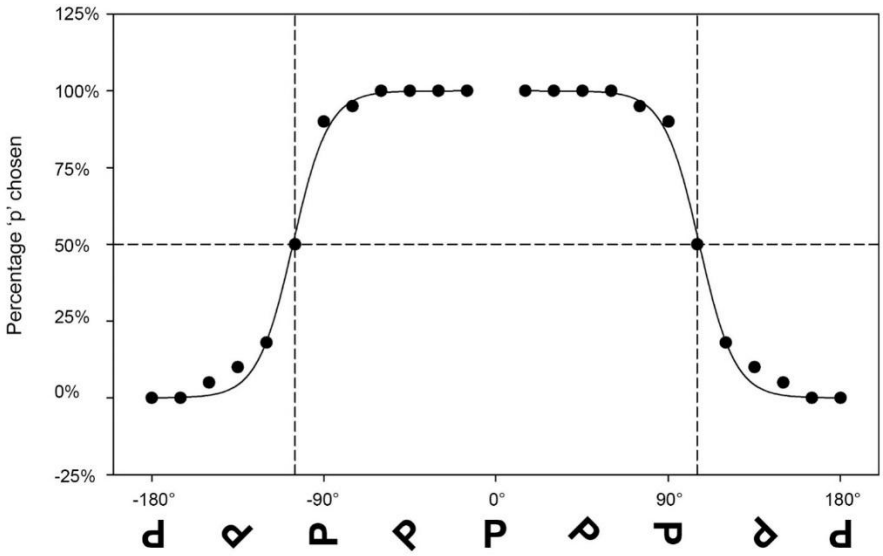


Figure 4.6: This figure illustrates how the PU was measured. Participants chose whether they thought the symbol was a 'p' or 'd.' Typical psychometric functions obtained from such responses are shown. The two 50% points were averaged to give the PU.

In addition to computing PU from psychometric fits, cue weights for vision, gravity, and body orientation were extracted using a vector sum model (Dyde et al., 2006). Statistical analyses included repeated-measures ANOVA with Greenhouse-Geisser corrections when sphericity was violated, as well as Pearson correlations to examine associations between PU shifts and cybersickness ratings.

Figure 4.7: Experimental visual backgrounds, rotated in 60° increments from 0° to 300°.

To assess the between-day test-retest reliability of baseline OCHART-derived PU, pre-exposure PU values from the two experimental sessions, Pre-noSTIM VR and Pre-STIM VR, were compared in the subset of participants with baseline data available at both timepoints. Because each participant completed baseline testing on separate days, this analysis was treated as a between-day repeatability assessment. Baseline PU values were analyzed using a 2 × 7 repeated-measures ANOVA with Session (Pre-noSTIM VR, Pre-STIM VR) and Background (grey, 0°, 60°, 120°, 180°, 240°, 300°) as within-subject factors. The primary question was whether there was a main effect of Session or a Session × Background interaction. The absence of these effects would indicate that baseline PU estimates were stable across days. As a

descriptive index of repeatability, an absolute-agreement intraclass correlation coefficient (ICC) was also calculated on the background-collapsed mean PU across the two baseline sessions.

### 4.3.9 Vector Sum Model

To determine how visual, bodily, and gravitational cues contribute to the perception of upright in this experiment, I will apply a vector sum model (Dyde et al., 2006; Mittelstaedt, 1983; Mittelstaedt & Fricke, 1988). In this approach, each sensory cue will be represented as a vector pointing in its respective direction, and each vector will have a length corresponding to its relative weight or influence. By fitting participants' OCHART responses to this model using the Marquardt-Levenberg optimisation algorithm, I will quantify the contribution of each cue to perceptual upright (Press et al., 1992). The relative weightings of vision, body, and gravity will be reflected in the lengths of their vectors. Specifically, vectors $\vec{v}$ and $\vec{b}$ will represent the orientations signalled by the visual and body cues, respectively, while $\vec{g}$ will denote the gravitational direction. As in the previous chapter, I will fix the length of the gravity vector to 1 and will fit the lengths of the visual and body vectors to best match the data. This modelling approach assumes that each vector accurately reflects the direction associated with its corresponding sensory input. Figure *4.6* illustrates this model, showing how the combination of these sensory cues determines the perceived upright according to the following equation:

$$\overrightarrow{PU} = \vec{v} + \vec{b} + \vec{g} \tag{4.3}$$

To evaluate the quality of individual model fits, the coefficient of determination ($R^2$) was computed for each participant and condition. Participants were excluded from the vector sum analysis for a given condition if (a) the fitted body weight exceeded 20, indicating that the optimiser was unable to converge on a physiologically interpretable solution, or (b) $R^2$ fell below 0.50, indicating that the model explained less than half of the variance in observed PU. These criteria were applied independently for each condition (pre-VR, no-EVS, EVS), meaning that a participant could be retained for one condition but excluded from another.

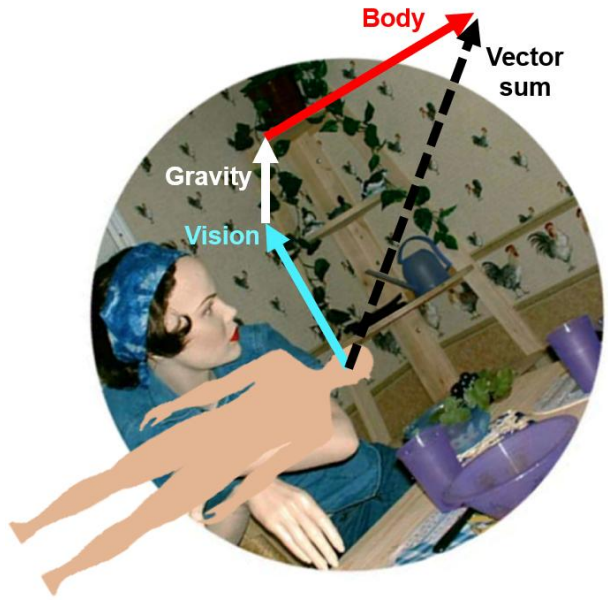


Figure 4.8: A weighted vector sum model, in which gravity, body orientation, and visual cues are summed to generate an estimate of vertical. The predicted outcome is depicted by the dotted black line, representing the vector sum.

This model conceptualizes each sensory input (vision, body, gravity) as a directional vector with a corresponding weight, and the resultant PU is the vector sum of these influences. Gravity is typically held constant (normalised to 1.0), while the relative weights of vision and body cues are adjusted to fit participant responses. This approach enables a direct comparison of sensory contributions and allows for pre vs. post VR change detection on an individual level.

(Eq 4.4)

$$\boldsymbol{vy} = \text{vis} \cdot \cos(v)$$

$$\boldsymbol{v_x} = \text{vis} \cdot \sin(v)$$

$$\boldsymbol{y} = \text{b} + v_y + g \cdot \cos(\text{tilt})$$

$$x = v_x + g \cdot \sin(\text{tilt})$$

$$f = \tan^{-1}\left(\frac{x}{y}\right)$$

Where *vis* = visual background orientation, b = body vector along y-axis, *g* = represents the otolith-mediated gravitational signal which is a constant on Earth = 1, *tilt* = body orientation, *f* = perceptual upright (PU).

# 4.4 Results

## 4.4.1 Effects of EVS and VR

To evaluate the impact of EVS on cybersickness and VR tolerance, we compared participants' FMS scores and time spent in the VR environment across EVS and no-EVS conditions. Time to stop the task was compared between the no-EVS and EVS conditions using a paired t-test; however, as the normality assumption was not met for either condition (Shapiro-Wilk, both $p < 0.001$), a Wilcoxon signed-rank test was performed instead. Median time to stop was 20.0 minutes for both no-EVS (IQR: 20.0 – 20.0) and EVS (IQR: 19.9 – 20.0). The Wilcoxon signed-rank test ($W = 1.0$, $Z = 2.00$, $p = 0.046$) indicated a statistically significant difference in task duration between conditions, with participants spending less time in VR during the EVS condition (Figure *4.9*).

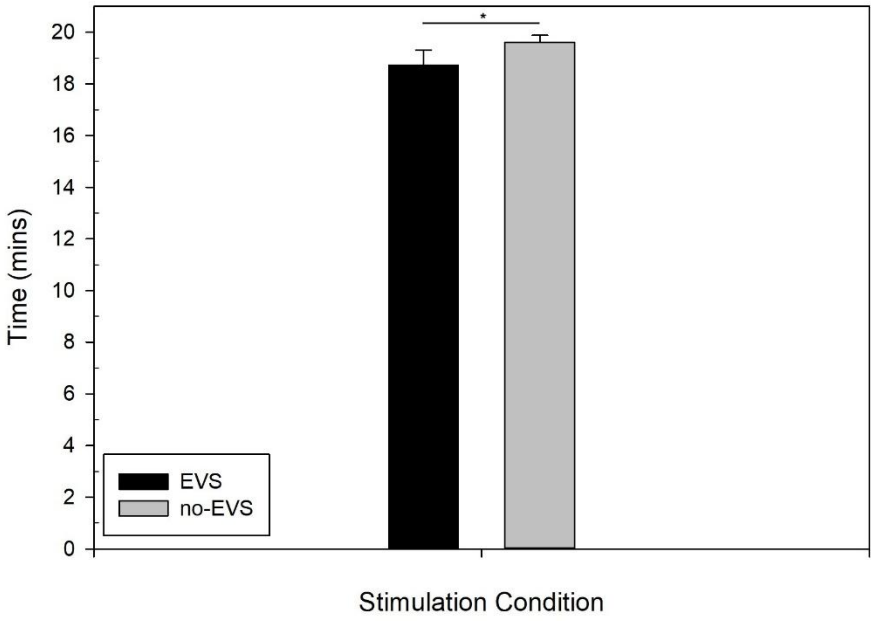


Figure 4.9: Mean time to stop the task in no-EVS and EVS conditions. Error bars represent the standard error of the mean. A significant difference was found between conditions (Wilcoxon signed-rank test, $p = 0.046$), indicated by an asterisk.

A paired t-test was performed to compare FMS scores between the no-EVS and EVS conditions. The data failed the normality assumption in the EVS condition (Shapiro-Wilk, $W = 0.911$, $p = 0.042$), so a Wilcoxon signed-rank test was conducted. Median FMS scores were 9.0 (IQR: 6.0 – 15.0) for no-EVS and 14.5 (IQR: 9.5 – 18.5) for EVS. The Wilcoxon signed-rank test indicated that FMS scores were

significantly higher in the EVS condition (W = 12.5, Z = 3.19, p = 0.001, r = 0.67), suggesting greater motion sickness severity when stimulation was applied (Figure *4.10*).

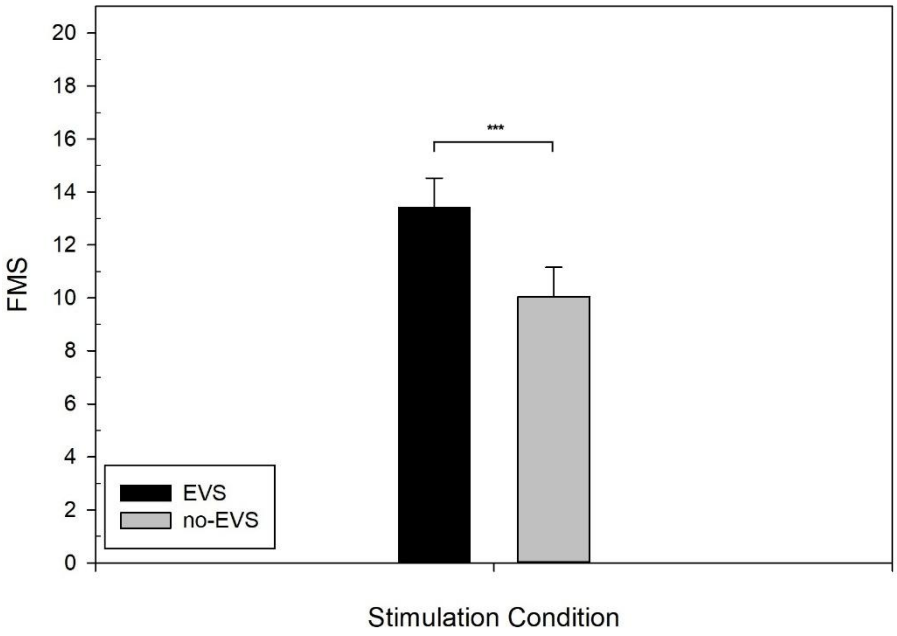


Figure 4.10: FMS scores by stimulation condition. Median FMS scores were significantly higher in the EVS condition compared to no-EVS (Wilcoxon signed-rank test, p = 0.001), indicated by asterisks.

### 4.4.2 Perceptual Upright

To determine whether there was an effect of EVS on performance in the OCHART task across all visual backgrounds, the average PU RSD for EVS (Figure *4.11*a) and no-EVS (Figure *4.11*b) conditions was plotted for each visual background. Prior to analysis, two data points were identified as extreme outliers in the EVS condition (P6 at 120°: PU = −146.1°; P8 at 60°: PU = −132.9°), likely reflecting psychometric fit failures, and were excluded. The effect of visual background orientation on PU can be seen under EVS [Grey (M = −26.07, SE = 3.75); 0° (M = −20.13, SE = 3.34); 60° (M = −9.92, SE = 4.41); 120° (M = −17.70, SE = 6.50); 180° (M = −33.79, SE = 5.32); 240° (M = −42.50, SE = 5.15); 300° (M = −32.02, SE = 4.00)]. A similar pattern was observed in the no-EVS condition [Grey (M = −31.59, SE = 3.35); 0° (M = −21.77, SE = 3.15); 60° (M = −12.92, SE = 3.52); 120° (M = −23.20, SE = 5.02); 180° (M = −38.09, SE = 3.72); 240° (M = −39.29, SE = 2.91); 300° (M = −35.35, SE = 3.34)].

A two-way repeated measures ANOVA was conducted on the 17 participants with complete data across both conditions and all seven backgrounds. Mauchly's test indicated that the sphericity assumption was violated for the Background factor (W ≈ 0.00, $\chi^2(20) = 151.61$, $p < 0.001$), so degrees of freedom were

corrected using the Greenhouse-Geisser estimate ($\varepsilon = 0.25$). The analysis revealed a significant main effect of visual background, $F(1.50, 23.97) = 19.34$, $p < 0.001$, $\eta^2p = 0.547$, but no significant main effect of stimulation condition, $F(1, 16) = 0.23$, $p = 0.640$, $\eta^2p = 0.014$, nor a significant interaction between condition and background, $F(1.50, 23.97) = 1.16$, $p = 0.317$, $\eta^2p = 0.067$. Bonferroni-corrected pairwise comparisons ($k = 21$, $\alpha = 0.0024$) revealed significant differences between multiple background pairs, most notably between the near-upright backgrounds (0° and 60°) and the more tilted orientations (180°, 240°, and 300°), reflecting the inverted-U influence of visual background on PU. These results indicate that while PU was strongly influenced by visual background orientation, there was no significant effect of EVS, nor any interaction between EVS and visual background orientation.

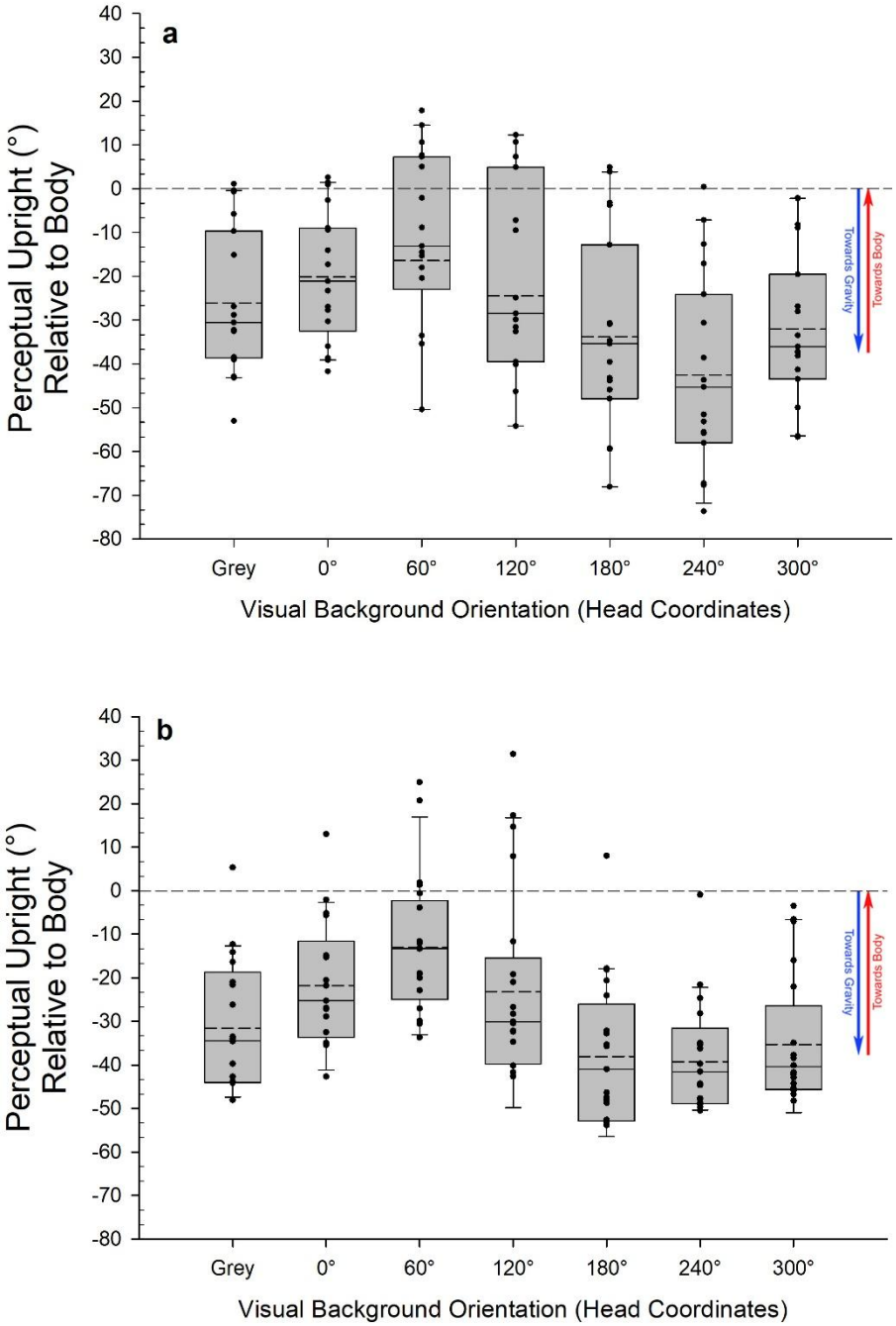


Figure 4.11: PU estimate of right side down a) EVS and b) no-EVS of different visual background orientations. Solid lines within the box plot are the median, and dashed lines are the mean. Error bars are standard error.

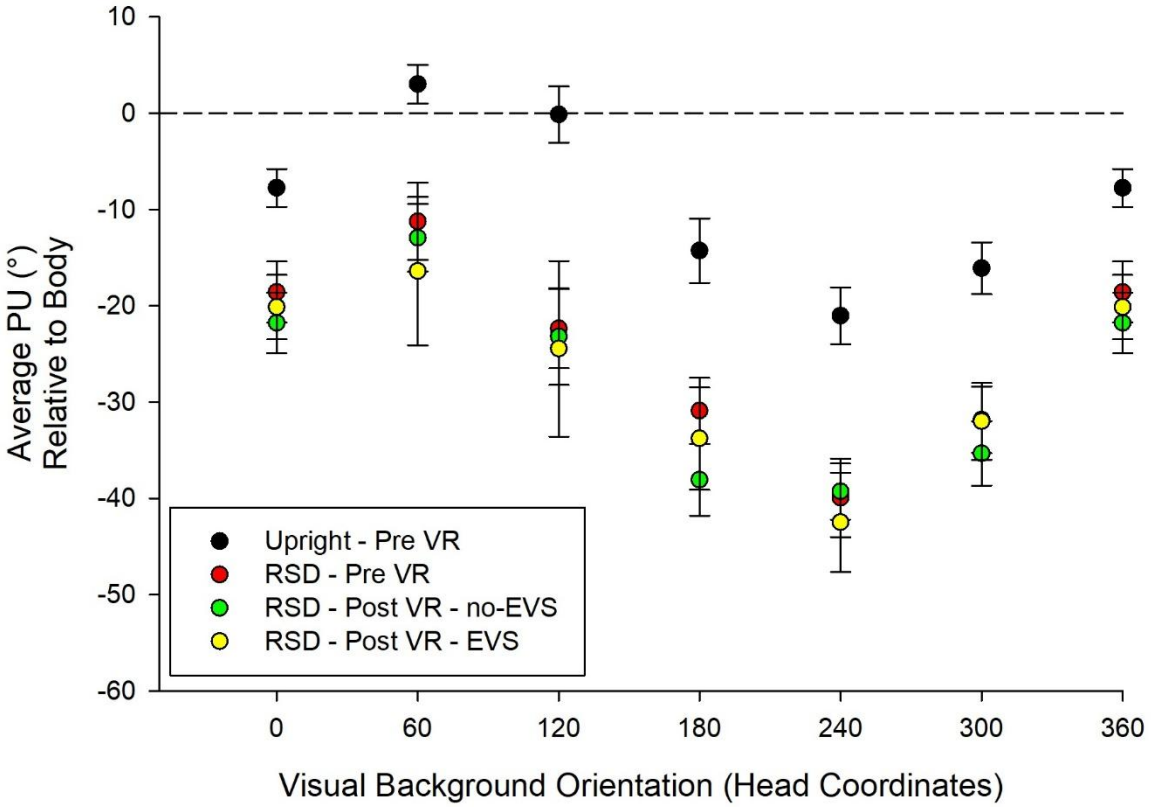


Figure 4.12: Average PU estimate of right side down of upright (black), pre-VR (red), post-VR EVS (green), and post-VR (no-EVS) of different visual background orientations. Error bars are standard error.

### 4.4.3 Baseline Repeatability of Perceptual Upright

To determine whether baseline OCHART-derived PU was stable across test days, pre-exposure PU values from the Pre-noSTIM VR and Pre-STIM VR sessions were compared for both sessions. There was no significant main effect of Session, $F(1, 27) = 0.24$, $p = 0.638$, indicating that baseline PU estimates did not differ systematically between days. As expected, there was a significant main effect of Background ($p < 0.001$) confirming that visual background orientation strongly influenced PU at baseline. Importantly, the Session × Background interaction was not significant ($p = 0.489$) indicating that the overall pattern of background-dependent PU was preserved across the two baseline sessions. Together, these findings suggest that baseline OCHART-derived PU showed acceptable between-day stability at the group level. Consistent with this interpretation, the absolute-agreement ICC for background-collapsed mean PU across the two baseline sessions was in the moderate range (ICC = 0.46), suggesting modest but meaningful between-day repeatability of the measure.

### 4.4.4 Sex Differences

To examine whether sex moderated the cybersickness response, FMS scores were compared between males and females using Mann-Whitney U tests, and the EVS effect was tested separately within each sex using Wilcoxon signed-rank tests. Females reported significantly higher FMS than males in the no-EVS condition (females Mdn = 15.0; males Mdn = 8.0; U = 30, p = 0.032), consistent with prior reports of elevated cybersickness susceptibility in females (Stanney et al., 2020). In the EVS condition, females again trended higher than males (females Mdn = 16.0; males Mdn = 12.0; U = 36, p = 0.079), though this difference did not survive correction.

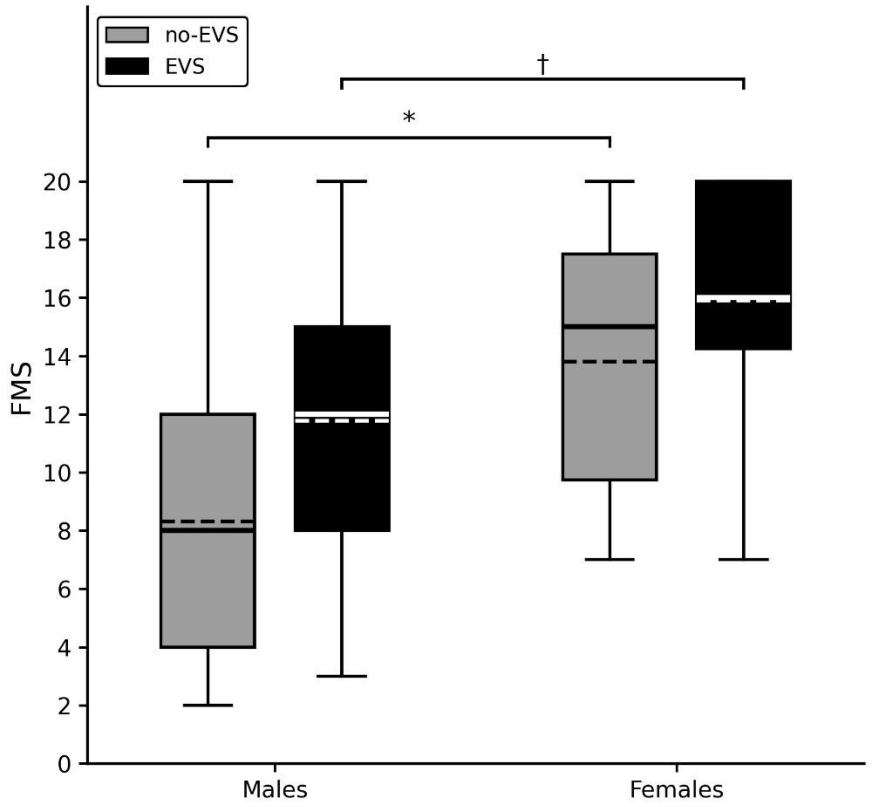


Figure 4.13: FMS scores by sex and stimulation condition. Solid lines indicate the median; dashed lines indicate the mean. Females reported significantly higher FMS scores than males in the no-EVS condition (p = .032). The equivalent comparison in the EVS condition approached but did not reach significance (p = .079).

Within-sex analyses revealed that the EVS effect was statistically significant in males (W = 4.0, Z = 2.71, p = 0.007, r = 0.750), with 10 males reporting higher FMS under EVS. The corresponding test in females did not reach significance (W = 3.5, Z = 1.68, p = 0.094, r = 0.530), with only 6 females showing higher FMS under EVS. However, the magnitude of the EVS-induced FMS change did not differ significantly

between sexes (males M = +3.46, females M = +2.10; U = 82, p = 0.317), indicating that EVS did not differentially affect sickness severity in males versus females. The pattern is most parsimoniously explained by a ceiling effect: females reported high sickness even without EVS, leaving less room for a further EVS-driven increase, whereas males had lower baseline sickness and therefore a larger observable margin for change.

### 4.4.5 Perceptual Upright and Cybersickness

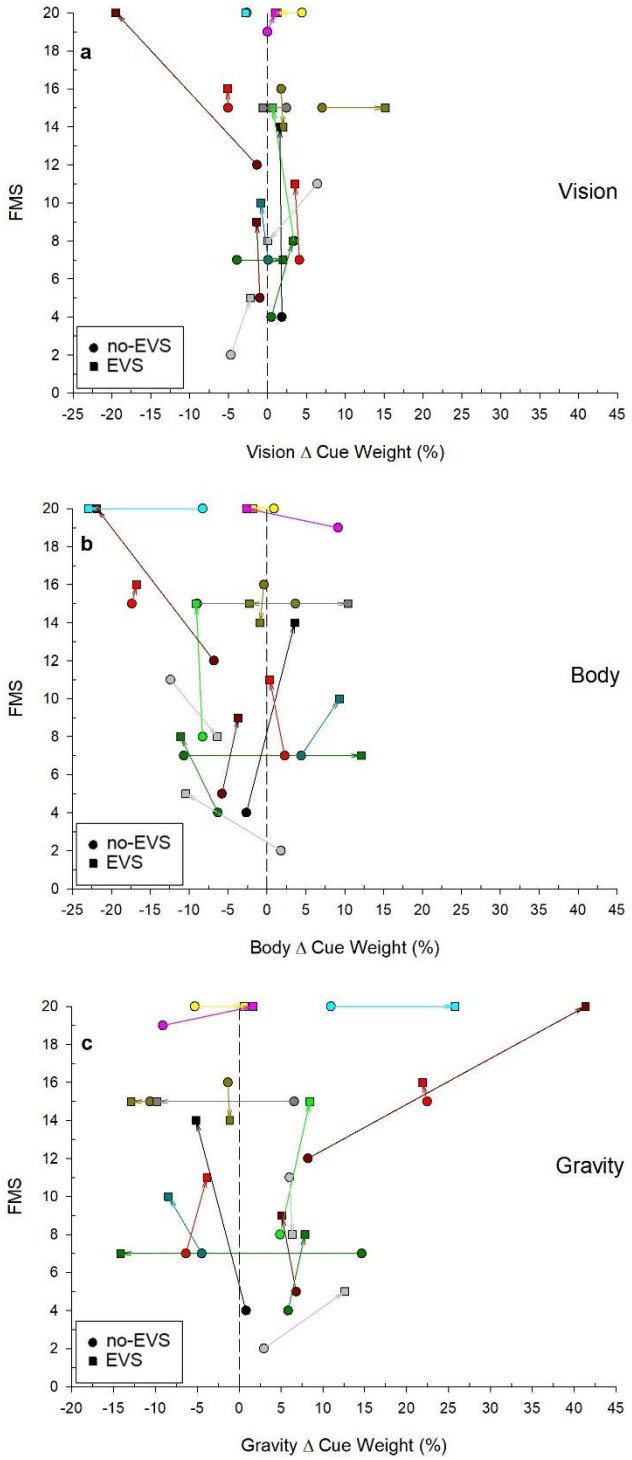


Figure 4.14: FMS ratings relative to the change in cue weight per cents for a) vision and b) body, and c) gravity. The dashed line indicates the upper specification of the regression of no difference in the PU from baseline.

Figure *4.13* illustrates the relationship between changes in sensory cue weights for vision, body, and gravity and cybersickness severity for individual participants across the no-EVS and EVS conditions. Overall, an inter-individual variability in cue reweighting can be seen in response to EVS, with no consistent trend across the group. For vision and body cues (Figure *4.14*a and Figure *4.14*b), most participants show small changes in weighting; however, for gravity cue weights (Figure *4.14*c), a wider range of change is observed, with some participants exhibiting large positive or negative shifts. While a flexible reweighting strategy does not guarantee lower sickness across all individuals, these data suggest that maladaptive or inflexible cue weighting, particularly regarding gravity, may be associated with heightened cybersickness severity.

### 4.4.6 Vector Sum Model Fitting and Data Quality

The vector sum model (Eq. 4.3) was fit to each participant's OCHART-derived PU values across 12 condition cells (6 visual backgrounds by 2 body orientations: upright and right-side-down) separately for the pre-VR, no-EVS, and EVS conditions. Model fit quality was evaluated using a convergence criterion on the fitted body weight parameter ($b \leq 20$). Of the 28 participants, 12 were excluded because their fitted body weight exceeded this threshold, either due to convergence at an implausibly large value or because the model failed to yield stable parameter estimates (Table *4.1*). This pattern is consistent with the degeneracy described above, in which minimal sensitivity to the 90° body tilt leads the fitted body weight to inflate to accommodate the data. The remaining 16 participants produced reliable fits across all three conditions and were retained for cue weight analyses.

| Participant | no-EVS v | no-EVS b | no-EVS $R^2$ | EVS v | EVS b | EVS $R^2$ |
|---|---|---|---|---|---|---|
| P1 | 0.521 | 1.576 | 0.193 | 0.528 | 1.622 | 0.164 |
| P2 | 0.358 | 1.319 | 0.965 | 0.259 | 1.184 | 0.953 |
| **P3** | **56720** | **325484** | **0.536** | **82.15** | **606.04** | **0.589** |
| P4 | 0.351 | 1.350 | -0.854 | 0.228 | 1.102 | -0.581 |
| **P5** | **0.589** | **1.324** | **0.895** | **10.23** | **89512** | **0.113** |
| P6 | 0.174 | 1.344 | 0.605 | 0.124 | 0.709 | 0.303 |
| P7 | 0.283 | 1.073 | 0.173 | 0.279 | 0.639 | 0.617 |
| P8 | 0.196 | 1.385 | 0.951 | 0.147 | 0.537 | 0.645 |
| **P9** | **0.335** | **1.833** | **0.477** | **5.962** | **29.133** | **0.306** |
| P10 | 2.049 | 3.305 | 0.449 | 2.993 | 3.413 | 0.730 |
| **P11** | **1556** | **831919** | **0.541** | **3055** | **208102** | **0.610** |
| **P12** | **0.339** | **1.719** | **0.468** | **29.64** | **114.08** | **0.407** |
| P13 | 0.410 | 1.196 | 0.946 | 0.430 | 1.478 | 0.952 |
| **P14** | **0.243** | **1.696** | **0.106** | **0.298** | **49.742** | **-4.911** |
| P15 | 0.850 | 4.875 | 0.517 | 1.392 | 8.846 | 0.463 |
| P16 | 0.893 | 1.712 | 0.129 | 0.799 | 1.506 | 0.207 |
| **P17** | **0.482** | **2.648** | **0.755** | **6.882** | **782.5** | **0.135** |

| | | | | | | |
|---|---|---|---|---|---|---|
| **P18** | **0.529** | **3.716** | **0.673** | **14570324** | **352163358** | **0.283** |
| **P19** | **0.573** | **3.375** | **0.656** | **23887853** | **547070534** | **0.293** |
| P20 | 0.623 | 3.579 | 0.715 | 0.433 | 1.903 | 0.922 |
| **P21** | **0.262** | **1.252** | **-1.999** | **1.862** | **33678** | **0.233** |
| P22 | 2.958 | 5.695 | 0.608 | 2.276 | 6.088 | 0.688 |
| P23 | 0.534 | 1.301 | 0.825 | 0.549 | 1.430 | 0.755 |
| P24 | 0.623 | 1.064 | 0.724 | 0.661 | 0.887 | 0.898 |
| P25 | 0.492 | 2.416 | 0.105 | 0.497 | 2.375 | 0.154 |
| P26 | 0.195 | 0.815 | 0.906 | 0.343 | 1.295 | 0.836 |
| **P27** | **1.154** | **0.578** | **0.753** | **99.99** | **662.9** | **-0.481** |
| **P28** | **0.728** | **1.843** | **0.732** | **6041** | **19527** | **0.4080** |

Table 4.1: Individual visual (v) and body (b) cue weights and model fit ($R^2$) estimated from the vector sum model for the no-EVS and EVS conditions across all 28 participants. Gravity weight was fixed at 1. Bold rows indicate participants excluded from group-level analyses due to fitted body weight exceeding the convergence criterion (b > 20).

Inspection of excluded cases revealed a consistent pattern: participants whose PU showed minimal sensitivity to the 90° body tilt produced unreliable or degenerate model solutions. In participants with reliable fits, tilting from upright to right-side-down typically shifted mean PU by several tens of degrees in the direction of gravity, consistent with a measurable graviceptive contribution to perceived upright. In excluded participants, however, this tilt-induced shift was markedly attenuated, often falling below 5°, and in some cases occurring in the opposite direction. When the observed tilt effect was near zero, the model could only reproduce the data by inflating the fitted body weight to very large values ($b \gg 20$), which numerically reduces the influence of the gravitational term in the predicted PU. In the most extreme case, one participant's EVS condition produced a body weight exceeding $5 \times 10^8$ with $R^2 = 0.28$, indicating numerical runaway consistent with poor parameter identifiability rather than a stable estimate of cue weighting.

The mathematical basis for this behaviour can be understood by considering the model at RSD tilt. When tilt = -90°, the gravitational contribution reduces to $g \times \sin(-90°) = -1$ in the numerator of the arctangent function, while $\cos(-90°) = 0$ eliminates the gravitational term from the denominator. The only mechanism by which the model can predict near-zero PU at -90° tilt is for the body weighting in the denominator to dominate the gravitational pull in the numerator. As body weight increases, predicted PU at -90° tilt converges asymptotically toward 0° (Table *4.2*).

| Body Weighting | Predicted PU (°) |
|---|---|
| 2 | -26.6 |
| 5 | -11.3 |
| 10 | -5.7 |

| | |
|---|---|
| 18 | -3.2 |
| 50 | -1.1 |
| 100 | -0.6 |

Table 4.2: Asymptotic effect of body weight on predicted PU under right-side-down tilt in the vector sum model.

Notably, excluded participants were not uniformly excluded across conditions. Several participants produced degenerate fits in the EVS condition but acceptable fits in pre-VR or no-EVS, and vice versa. This condition-dependent pattern indicates that model stability varied across experimental contexts for some individuals. While this could reflect genuine context-dependent changes in cue use, it may equally arise from increased noise or reduced parameter identifiability in specific conditions. These exclusions are therefore treated as a modelling limitation rather than evidence for a specific strategy shift.

### 4.4.7 Cue Weights and Associations with Cybersickness

To examine whether EVS was associated with changes in OCHART-derived cue contributions to the perceptual upright, and whether individual differences in cue weight change were associated with cybersickness severity, the vector sum model was applied to the 15 participants with reliable fits across all three conditions.

Pre-VR vector sums (Mdn = 3.49, IQR: 3.04-3.73) did not differ significantly from post-VR vector sums in either the no-EVS condition (Mdn = 2.89, IQR: 2.66-4.23; $W = 44.0$, $Z = 1.20$, $p = 0.231$, $r = 0.30$) or the EVS condition (Mdn = 3.07, IQR: 2.42-4.32; $W = 51.0$, $Z = 0.84$, $p = 0.404$, $r = 0.21$), indicating that VR exposure did not produce a significant overall shift in sensory cue weighting in either condition. Post-VR vector sums also did not differ significantly between the two conditions ($W = 57.0$, $Z = 0.53$, $p = 0.597$).

Examination of individual cue weight percentage changes revealed small and variable shifts following VR exposure. Vision weighting showed a small average increase in the no-EVS condition (M = +1.07%, SD = 3.56) and a negligible change under EVS (M = -0.25%, SD = 6.75). Body weighting decreased slightly in both conditions (noEVS: M = -3.45%, SD = 7.00; EVS: M = -5.39%, SD = 9.72), and gravity weighting increased slightly (noEVS: M = +2.38%, SD = 8.48; EVS: M = +5.64%, SD = 14.34). None of these changes differed significantly between EVS and no-EVS for vision ($W = 47.0$, $Z = 1.04$, $p = 0.298$), body ($W = 52.0$, $Z = 0.78$, $p = 0.433$), or gravity ($W = 48.0$, $Z = 0.99$, $p = 0.323$).

Spearman correlations between FMS scores and cue weight percentage changes were non-significant across all three cues in both conditions (all $\rho$ between -0.27 and +0.10, all $p > 0.30$, N = 15-16). These results suggest that, within the subset of participants with reliable model fits, individual differences in cybersickness severity were not clearly explained by post-exposure differences in OCHART-derived cue weighting in either condition.

### 4.4.8 Model-Free Tilt Sensitivity

To supplement the vector sum modelling and recover interpretable data from participants whose model fits were degenerate, a model-free tilt sensitivity metric was computed. For each participant and condition, tilt sensitivity was defined as the difference between mean PU in the upright body position and mean PU in the right-side-down position, averaged across all visual background orientations. Larger positive values indicate greater sensitivity of PU to the 90-degree body tilt, which is the manipulation that the vector sum model relies upon to separate body and gravity contributions.

In the no-EVS condition, tilt sensitivity showed a negative association with cybersickness severity that approached but did not reach conventional significance (Spearman $\rho = -0.381$, $p = 0.088$). Participants with greater tilt sensitivity tended to report lower FMS scores. This trend is directionally consistent with the reweighting-cybersickness relationship reported in Chapters 2 and 3, where greater perceptual flexibility was associated with reduced sickness. In the EVS condition, no association was observed between tilt sensitivity and FMS (Spearman $\rho = 0.040$, $p = 0.875$), suggesting that EVS may have disrupted the relationship between tilt responsiveness and cybersickness outcomes.

Within-subject comparison of tilt sensitivity between no-EVS and EVS conditions revealed no significant difference (no-EVS Mdn = 16.58°; EVS Mdn = 15.64°; Wilcoxon W = 81.0, $p = 0.595$, N = 19; Figure 4.15). Although 12 of 19 participants showed numerically larger tilt sensitivity under EVS, this was offset by several participants showing substantially reduced sensitivity, resulting in high variability and no net group-level effect. Pre-VR baseline tilt sensitivity did not significantly predict FMS in either the no-EVS ($\rho = -0.206$, $p = 0.370$) or EVS ($\rho = 0.135$, $p = 0.570$) conditions.

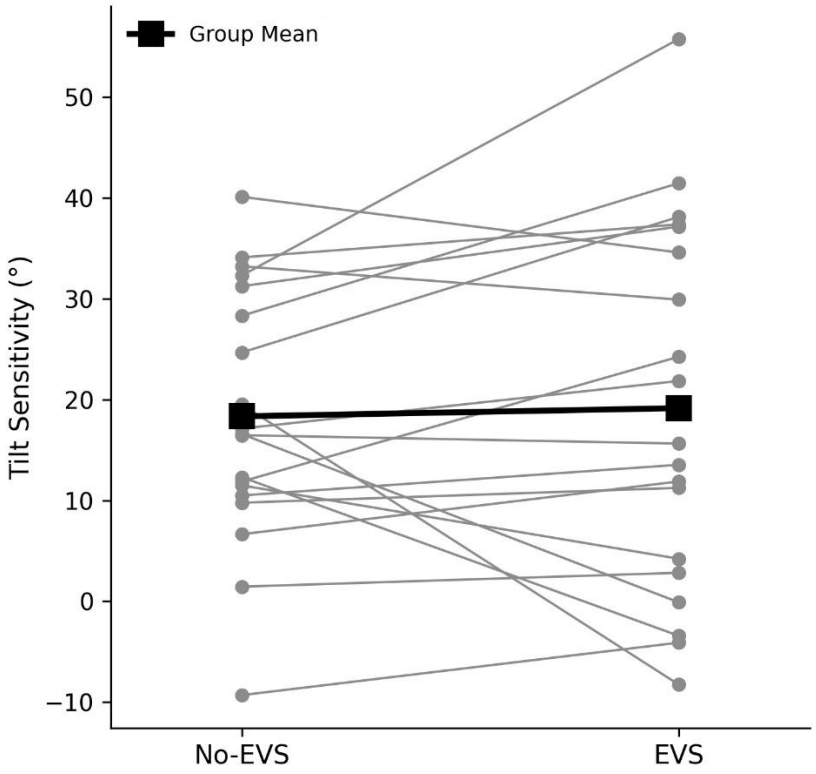


Figure 4.15: Within-subject tilt sensitivity in the no-EVS and EVS conditions for all participants with paired data. Grey lines represent individual participants. Black squares indicate the group mean. The Wilcoxon signed-rank test indicated no significant difference between conditions (W = 81.0, p = 0.595).

## 4.5 Discussion

The primary aim of this chapter was to determine whether bilateral stochastic EVS would alter vestibular contributions to multisensory orientation and reduce cybersickness during immersive VR exposure. The results were clear in direction, but contrary to the hypothesis. EVS significantly increased cybersickness severity and reduced VR tolerance, while producing no detectable change in perceptual upright or sensory cue weighting. The following sections interpret these findings in the context of stimulation parameter sensitivity, the dissociation between vestibular perturbation and perceptual upright, and the implications for the sensory reweighting account of cybersickness.

### 4.5.1 Effect of EVS on Cybersickness

The finding that EVS significantly elevated FMS scores (no-EVS Mdn = 9.0 vs EVS Mdn = 14.5; W = 12.5, p = 0.001, r = 0.67) and increased the rate of early task termination (6 vs 2 participants; W = 1.0, p = 0.046) was directly opposite to the predicted outcome. The strong correlation between FMS and task

duration in the EVS condition ($\rho = -0.758$, $p < 0.001$) confirms that these measures reflect the same underlying sickness construct, and that EVS worsened the overall sickness experience rather than selectively affecting one aspect of it.

The simplest explanation is that the stimulation amplitude of ±2500 μA placed the EVS signal outside the SR regime in which subthreshold noise can improve detection and stabilize sensory estimates. In SR theory, noise-related benefits are confined to a narrow range in which the added noise remains below the system's detection threshold. Once stimulation becomes suprathreshold, the added signal degrades the effective signal-to-noise ratio and acts as an interferent rather than a facilitator (Moss et al., 2004). In the vestibular SR literature, reported optimal amplitudes for GVS tend to fall in the low hundreds of microamps, commonly on the order of 100-500 μA (Galvan-Garza et al., 2018; Keywan et al., 2018; Wuehr et al., 2018). A systematic review of 31 GVS studies targeting postural outcomes similarly noted that effects become more variable at higher intensities, with outcomes often diminishing or becoming detrimental as current approaches and exceeds the subthreshold range (McLaren et al., 2023).

Galvan-Garza et al. (2018) demonstrated stochastic resonance benefits at amplitudes up to ±700 μA for vestibular tilt motion perception. Goel et al. (2015) found balance improvements at ±100 to 500 μA, with the optimal range falling near 42 to 55% of perceptual threshold. Weech et al. (2020), using ±1750 μA in VR, observed a modest and transient reduction in FMS ratings, though the effect did not persist beyond 3 to 6 minutes. The present study used ±2500 μA, which is approximately 5 to 25 times the reported stochastic resonance optimum and 43% above the amplitude used by Weech et al. (2020). This positions the current protocol firmly outside the stochastic facilitation regime and within a range where vestibular noise is more likely to function as an additional conflict signal than as a reliability degrader.

This interpretation is supported by converging evidence at comparable intensity ranges. Research has shown that sinusoidal GVS at ±2 mA could induce nausea in a substantial subset of participants even in the absence of VR exposure, indicating that high-amplitude vestibular stimulation can carry intrinsic nauseogenic effects (Klingberg et al., 2015). Weech and colleagues (2020), using noisy GVS at ±1750 μA, reported only a modest and transient reduction in FMS ratings, and only during the most intense VR content. Allred and colleagues (2025) further demonstrated that the direction of GVS effects on sickness is parameter dependent, with computationally optimised detrimental waveforms increasing sickness while beneficial waveforms reduced it. Taken together, these findings align with a framework in which fixed bilateral stochastic stimulation at higher intensities is more likely to add vestibular conflict than to resolve it, particularly during highly provocative VR.

A more complete interpretation is that the present EVS waveform may not have been treated by the nervous system as neutral background noise at all. Although stochastic GVS is often discussed as a method for modifying vestibular reliability, galvanic stimulation also produces lawful modulation of vestibular afferents and can evoke self-motion and tilt-related percepts, with effects that depend on waveform and motion context (Fitzpatrick & Day, 2004; Allred et al., 2024). Thus, at ±2500 μA, the bilateral stochastic waveform may have functioned as an additional vestibular motion-like signal rather than a simple reliability degrader. In ADR1FT, where strong optic flow was present in the absence of matching inertial cues, such a signal would be expected to increase the mismatch with the visual scene unless it was explicitly coupled to the displayed self-motion. This interpretation is consistent with the present increase in FMS and reduced tolerance, and with prior work showing that uncoupled GVS can aggravate disturbance or delay habituation in visually provocative environments, whereas motion-coupled GVS can reduce discomfort when the vestibular signal is aligned with the visual motion profile (Qi et al., 2021; Groth et al., 2022).

A secondary mechanism may involve cortical engagement of vestibular networks. Noisy GVS can evoke stronger activation in vestibular-related cortical regions than conventional deterministic stimulation, suggesting that suprathreshold stochastic input may increase the conflict signal reaching multisensory areas rather than attenuating it (Mitsutake et al., 2021). Bilateral mastoid stimulation provides an artificial vestibular input that is lawful enough to be interpreted as self-motion, but that input is not necessarily congruent with the direction, timing, or dynamics of the visually specified motion in VR (Dlugaiczyk et al., 2019). In ADR1FT, participants experienced sustained first-person navigation in a zero-gravity setting with strong optic flow and no matching inertial cues, creating an intense and continuous visuo-vestibular mismatch. Under these circumstances, adding overt vestibular noise at ±2500 μA likely compounded the mismatch and increased cybersickness, rather than facilitating adaptive sensory reweighting within a single 20-minute exposure.

One way to address the ambiguous nature of fixed binaural mastoid stimulation is to move from open-loop stochastic EVS toward adaptive, motion-coupled EVS, in which the stimulation waveform is updated in real time based on the user's head kinematics or the VR camera motion so that the induced vestibular signal is coherent with the intended self-motion. Motion-synchronized GVS approaches in VR have shown that coupling stimulation to the visual motion profile can reduce discomfort and markers of sickness, supporting the broader idea that congruence-based stimulation may be more effective than adding non-specific vestibular noise (Groth et al., 2022; Pradhan et al., 2022). More generally, head-coupled EVS work demonstrates that the nervous system treats coherent, movement-linked vestibular perturbations differently from the same waveform delivered without behavioural coupling, suggesting that

real-time coupling may reduce the interpretive ambiguity that likely contributed to the adverse effects observed here (Héroux et al., 2015; Chen et al., 2022).

### 4.5.2 Perceptual Upright and OCHART

Perceptual upright was strongly modulated by visual background orientation in both stimulation conditions, replicating the characteristic non-linear background effect typically observed in OCHART paradigms (Dyde et al., 2006). In contrast, EVS produced no detectable main effect on PU and no EVS by background interaction. If stochastic EVS reduced cybersickness by degrading vestibular reliability and thereby encouraging a compensatory shift toward other cues, one plausible prediction is that post-VR PU would show a systematic change in mean bias and or a change in how background orientation shapes the upright estimate. The present data instead indicate that, under these parameters, EVS can meaningfully increase symptom burden without producing a stable post-exposure shift in PU as captured by OCHART.

Several factors could explain this dissociation. First, OCHART is a bias-focused measure of the post-exposure upright estimate, and a vestibular manipulation can increase internal variability or reduce moment-to-moment stability without shifting the mean. Increased response instability under EVS would reduce power to detect subtle mean shifts and could manifest primarily as noisier psychometric functions rather than systematic re-anchoring. Second, EVS effects may be time-locked to stimulation and to the conflict state during exposure rather than persisting as a durable aftereffect. This possibility is especially relevant here because beneficial effects of imperceptible nGVS on vestibular motion perception have been observed primarily during active stimulation and do not appear to persist reliably once stimulation stops (Keywan et al., 2020). In noisy GVS work, reductions in cybersickness can be transient and can extinguish within minutes after stimulation terminates, implying that the key mechanism may operate online rather than producing a long-lasting recalibration detectable after VR (Weech et al., 2020). Third, the physiology of GVS complicates the assumption that it functions purely as additive noise. GVS can evoke directional biases in verticality judgements with rapid time constants, and these biases reflect a vestibular signal that mimics a deterministic tilt-like input rather than an unstructured reduction in vestibular reliability (Mars et al., 2001; Niehof et al., 2019). In the present study, the stimulation waveform was stochastic and bilateral, which may have produced strong perturbation during VR while yielding limited net bias in the post-VR estimate once stimulation ceased.

More broadly, the null PU effect under EVS is compatible with the idea that this protocol primarily increased conflict during exposure, rather than shifting the post-exposure perceptual set point. ADR1FT generates sustained optic flow and controller-driven translation without matching inertial cues, and the

resulting visuo-vestibular mismatch is both continuous and difficult to predict. Under such conditions, adding vestibular input that is not tightly coupled to the visual motion may increase conflict without producing a coherent post-exposure remapping of upright. This interpretation aligns with the broader observation that vestibular stimulation can either attenuate or exacerbate sickness depending on waveform design, coupling to motion, and the underlying sensory-conflict geometry (Allred et al., 2025; Groth et al., 2022; Qi et al., 2021).

An additional point that strengthens interpretation of the present PU findings is that the OCHART measure showed no systematic drift between the two pre-exposure sessions. The absence of a main effect of session and of a session by background interaction indicates that the overall PU function was reproducible across days at the group level. This is important because it suggests that the null EVS effect on PU is unlikely to reflect gross instability of the baseline measure itself. Rather, OCHART appears to provide a reasonably stable behavioural estimate of perceived upright across repeated testing in this paradigm.

This interpretation is broadly consistent with previous work showing that perceptual upright is a valid multisensory estimate derived from combined body, visual, and gravitational cues. Dyde et al. (2006) showed that PU, as measured with OCHART, is influenced by all three cues and can be described by a weighted vectorial sum, with PU tending to align more strongly with the body axis than the subjective visual vertical. Earlier work by Jenkin et al. (2004) similarly showed that perceived up varies systematically when visual, body, and gravitational frames are dissociated, and that a weighted-sum model predicts both the direction and variability of this estimate. The present repeated-baseline findings extend that logic by showing that, despite substantial inter-individual variability, OCHART-derived PU remains sufficiently stable across days to support its use as a repeated measure in studies of sensory conflict and reweighting.

### 4.5.3 Cue Weights and Sensory Reweighting

Model-derived cue weights provided no evidence that EVS promoted adaptive sensory reweighting in this paradigm. Vector sums did not differ from pre-VR to post-VR in either condition, and post-VR vector sums did not differ between EVS and no-EVS. Individual cue-weight changes were small on average, highly variable, and not correlated with cybersickness severity. These outcomes suggest that, at least as measured by post-exposure OCHART and the weighted vector-sum model, EVS did not systematically alter the balance of visual, body, and gravitational contributions to the upright estimate.

At the same time, the cue-weight nulls should be interpreted with two important constraints. First, the feasibility limits of the vector-sum model were substantial in this dataset. Nearly half of the recruited sample was excluded from cue-weight analyses due to degeneracy in which minimal sensitivity to the 90° body tilt led to implausibly large fitted body weights. This is not merely a statistical inconvenience. It implies that, for a sizeable subset of participants, the experimental design did not elicit a tilt effect that reliably separates body and gravitational contributions, undermining identifiability of individual cue weights. The resulting cue-weight estimates therefore characterize a selected subset with measurable tilt sensitivity and stable psychometric structure, rather than the full population exposed to EVS. Second, early termination under EVS likely introduced informative missingness, because participants who were most affected by stimulation contributed less VR exposure and fewer usable trials. This can reduce the probability of observing stable post-exposure perceptual changes, even if online changes occurred during exposure.

Sensory reweighting is not instantaneous, and dynamics can differ depending on whether reliabilities increase or decrease and on whether reweighting occurs within a modality or across modalities (Assländer & Peterka, 2014; Peterka, 2002). In visual-vestibular heading tasks, cue weights can update rapidly on a trial-by-trial basis as reliabilities vary, but observers also show stable idiosyncrasies and deviations from optimality that can mask simple group-level signatures (Fetsch et al., 2009). These principles imply that EVS could disrupt online inference during the most provocative phases of VR without producing a consistent post-exposure shift detectable in a subsequent task performed outside the VR environment.

Critically, the stimulation amplitude used here likely placed the protocol outside the stochastic facilitation regime that motivates the “noise improves function” hypothesis. Stochastic resonance and related stochastic facilitation effects are expected when noise is subthreshold and tuned to a regime where it improves detectability or stabilizes weak signals. Across vestibular stochastic stimulation studies, optimal amplitudes often fall in the low hundreds of microamps and are frequently calibrated relative to perceptual threshold (Galvan-Garza et al., 2018; Goel et al., 2015; Mulavara et al., 2011; Wuehr et al., 2018). At higher amplitudes, benefits can become inconsistent or can reflect alternative processes such as stiffening responses rather than genuine sensory facilitation, and in young healthy adults SR-like effects are not always observed even when amplitudes are varied (Assländer et al., 2021; McDonnell & Ward, 2011). In the present study, the combination of suprathreshold bilateral stimulation and intense VR conflict is therefore more consistent with a perturbation account than a facilitation account. This perspective is compatible with evidence that GVS can aggravate cybersickness and delay habituation across repeated exposures in visually induced ship-motion VR (Qi et al., 2021), and with demonstrations

that vestibular stimulation can bidirectionally modulate sickness depending on whether the waveform is matched to a sensory-conflict target signal (Allred et al., 2025).

#### 4.5.3.1 Excluded Model Fits and Extreme Cue Weights

Twelve participants were excluded from the vector sum analysis because the optimisation converged on implausibly large body cue weights. In this framework, separating body and gravity contributions depends on the tilt manipulation changing the body vector relative to gravity. When PU shows little or no shift with body tilt, the model can only fit the pattern by letting one term, typically the body term, absorb variance in an extreme way (Mittelstaedt, 1983; Dyde et al., 2006). In this situation, the fitted body weight is not interpretable as a literal estimate of sensory reliability. It mainly indicates that the data do not contain enough information to uniquely identify the body and gravity parameters under an obligatory-integration, fixed-weight model.

There are at least three non-exclusive explanations. First, this may reflect measurement instability. Elevated trial noise, lapses, fatigue, or reduced compliance can compress condition differences and attenuate tilt effects, particularly when participants are uncomfortable. Second, it may reflect a real individual difference in task strategy. OCHART is a categorical upright judgement and may allow solutions that rely more on body-referenced heuristics or scene-based anchors than on graviceptive information, especially when vestibular reliability is uncertain. Training history could plausibly contribute to this. For example, dancers and gymnasts often show reduced visual field dependence and stronger reference-frame switching in rod-and-frame style tasks, although expertise effects are task dependent and not uniformly protective across verticality measures (Beck et al., 2020). Third, the pattern may reflect a change in inferred causal structure rather than a smooth redistribution of weights. Causal inference accounts predict that when cues are unlikely to share a common cause, the nervous system can move toward partial integration or segregation, which can appear as tilt insensitivity within a vector-sum parameterization even if body cue reliability has not increased (Körding et al., 2007; de Winkel et al., 2021).

Critically, these exclusions likely identify a subgroup for whom the current measurement and modelling pipeline does not yield interpretable estimates, and that subgroup may be mechanistically important. Weak tilt sensitivity could reflect reduced responsiveness to gravito-body conflict in this task, a regime switch under conflict, or elevated response variability that obscures post-exposure effects. Each possibility is relevant to cybersickness heterogeneity. Accordingly, future work should treat tilt sensitivity and model identifiability as outcomes rather than only exclusion criteria. Concretely, this includes reporting a model-free upright-to-right-side-down PU shift and testing its relation to sickness and

tolerance, recording training history and a brief visual dependence screen, and adopting modelling approaches that can accommodate partial integration and cue segregation rather than forcing a single fixed-weight solution (Acerbi et al., 2014; Clemens et al., 2011; Körding et al., 2007).

The vector sum model assumes obligatory integration of all available cues (forced fusion). However, de Winkel and colleagues (2018) directly compared forced fusion, cue capture, and Bayesian causal inference models for the perception of verticality and found that causal inference outperformed forced fusion, demonstrating that participants can and do segregate cues when conflict is large. In the causal inference framework, the nervous system infers whether sensory signals share a common cause and, when they likely do not, downweights or discards the discrepant signal rather than integrating it linearly. Participants whose vector sum fits were degenerate in the present study may have been operating in a segregation regime, where the 90-degree body tilt was treated as arising from a different source than the visual background, rendering the tilt manipulation ineffective for separating body and gravity parameters. If so, these participants are not simply "noisy" but may represent a mechanistically informative subgroup whose perceptual strategy cannot be captured by a fixed-weight model.

The model-free tilt sensitivity analysis (Section 4.4.8) provided partial insight into the excluded participants. Contrary to the expectation that excluded participants would show uniformly low tilt sensitivity, their median tilt sensitivity in the no-EVS condition (Mdn = 15.86°) did not differ significantly from the retained group (Mdn = 16.58°; $p = 0.635$). This suggests that at least some excluded participants exhibited meaningful PU shifts with body tilt that the vector sum model nevertheless failed to capture, possibly because tilt sensitivity was inconsistent across visual backgrounds or because the relationship between PU and background orientation did not conform to the sinusoidal pattern the model assumes.

### 4.5.4 Sex Differences

Sex differences in cybersickness remain a contested topic, and the present study was not designed or powered to adjudicate these effects. Nonetheless, it is important to situate the Chapter 4 findings within this literature because sex-linked variability can influence both cybersickness severity and the apparent effectiveness of countermeasures. Meta-analytic and review evidence is mixed. Some syntheses report no reliable sex difference in cybersickness severity across modern HMD studies (Saredakis et al., 2020; Grassini & Laumann, 2020), while other meta-analytic work concludes that women report greater sickness on average (Howard & Van Zandt, 2021). Recent large-sample studies further suggest that women may terminate VR exposure more often and report higher symptom intensity, although the

mechanisms remain unresolved and may be moderated by susceptibility history and task characteristics (Kelly et al., 2023; Munafo et al., 2017).

One mechanistic hypothesis is that apparent sex differences can be driven by hardware fit rather than intrinsic biological susceptibility. Interpupillary distance mismatch has been proposed as a key contributor, with evidence that non-fit to a headset's interpupillary distance (IPD) range can disproportionately affect women and can increase symptom burden and prolong recovery (Stanney et al., 2020). However, other work has not found a straightforward relationship between IPD mismatch and immediate cybersickness, indicating that IPD is not a universal explanation and may interact with additional factors such as prior screen sickness and susceptibility (Doty et al., 2023). Hormonal influences represent another plausible contributor. Both classical and recent studies report fluctuations in motion sickness and simulation sickness susceptibility across menstrual cycle phases in naturally cycling women, although effects are inconsistent and adaptation with repeated exposure can be substantial (Bannigan et al., 2024; Clemes & Howarth, 2005; Matchock et al., 2008). Finally, susceptibility questionnaires that target visually induced motion sickness-like symptoms tend to show higher average scores in females, though effect sizes are often weak to moderate and are unlikely to fully explain individual outcomes (Keshavarz et al., 2023).

Sex-linked variability is a plausible contributor to heterogeneity in cybersickness and tolerance outcomes, but it cannot be separated from hardware fit, prior susceptibility, and adaptation history without targeted measurement. The sample included both males and females, but sex was not incorporated as a planned factor in the inferential analyses, and key moderators such as IPD fit, menstrual cycle phase, and baseline susceptibility were not measured. Therefore, the present results should be interpreted as establishing a robust within-subject behavioural effect of EVS under these parameters, while leaving open whether sex-linked factors moderate the magnitude of that effect. A concrete future direction is to include sex as a planned moderator with adequate power, and to measure candidate mediators such as baseline susceptibility (for example MSSQ scores), IPD and headset fit metrics, and exposure history, to determine whether any observed sex differences are primarily ergonomic, hormonal, or mechanistic (Keshavarz et al., 2023; Stanney et al., 2020).

### 4.5.5 Limitations and Future Directions

A primary limitation of this study is that the EVS protocol was not designed to sample the parameter space where stochastic facilitation is most likely to occur. Stochastic resonance predicts an inverted-U relationship between noise intensity and performance, where subthreshold noise can enhance processing

but suprathreshold noise degrades it (McDonnell & Abbott, 2009; Moss et al., 2004). In vestibular applications, benefits of noisy GVS are most often reported at low, individually optimised amplitudes in the few-hundred-microamp range (roughly 100–500 μA), frequently defined relative to perceptual threshold (Goel et al., 2015; Mulavara et al., 2011; Wuehr et al., 2018). Here, EVS was delivered at a fixed amplitude (±2500 μA), which likely placed many participants outside any facilitative window and into a perturbation regime. This aligns with evidence that SR-like effects are not reliably observed in young healthy samples and that higher intensities can reflect compensatory stiffening rather than improved vestibular inference (Assländer et al., 2021; McDonnell & Ward, 2011). A clear next step is threshold assessment and within-participant amplitude titration, with a priori tests for inverted-U behaviour and responder subgroups (Galvan-Garza et al., 2018; Goel et al., 2015).

A second limitation is the ecological validity of the vestibular signal under GVS. Bilateral mastoid stimulation drives a non-physiological afferent pattern with strong dependence on afferent class and nonlinearities that can introduce biases rather than unstructured noise (Fitzpatrick & Day, 2004; Goldberg et al., 2012). GVS also engages cortical vestibular networks and can reduce visual cortical activity, consistent with reciprocal visual–vestibular interactions. In an optic-flow-dominant VR context, suprathreshold vestibular drive may therefore add competing signals rather than resolve conflict. This motivates motion-matched or adaptive stimulation that is coupled to camera motion or ego-motion, which has been associated with reduced discomfort in VR and simulator contexts (Cevette et al., 2012; Groth et al., 2022; Pradhan et al., 2022). Computational waveform design provides a parallel route to targeting the predicted conflict signal, with recent work demonstrating bidirectional sickness effects using designed beneficial versus detrimental waveforms (Allred et al., 2025; Lonner et al., 2025). An important future direction is therefore to dissociate reliability-degradation effects from added-conflict effects by directly comparing individually thresholded nGVS, fixed suprathreshold stochastic stimulation, and motion-coupled stimulation that is synchronised to the visual motion trajectory.

A third limitation is that the primary outcomes index post-exposure perceptual status. Reweighting dynamics can be time dependent, and the effects that matter for cybersickness may occur during exposure without persisting as stable aftereffects (Assländer & Peterka, 2014, 2016; Peterka, 2002). This is compounded by differential exposure duration. EVS increased early termination, and termination tracked symptom severity, so the most affected participants contributed the least exposure time and the least usable psychometric data. This informative missingness reduces sensitivity to detect post-exposure change and complicates interpretation of null PU and cue-weight effects. Future designs should treat tolerance time as a primary endpoint and use analyses that model time-to-termination.

A fourth limitation concerns model identifiability. Twelve participants were excluded from vector-sum modelling because optimisation converged on implausibly large body cue weights, most consistently when PU showed minimal sensitivity to the 90° body-tilt manipulation. In these cases, fitted weights are not interpretable as sensory reliabilities, and the exclusion pattern indicates that the current pipeline does not yield identifiable estimates for a sizeable subgroup. This subgroup may reflect noise and compliance limits, but it could also reflect differences in reference-frame strategy linked to training history or regime shifts such as partial integration or cue segregation under conflict (Körding et al., 2007; de Winkel et al., 2021). Future work should record training history, include a brief visual dependence screen, and use modelling approaches that can accommodate segregation rather than forcing a single fixed-weight solution (Acerbi et al., 2014; Clemens et al., 2011).

Finally, the study was not powered to test key moderators of cybersickness, including sex-linked variability and headset fit. Given mixed evidence for sex differences and evidence that ergonomic factors such as interpupillary distance mismatch can drive apparent sex effects, future EVS work should measure these factors directly and treat them as planned moderators (Grassini & Laumann, 2020; M. C. Howard & Van Zandt, 2021; Saredakis et al., 2020; Stanney, Lawson, et al., 2020).

## 4.6 Conclusion

This chapter tested whether bilateral stochastic EVS delivered during a highly provocative VR exposure would reduce cybersickness and promote adaptive sensory reweighting, as indexed by post-exposure perceptual upright and model-derived cue weights. The behavioural outcome was clear and opposite to the mitigation hypothesis. EVS increased cybersickness severity and reduced VR tolerance, indicating that under the present parameters stimulation acted as an additional perturbation rather than a countermeasure. In contrast, perceptual upright remained strongly modulated by visual background orientation, and EVS produced no detectable main effect on PU and no interaction with background at the post-exposure time point. Model-derived cue weights likewise showed no consistent EVS effect within the subset of participants whose data supported identifiable fits.

A second contribution of this chapter is that the modelling failures were systematic and potentially informative. Twelve participants produced degenerate vector-sum solutions characterized by extreme body cue weights, most consistently when PU showed minimal sensitivity to the body-tilt manipulation. This constrains strong claims about individual cue weights from OCHART in small samples, but it also highlights a subgroup that may reflect differences in task strategy, reference-frame selection, or integration versus segregation regimes. Because training history can shift visual dependence and

reference-frame flexibility, and because expertise effects can be task-specific rather than uniformly protective, the “tilt-insensitive” phenotype warrants targeted examination rather than being treated only as exclusion noise (Beck et al., 2020).

Taken together, these findings support a parameter-sensitive interpretation of vestibular stimulation in VR. Stochastic vestibular stimulation may be facilitative under specific subthreshold regimes and when aligned with the user’s motion and conflict geometry, but suprathreshold, motion-uncoupled stimulation can worsen cybersickness and reduce tolerance. Future work should prioritize amplitude titration relative to perceptual threshold, motion-attuned or computationally designed waveforms, online measurement of adaptation during exposure, and modelling frameworks that address identifiability and allow cue segregation when causal structure is uncertain (Allred et al., 2025; Körding et al., 2007; Lonner et al., 2025; Moss et al., 2004).

# Chapter 5
# General Discussion

Cybersickness remains a practical barrier to VR adoption even in current generation head mounted displays. Meta analytic and systematic review evidence indicates that symptom severity has decreased on average with newer hardware, but that clinically meaningful nausea, disorientation, and oculomotor disturbance remain common and are still strongly linked to sensory mismatch and perceived self motion (Caserman et al., 2021; Saredakis et al., 2020; Tian et al., 2022). This persistence motivates a mechanistic focus on individual susceptibility, because group mean symptom levels can mask large between participant differences in how the nervous system detects and resolves multisensory conflict (Biswas et al., 2024; Tian et al., 2022). In this dissertation, upright perception is treated as a tractable behavioural window into that problem, because it provides measurable outputs that can be decomposed into bias, precision, and model based cue influence.

## 5.1 Summary of Findings

The central aim of this dissertation was to characterize how the nervous system resolves persistent multisensory conflicts in immersive virtual reality, and to test whether individual differences in this adaptive process help explain susceptibility to cybersickness. Across three studies, I used perceptual measures of upright as a tractable read-out of cue weighting, focusing on how visual, gravitational (vestibular), and body-based cues are integrated and updated under conflict. This approach treats upright perception as a practical window into broader multisensory inference, because it requires the nervous system to combine partially redundant signals with different reliabilities and different noise structures, and to adjust their relative influence when environmental statistics change.

In this dissertation, reweighting capacity is not treated as a purely fixed trait, nor as a fully transient state. A more defensible interpretation is that cybersickness susceptibility reflects an interaction between trait-like baseline sensory tendencies and state-dependent updating processes. Baseline cue preferences, prior motion sickness history, and sensorimotor experience likely shape the starting point from which an individual enters VR, but the more mechanistically relevant factor may be how efficiently those cue weights can be updated when conflict persists. In that sense, susceptibility is best viewed as a dynamic property of the observer-environment system rather than as a single immutable sensory style.

In Experiment 1, I hypothesized that larger changes in SVV following intense VR exposure would be associated with lower cybersickness scores, and that the relative weighting of visual and body cues would decrease while the weighting of gravitational cues would increase. The results partially supported these predictions. A significant main effect of Condition confirmed that SVV shifted toward the gravitational upright after VR ($F(1, 18) = 8.07$, $p = 0.012$, partial $\eta^2 = 0.34$), replicating the direction of the effect reported by Chung and Barnett-Cowan (2023). Bootstrap confidence intervals indicated a reliable decrease in body cue weighting and increase in visual cue weighting, though the overall Condition × Cue Weight interaction approached but did not reach conventional significance ($p = 0.053$). The hypothesized link between reweighting magnitude and cybersickness severity was best supported for the grey background SVV condition, where a Gaussian relationship indicated that participants with minimal SVV change reported the highest sickness. However, this relationship was inconsistent across visual background conditions, and the cue weight to cybersickness associations were weaker than anticipated.

In Experiment 2, I hypothesized that OCHART-derived PU would shift following VR exposure and that larger PU shifts would predict lower cybersickness. Results confirmed that OCHART was sensitive to changes in visual background orientation and body posture, with a significant main effect of Background on PU. However, the hypothesized PU to cybersickness relationship was not reliably supported; rank-based correlations between PU change and FMS scores were non-significant for both grey and room background conditions. The Gaussian model did not outperform an intercept-only model. These findings indicate that while OCHART can detect sensory weighting changes, PU shifts alone are an incomplete predictor of cybersickness severity at the individual level.

In the pooled OCHART analysis combining Experiment 2 and Experiment 3 no-EVS participants ($N = 36$), the clearest candidate individual-differences signal was a positive association between post-VR change in the model-derived vision component and cybersickness severity (Spearman $\rho = 0.37$, $p = 0.027$). This relationship was better described by a linear term than by a Gaussian form in this pooled dataset ($R^2 = 0.120$), but the effect size was modest and should be treated as a hypothesis-generating pattern rather than confirmatory evidence about functional form. Two constraints are particularly important. First, because vector-sum cue weights are compositional, an apparent increase in vision necessarily implies a relative decrease in at least one other component, so the interpretation is about redistribution within the model rather than an isolated change in a single cue. Second, the association differed across the contributing study samples and did not remain robust when study group was included as a covariate. Accordingly, the safest interpretation is that increased post-VR visual reliance is a plausible marker of maladaptive adaptation under conflict in some contexts, but it requires preregistered replication in a larger, more uniform dataset before being used to motivate strong mechanistic claims.

Framing the problem as upright estimation also provides a concrete bridge to computational motion-sickness models grounded in Subjective Vertical Conflict (SVC) theory. Contemporary SVC-based models treat sickness risk as arising from persistent disagreement between an internal estimate of vertical and the gravito-inertial context, with recent analyses explicitly evaluating how 6DOF formulations can unify self-motion perception and sickness predictions within a single dynamical framework (Wada & Bos, 2025). In parallel, recent modelling work has begun moving beyond incidence-style outcomes toward predicting individual symptom progression by fitting model parameters to each participant's early symptom history, improving prediction of later symptom trajectories (Inoue et al., 2025). In this dissertation, SVV and PU are not treated as direct readouts of a single latent model state. Instead, upright tasks are used to operationalize conflict resolution as a measurable perceptual computation that can, in principle, provide independent constraints on the vertical-estimation processes that SVC-style models assume. A practical next step is therefore to combine upright-task measures with within-exposure symptom time courses, enabling explicit tests of whether individuals who show particular post-VR re-anchoring patterns also show different fitted conflict dynamics or symptom progression parameters.

In Experiment 3, I hypothesized that stochastic EVS applied during VR would reduce cybersickness by degrading vestibular reliability and promoting adaptive sensory reweighting. The behavioural results were clear in direction but contrary to prediction. EVS significantly increased cybersickness severity (no-EVS Mdn = 9.0 vs EVS Mdn = 14.5; $W = 12.5$, $p = 0.001$, $r = 0.67$) and increased the rate of early task termination (6 vs 2 participants; $W = 1.0$, $p = 0.046$). FMS was strongly correlated with task duration in the EVS condition ($\rho = -0.758$, $p < 0.001$), indicating that EVS worsened the overall sickness experience rather than selectively affecting one outcome. Despite this behavioural effect, EVS did not produce a reliable group-level change in perceptual upright or overall cue weighting, and model-based cue weight estimation was not stable for a subset of participants. Taken together, Experiment 3 constrains any simple claim that vestibular noise reduces cybersickness and instead supports a parameter-sensitive interpretation in which vestibular stimulation can either attenuate or exacerbate conflict depending on waveform, intensity, and VR context.

## 5.2 Convergence and Divergence Across Studies

Across the three studies, a consistent theme emerged: cybersickness vulnerability is not well captured by group-average perceptual shifts alone, but is more clearly expressed in the interaction between baseline sensory tendencies and the way individuals adapt, or fail to adapt, their cue weighting under conflict. Experiments 1 and 2 converged in showing that upright-related judgements in VR are shaped by multisensory cue interactions, but they differed in the strength of post-exposure change detected at the

group level. Experiment 1 showed a clear post-VR shift in SVV toward gravitational upright, consistent with reduced body weighting and greater reliance on gravity-related vestibular information. Experiment 2 showed robust modulation of perceptual upright by visual background orientation, confirming that OCHART is sensitive to visual contextual influences, but it did not show a reliable group-level pre to post shift in PU. In pooled exploratory analyses, however, the direction of OCHART-derived cue weight change was more consistent with Experiment 1, suggesting modest body down-weighting and gravity up-weighting after VR. These complementary findings support the broader conclusion that the CNS actively recalibrates its internal model of upright in response to VR-induced conflict, consistent with Bayesian models of multisensory integration (Ernst & Banks, 2002; Fetsch et al., 2013).

However, the studies differed in how strongly sensory reweighting related to cybersickness. Study 1 provided the clearest evidence for this relationship, particularly in the grey background condition, whereas Experiment 2 yielded weaker and less consistent associations. One likely explanation is that the SVV and OCHART tap into partly different aspects of orientation perception. The SVV is heavily gravity-dominated, with typical weighting ratios of approximately 77% gravity, 15% body, and 8% vision (Dyde et al., 2006), making it sensitive to changes in graviceptive and body weighting. OCHART, by contrast, distributes cue influence more evenly (approximately 21% gravity, 54% body, 25% vision; Dyde et al., 2006; Harris et al., 2022), making it more responsive to visual manipulations but potentially less direct as a readout of the vestibular to body weighting shift that appears to drive the cybersickness relationship. This dissociation is informative because it suggests that the protective reweighting mechanism described by Chung and Barnett-Cowan (2023) may be easiest to detect when the perceptual task preferentially reflects the cues most disrupted by VR exposure (vestibular relative to body), rather than when visual, body, and graviceptive cues contribute more evenly to the upright estimate. This task-dependent sensitivity is further supported by recent work examining visual and vestibular reweighting after cybersickness and space sickness. Bonnard and colleagues (2026) found that VR exposure specifically decreased VOR gain (reflecting reduced vestibular reliance) while parabolic flight reduced optokinetic nystagmus (reflecting reduced visual reliance), demonstrating that the modality downweighted depends on which cue fails to detect actual displacement. Because VR generates visual motion without matching inertial cues, the SVV, which is heavily vestibular-dominated, may be more sensitive to the specific body-to-gravity reweighting that VR conflict provokes, whereas the PU distributes cue influence more evenly and may require stronger or more prolonged conflict to produce detectable shifts (Bonnard et al., 2026).

This task dependence is also consistent with Bayesian causal inference accounts, in which the nervous system does not simply reweight cues continuously, but also arbitrates between integrating and

segregating cues based on whether they are inferred to share a common cause. Under sustained VR conflict, visual motion can be highly informative yet simultaneously less trustworthy for self-motion, so the optimal strategy may vary by computation and task demands rather than producing a single uniform reweighting signature. This interpretation also helps clarify that the relevant individual difference may not be a fixed dominance profile alone, but the flexibility with which that profile can be updated across tasks, contexts, and repeated exposures. This framing helps explain why SVV-based effects can be cleaner and more predictive in some contexts, while PU-based effects can be weaker, less stable, or expressed primarily in model-derived components rather than in raw perceptual shifts.

Experiment 3 introduced an important contradiction. Rather than reducing cybersickness, EVS increased it. This finding challenges the straightforward prediction that degrading vestibular reliability should attenuate visuo-vestibular conflict. Several factors may explain this discrepancy. First, the EVS parameters (continuous stochastic bilateral stimulation) may have introduced perceptual disturbance that itself contributed to nausea, rather than simply reducing vestibular precision as intended (Qi et al., 2021). Second, the VR content differed across studies: Studies 1 and 2 used Echo VR (an active, multiplayer game with user-initiated motion), whereas Experiment 3 used ADR1FT (a passive exploration game with controller-driven translation). The degree of user agency has been shown to modulate cybersickness (Stanney et al., 2020), and the combination of reduced control and vestibular perturbation may have compounded rather than attenuated conflict. Third, recent work by Allred et al. (2025) demonstrated that the effectiveness of GVS for mitigating motion sickness depends critically on the computational design of the stimulation waveform relative to the expected sensory conflict; "beneficial" GVS reduced sickness by 26%, while poorly matched "detrimental" GVS increased it by 56%. This suggests that the stochastic EVS used in the present study may not have been optimally configured to produce the intended reweighting effect (Allred et al., 2025).

The direction of the EVS effect is also consistent with a broader literature showing that vestibular stimulation in VR is highly parameter- and context-sensitive, including the possibility of sign reversals. Noisy-current GVS has been reported to reduce cybersickness during intense VR exposure and immediately after stimulation, but these benefits can extinguish within minutes once stimulation ends and exposure continues, implying limited persistence and strong dependence on stimulation timing and task context (Weech et al., 2020). In contrast, other protocols show that vestibular stimulation can aggravate balance disturbance and can delay habituation to repeated VR exposures, suggesting that added vestibular perturbation can interfere with adaptation rather than facilitate it (Qi et al., 2021). More recently, computationally designed GVS waveforms have been shown to reduce motion sickness when tuned to the predicted sensory-conflict signal, while an equal-and-opposite “detrimental” waveform increased

symptoms substantially, demonstrating that waveform design can push the system toward mitigation or exacerbation under otherwise matched conditions (Allred et al., 2025). In VR specifically, motion-attuned GVS that is explicitly coupled to displayed camera motion has been reported to reduce discomfort in moving-camera content, consistent with the idea that stimulation may be most effective when it increases multisensory congruence rather than adding generic vestibular noise (Groth et al., 2022). Together, these findings support a conservative interpretation of Experiment 3: the observed increase in cybersickness under stochastic EVS is not anomalous, but instead reflects the fact that vestibular stimulation can worsen conflict when its statistics or timing are mismatched to the VR motion context.

An important interpretive constraint is that the upright measures in this dissertation primarily index post-exposure aftereffects, not the ongoing computations that unfold while symptoms accumulate during VR. Post-VR shifts in SVV or PU can reflect a mixture of sensory recalibration, re-anchoring to gravity after conflict, and changes in how participants interpret task-relevant cues once the VR scene is removed. These aftereffects are informative, but they are not guaranteed to covary with the within-exposure processes that drive nausea and disorientation in real time. This distinction provides one plausible explanation for why robust group-level changes in upright can coexist with weak or conditional relationships to cybersickness severity across studies.

Recent work on Cybersickness Abatement from Repeated Exposure (CARE) reinforces that cybersickness is strongly shaped by adaptation over repeated sessions and by the structure of exposure, not only by the size of a single post-VR aftereffect. CARE can be induced while reducing discomfort during training, and its benefits can generalize across distinct VR experiences, suggesting that adaptation involves learning and context generalization rather than a single stimulus-bound recalibration process (Doty et al., 2025; Kelly et al., 2026). In this framework, post-VR upright shifts can be treated as one behavioural readout of how an individual's internal model changes after conflict, but future work should also quantify the time course of sickness and cue use during exposure to better dissociate online regulation from post-exposure aftereffects.

## 5.3 Individual Variability

### 5.3.1 Individual Variability in Sensory Reweighting and Cybersickness

A consistent finding across all three studies was the substantial inter-individual variability in both perceptual outcomes and cybersickness severity. SVV shifts ranged from near zero to more than 10 degrees, and cue-weight changes varied markedly in both direction and magnitude. This heterogeneity is

not unique to the present work, it is a well-documented feature of the cybersickness literature (Garrido et al., 2022; Kourtesis et al., 2024; Tian et al., 2022). The present dissertation adds a more mechanistic lens for interpreting this variability by treating it as, at least in part, an expression of individual differences in the capacity to reweight sensory cues under conflict.

These findings suggest that the relationship between sensory reweighting and cybersickness is plausible, but more conditional and task dependent than originally framed. Across studies, the strongest evidence emerged when the perceptual measure most directly indexed the specific conflict VR disrupts, whereas measures with stronger visual dominance, or those requiring more complex model fitting, yielded weaker or less stable links to symptom severity. This pattern is consistent with broader work showing that cybersickness is heterogeneous in both magnitude and underlying drivers, and that single behavioural markers rarely explain a large proportion of variance on their own (Garrido et al., 2022; Saredakis et al., 2020).

A further reason that single reweighting metrics can show modest predictive power is that cybersickness is not a unitary outcome. Recent work using latent profile analysis has identified subgroups with different relative patterns of nausea, oculomotor disturbance, and disorientation, even when overall severity differs in expected ways (Kelly et al., 2024). This matters for interpretation because upright tasks and cue-weight estimates are plausibly more proximal to disorientation and spatial-orientation symptoms than to nausea-specific pathways in some individuals. Accordingly, the same change in post-VR upright could map onto different symptom compositions across participants, which would weaken associations when severity is summarized with a single global score.

First, cybersickness susceptibility is multifactorial, and several established moderators were not controlled or stratified in the present datasets. Recent synthesis work indicates that pre-existing motion sickness susceptibility is one of the most consistent predictors of cybersickness outcomes, and prior interactive media or gaming experience can moderate symptom severity and adaptation rates (Kourtesis et al., 2024; Saredakis et al., 2020). Sex and age effects are also reported in the literature, but their interpretation is complicated by hardware and design factors. In particular, headset fit and interpupillary distance mismatch can differentially affect comfort and visual-vestibular strain, which can masquerade as biological sex differences if not explicitly measured and adjusted (Stanney et al., 2020). Meta-analytic evidence suggests that demographic effects are variable across studies and may depend on task demands, exposure duration, and measurement choices (Saredakis et al., 2020). With sample sizes of approximately 18 to 20 participants, the present work was not powered to test interactions between reweighting indices

and these moderators, so null or weak associations should be interpreted as inconclusive rather than definitive evidence of absence.

Second, treating "sensory dominance" as a stable trait may be an oversimplification. The present findings are more consistent with a hybrid account in which baseline cue preferences are partly trait-like, but the capacity to update those preferences under conflict is state-dependent and likely modifiable with experience. Screening-style measures of visual dependence, such as rod-and-frame performance, show context-dependent and generally modest relationships with cybersickness, indicating that baseline cue preferences alone are unlikely to be sufficient predictors (Josupeit, 2024). At the same time, repeated exposure, training history, and altered sensory environments can shift verticality judgements and perceptual upright, demonstrating that cue use is not fixed and may change meaningfully over time (Beck et al., 2020; Harris et al., 2017, 2022).

A useful way to reconcile these findings is to distinguish three levels of analysis. The first is a baseline sensory profile, which may reflect relatively stable tendencies such as greater visual dependence, stronger body anchoring, prior susceptibility, or training history. The second is online state-dependent reweighting during VR exposure, which may determine whether conflict is resolved or amplified in the moment. The third is longer-term adaptation across sessions, where repeated exposure may train more efficient conflict resolution rather than simply revealing an intrinsic trait. In this framework, the most important individual difference is not whether a participant is inherently "visual dominant" or "body dominant," but whether they can shift cue influence appropriately when the reliability structure of the environment changes.

This distinction also helps separate conflict detection from conflict resolution. Some participants may be more sensitive to mismatch because they process specific visual cues with higher fidelity. For example, sensitivity to motion parallax cues predicts motion sickness severity in VR, consistent with the idea that more precise cue processing can act as a risk amplifier under conflict (Fulvio et al., 2021). At the same time, state-based measures collected during exposure may capture emerging sickness more directly than static pre-screening measures. Natural gaze behaviour during HMD VR can predict cybersickness risk, and head rotation features combined with field dependence measures can support predictive pipelines, even if prediction does not uniquely identify mechanism (Chang et al., 2021). These findings motivate future work that links in-exposure behavioural predictors back to mechanistically interpretable outcomes, such as upright-task cue influence and within-person change after conflict.

Finally, the feasibility limits of cue-weight modelling are themselves an important finding that constrains interpretation. In Experiment 3, reliable parameter estimation was not achievable for a substantial subset

of participants, which implies that modelling-based inferences about individual reweighting must be bounded by data quality, identifiability, and the stability of psychometric fits. In practice, this means that the reweighting-cybersickness link should be framed as strongest when both the perceptual task and the modelling pipeline provide stable individual estimates, and more tentative when additional estimation layers introduce variance that can overwhelm true effects.

### 5.3.2 Generalizability and Sample Representativeness

The samples recruited for all three studies were drawn from the University of Waterloo undergraduate and graduate student population, with mean ages between 21.7 and 23.2 years and predominantly healthy adults with no reported vestibular or neurological disorders. This recruitment approach is broadly consistent with the cybersickness literature, which relies heavily on young adult convenience samples and often prioritizes experimental control over population representativeness (Caserman et al., 2021; Saredakis et al., 2020; Tian et al., 2022). However, because cybersickness susceptibility and multisensory weighting vary systematically with participant characteristics and prior experience, the extent to which our findings generalize beyond this sampling frame requires careful qualification (Biswas et al., 2024; Saredakis et al., 2020).

Age-related patterns in cybersickness are not uniform across studies, but the available evidence suggests that younger adults frequently report higher cybersickness than older adults under comparable exposures, while older-adult VR studies often report lower symptom severity (Dilanchian et al., 2021; Saredakis et al., 2020). This matters for interpretation because our restricted age range may inflate apparent between-participant variability while simultaneously limiting our ability to test age as a moderator of reweighting. In addition, university samples can over-represent individuals with at least moderate gaming or VR familiarity, which is often associated with reduced cybersickness through habituation or adaptation, and may compress the upper tail of sickness severity (Biswas et al., 2024; Saredakis et al., 2020). As a result, the observed reweighting-to-cybersickness associations may reflect a narrower portion of the broader susceptibility distribution than would be expected in more heterogeneous community samples.

A second representativeness issue is that student samples may underrepresent individuals with extreme or specialized sensory calibration. For example, expert dancers show measurable differences in verticality perception and somatosensory contributions relative to non-experts, consistent with the idea that intensive training can alter baseline cue use (Beck et al., 2020). Importantly, these baseline sensory profiles do not necessarily imply reduced flexibility. A participant can be relatively body-dominant at baseline yet still exhibit rapid context-dependent reweighting, which is precisely the construct this dissertation aimed to

interrogate. Therefore, the restricted range of baseline sensory dominance likely limited our ability to observe how baseline profiles moderate reweighting dynamics and cybersickness outcomes.

Future work would benefit from recruiting stratified samples that include individuals with known differences in sensory dominance and sensorimotor training histories (e.g., athletes, dancers), as well as broader age ranges, to test whether the reweighting–cybersickness relationship is moderated by baseline cue profiles and adaptation capacity. More broadly, work comparing altered-gravity analogues shows that the specific sensory context can qualitatively change spatial-orientation outcomes. For instance, neutral buoyancy does not fully reproduce microgravity-related perceptual effects relevant to self-motion and orientation, and perceptual upright measures themselves can shift across analogue environments and task contexts (Bury et al., 2023; H. Jenkin et al., 2023). This reinforces that "normative" cue-weight patterns depend on both the population and the sensory ecology in which upright perception is probed.

## 5.4 Limitations

A recurring challenge across this dissertation was the feasibility and interpretability of applying the weighted vector sum model to estimate cue weights at the individual participant level. The model is well suited for describing condition-level and group-level shifts in perceived upright, but individual-level estimation can be unstable when the underlying PU or SVV functions are noisy or weakly modulated across visual backgrounds (Mittelstaedt, 1983; Dyde et al., 2006).

This framework has been widely applied to characterise how gravity, body, and visual cues contribute to upright perception across contexts, including tilted-body paradigms and altered graviceptive environments, and it remains a useful descriptive baseline for comparing conditions (Dyde et al., 2006; Jenkin et al., 2023). This failure rate reduced the effective sample available for cue-weight inference and may have introduced selection bias, insofar as participants with cleaner psychometric functions and stronger visual modulation are more likely to yield stable parameter estimates. Accordingly, cue-weight results in Experiment 3 should be interpreted as characterizing the subset of participants whose data supported stable model fits, rather than the full sample.

Several factors likely contributed. First, the vector sum approach assumes that the visual contribution to PU varies systematically with background orientation, which in practice requires a sufficiently strong and regular visual modulation across orientations to identify the visual parameters (Dyde et al., 2006). When an individual's PU shows weak modulation, high trial noise, fatigue-related lapses, or idiosyncratic background responses, the visual parameters become poorly identifiable, and optimisation can fail to

converge on a stable solution. Second, the model is effectively parameter-coupled: because cue contributions compete to explain the same PU outcome, uncertainty in one component (often vision) propagates to the body and gravity estimates, producing unstable individual fits when the dataset does not cleanly separate these influences. A related constraint is that vector-sum cue weights are compositional by definition. Vision, body, and gravity contributions compete to explain the same perceptual outcome and must sum to a constant, so changes in one component necessarily imply changes in at least one other component. Accordingly, these cue weights are best interpreted as descriptive indices of how behaviour is captured by the model within a given task context, rather than as direct independent readouts of neural reliabilities.

Third, the vector sum model assumes obligatory integration of all available cues with fixed weights within a condition. However, a large body of work shows that multisensory perception can switch between integration and segregation depending on inferred common cause, meaning that under large conflicts the nervous system may effectively discount an outlying cue rather than combine it linearly (de Winkel et al., 2018; Körding et al., 2007; Shams & Beierholm, 2022). In these cases, a fixed-weight vector sum model can mischaracterize behaviour and can also become difficult to fit, because the best description may be a regime shift (for example, near-discarding of one cue) rather than a smooth change in weights.

Next, post-exposure changes in upright-related judgements should not be assumed to reflect a single reweighting process. Sensorimotor adaptation work in VR shows that the magnitude and persistence of aftereffects can depend on instruction and error attribution, implying that post-VR shifts can reflect a mixture of perceptual recalibration, strategy, and beliefs about causal structure (Wähnert & Schäfer, 2024).

Finally, a related limitation is that headset fit and interpupillary distance (IPD) calibration were not systematically measured or reported across the three studies. This is relevant because Stanney and colleagues (2020) demonstrated that IPD non-fit to headset specifications can differentially affect female users and significantly increase cybersickness severity. Kourtesis and colleagues (2024) confirmed that with proper eye-tracking IPD calibration, sex did not independently predict cybersickness outcomes. The Oculus Quest used in all three studies offers a fixed-lens IPD adjustment with three settings (56, 63, and 68 mm), which may not accommodate the full range of participant IPD values. Future work should report participant IPD, headset IPD setting, and any mismatch, and include these as covariates in cybersickness analyses to disentangle ergonomic from biological contributions to individual differences.

These limitations do not negate the usefulness of the vector sum framework as a descriptive model, but they do constrain what can be claimed from individual-level estimates in small samples. In this dissertation, we therefore interpret cue-weight outputs as model-based indices when fits are stable and when the underlying PU or SVV modulation is well characterized, and we treat non-convergent or unstable fits as non-interpretable rather than forcing inference. More flexible approaches, including causal inference models and hierarchical frameworks that pool information across participants, provide a principled direction for future work aimed at individual-level prediction under conflict (de Winkel et al., 2018; Irmak et al., 2023; Körding et al., 2007).

## 5.5 Implications and Future Directions

Despite the limitations discussed above, this dissertation contributes convergent evidence on how VR exposure relates to multisensory orientation judgements and cybersickness, while also clarifying where current reweighting metrics and modelling approaches are least stable. First, it extends Chung and Barnett-Cowan (2023) by introducing visual backgrounds into the SVV task (Chapter 2) and by applying OCHART as a complementary measure of multisensory integration (Chapter 3). As discussed in Section 5.2, the task dependence of the reweighting–cybersickness relationship is itself informative, because SVV and OCHART place different constraints on cue geometry and reliability, which changes how strongly each task expresses the specific vestibular–body conflict that VR likely perturbs.

Second, the EVS study (Chapter 4) is best interpreted as a feasibility and methods study of vestibular reliability manipulation during VR, rather than as a definitive test of EVS as a cybersickness countermeasure. Although cue-weight modelling was not stable for a substantial subset of participants, the behavioural outcome was clear in showing increased cybersickness during EVS, which is consistent with reports that some vestibular stimulation protocols can exacerbate discomfort and disrupt adaptation under visually induced motion conditions (Qi et al., 2021). More broadly, these outcomes suggest that vestibular stimulation effects are highly parameter and context dependent. For example, Allred et al. (2025) showed that galvanic vestibular stimulation waveforms designed using a sensory-conflict metric can reduce motion sickness more effectively than generic stimulation, highlighting that efficacy is strongly parameter and model dependent.

Third, this work reinforces the need to move beyond group-level summaries when studying cybersickness mechanisms. A complementary design direction is to reduce ambiguity by coupling stimulation to the user's motion or to the VR camera motion, rather than delivering fixed bilateral stochastic input. Motion-synchronized GVS approaches in VR report reductions in discomfort and sickness, consistent with the

idea that stimulation may be most useful when it increases sensory congruence rather than adding non-naturalistic vestibular perturbations (Groth et al., 2022; Pradhan et al., 2022). Neurophysiological measures are a promising complement to behavioural and modelling approaches, but the current evidence base supports a cautious framing. A recent scoping review of brain activity during cybersickness highlighted substantial heterogeneity in paradigms, outcomes, and analytic choices across EEG studies, with relatively limited convergence on a single robust neural signature that generalizes across VR contexts (Chang et al., 2023). Related work using other modalities, including fNIRS during cybersickness-inducing VR content, suggests that symptom severity can correlate with cortical activation patterns, but these findings remain context-dependent and are not yet sufficient to adjudicate between competing mechanistic accounts of conflict detection versus conflict resolution (Yeo et al., 2024). For the present dissertation, the most defensible role of neural measures is therefore not as a standalone explanation, but as a way to triangulate whether individuals who show specific behavioural adaptation patterns (for example, post-VR re-anchoring in upright judgements) also exhibit consistent physiological or cortical correlates during exposure.

Looking forward, several specific directions emerge from this work. First, future studies should recruit larger and more diverse samples, including stratified groups with known differences in baseline sensory profiles (for example, dancers or athletes with high proprioceptive precision, and older adults with age-related vestibular decline), to test whether the reweighting to cybersickness relationship is moderated by baseline cue dominance. Second, the vector sum model should be complemented by Bayesian causal inference approaches that can explicitly accommodate cue segregation under large conflicts, which is increasingly recognised as a realistic mode of multisensory processing when cues become unreliable or mutually inconsistent (de Winkel et al., 2018; Körding et al., 2007). Third, EVS parameters should be optimised using computationally guided, participant-specific estimates of vestibular sensitivity and conflict, following recent work showing that vestibular stimulation can reduce sickness when the waveform is designed to target the predicted conflict signal rather than applied generically (Allred et al., 2025). Fourth, repeated-exposure paradigms that reduce discomfort early and then probe generalization can help disentangle gradual adaptation (habituation) from acute, session-specific reweighting, as demonstrated in CARE-style training protocols (Doty et al., 2025; Kelly et al., 2026). More broadly, future repeated-exposure designs should test whether reweighting capacity itself can be trained over time. If so, the clinically relevant target would shift from identifying fixed “susceptible” individuals to identifying which baseline profiles show the greatest capacity for adaptive improvement across sessions.

# Chapter 6
# Conclusion

This dissertation tested whether individual differences in sensory reweighting help explain variability in cybersickness during virtual reality exposure. Across three studies using the Subjective Visual Vertical (SVV), The Oriented CHAracter Recognition Test (OCHART), and Electrical Vestibular Stimulation (EVS), the results provide partial and conditional support for this hypothesis. In Experiment 1, SVV estimates showed a robust gravity-ward shift following VR exposure, and the grey (vision-free) condition produced the clearest evidence that post-exposure changes in verticality perception relate to cybersickness severity, consistent with prior work linking sensory reweighting to cybersickness susceptibility (Chung & Barnett-Cowan, 2023). In contrast, Experiment 2 showed that Perceptual Upright (PU) was strongly modulated by visual background orientation, but VR exposure did not produce a consistent group-level PU shift, suggesting that OCHART was strongly influenced by visual context, which may reduce sensitivity to subtle post-VR changes under some conditions (Dyde et al., 2006). Experiment 3 further highlighted feasibility constraints and an important boundary condition. EVS increased cybersickness severity and reduced VR tolerance time, while producing no reliable group-level change in perceptual upright or cue weighting, and cue weight estimation was unstable for a subset of participants. This outcome supports the conclusion that vestibular stimulation effects are parameter sensitive and context dependent rather than uniformly beneficial and, under the present stimulation parameters, are more consistent with EVS acting as an additional source of visuo-vestibular conflict than as a simple degrader of vestibular reliability. At the same time, Chapter 3 identified a key individual-differences pattern in the pooled OCHART analysis, where increased post-VR visual weighting was associated with higher cybersickness severity, consistent with a maladaptive reliance on vision under conflict, although this association differed across study samples and requires replication.

Taken together, these findings support a conservative conclusion: VR-induced conflict can alter upright-related perceptual judgements and can be accompanied by changes consistent with cue redistribution, but the extent to which these changes predict cybersickness depends on task demands, visual context, and model stability. This pattern aligns with broader evidence that cybersickness is multifactorial and highly heterogeneous, even within typical young-adult convenience samples (Caserman et al., 2021; Saredakis et al., 2020), and that stable baseline “sensory dominance” measures may be less informative than dynamic, context-dependent adaptation processes (Kourtesis et al., 2024). In this sense, susceptibility is best understood not as a purely fixed trait, but as an interaction between trait-like baseline sensory tendencies

and state-dependent reweighting processes that unfold under conflict. Baseline cue preferences may shape the starting point from which an individual enters VR, but the more relevant factor may be how efficiently those preferences can be updated when the sensory reliability structure changes. This framing also leaves open the possibility that adaptive reweighting can be strengthened over time through repeated exposure or training. It also suggests that simple obligatory-integration models, such as weighted vector summation, may be insufficient for some participants and conditions, particularly when sensory conflict is large and the nervous system may instead engage cue segregation or causal inference strategies (de Winkel et al., 2018; Körding et al., 2007; Shams & Beierholm, 2022).

This dissertation provides a framework for understanding cybersickness that goes beyond simply describing symptoms and instead focuses on the processes that shape how people respond to sensory conflict. Across three experiments, the findings show that upright perception tasks can be used to examine multisensory cue weighting, that VR-induced conflict can shift the relative contributions of visual, bodily, and gravitational cues, and that individual differences in how flexibly those cues are reweighted are meaningfully related to cybersickness outcomes. At the same time, this work also shows that these relationships are not universal. They depend on the task being used, the assumptions of the model, and the parameters of the manipulation, as illustrated by the negative effects of suprathreshold EVS in Experiment 3.

Several clear directions follow from these findings. First, future studies should include standardised measures of susceptibility, such as the Motion Sickness Susceptibility Questionnaire and the Simulator Sickness Questionnaire, alongside repeated FMS ratings collected throughout VR exposure. This would allow both trait-level susceptibility and moment-to-moment symptom progression to be measured, rather than relying on a single post-exposure score. Second, future work should recruit participants with more diverse sensory backgrounds, including athletes, dancers, individuals with vestibular dysfunction, and a wider age range, to determine whether baseline cue weighting profiles moderate the relationship between reweighting and cybersickness, or whether the more important factor is the ability to update those weights adaptively. Third, vestibular stimulation protocols should move toward individually calibrated, threshold-referenced amplitudes, where each participant's perceptual threshold is measured first and the stimulation is then set to a fixed proportion of that threshold, such as 40 to 50%, in line with the stochastic resonance range where beneficial effects have been reported (Galvan-Garza et al., 2018; Goel et al., 2015). When combined with motion-coupled waveform designs that better match the vestibular input to the VR motion profile (Allred et al., 2025; Groth et al., 2022), this would provide a stronger test of whether vestibular noise can support adaptive reweighting rather than add further conflict. Fourth, the weighted vector sum model should be complemented by hierarchical and causal-inference approaches that are better able to

account for cue segregation under large conflicts (de Winkel et al., 2018; Körding et al., 2007), which may allow more interpretable estimates to be recovered from participants whose data are currently excluded by forced-fusion assumptions. Finally, repeated-exposure designs based on the CARE paradigm (Doty et al., 2025; Kelly et al., 2026) would help determine whether the single-session reweighting patterns observed here predict how quickly participants adapt across repeated sessions, and whether reweighting capacity itself can be strengthened over time. Overall, these findings move the field beyond identifying who gets sick in VR and closer to understanding why, with the longer-term goal of developing personalised and mechanistically informed countermeasures for cybersickness.

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